

Looped_Transformer

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1 Looped Transformer

1.1 理论

Looped Transformer 所有层共享相同的权重 θ ，每层的输入是前一层输出与初始输入的和：

$$\begin{aligned} h_0 &= \text{input} \\ h_l &= \text{TransformerBlock}(h_{l-1} + h_0 | \theta) \quad \text{for } l = 1, 2, \dots, L \end{aligned}$$

1.1.1 截断损失函数

$$\text{Loss}(\theta) = \mathbb{E}_P \left[\frac{1}{b - b_0} \sum_{t=b_0}^b \frac{1}{k+1} \sum_{i=0}^k (Y_t(P^i | \theta) - f(x_{i+1}))^2 \right]$$

其中 - $\min_{\theta}$ ：这是我们在寻找一组最优的物理定律（模型参数 θ ）。 - $\mathbb{E}_P$ ：对所有生成的 Prompt 分布求期望。在统计物理中，这相当于求“系综平均”，即我们在无数种不同的初始条件（数据集）下测试这个定律是否普适。 - $\frac{1}{b-b_0} \sum_{t=b_0}^b$ ：这是时间平均。 b 是总迭代次数， T 是窗口大小， $b_0 = \max(b - T, 0)$ 。我们只关心系统演化到“稳态”附近的这段时间内（也就是 b_0 到 b 的表现），前面瞬态的误差我们不计较。 - $\frac{1}{k+1} \sum_{i=0}^k$ ：这是空间/序列平均。也就是在给定不同长度的上下文 P^i 时，模型预测下一个点的值与真实函数 $f(x_{i+1})$ 之间的均方误差。 - $Y_t(P^i | \theta)$ ：在时间 t 、给定历史信息 P^i 和系统参数 θ 时，系统当前的状态输出。

1.1.2 下游任务

- 数据生成方式：这篇论文解决的是纯粹的数学数据拟合问题。数据集是在训练循环中即时生成的浮点数向量。
- 以最基础的 **线性回归（Linear Regression）** 为例：
 - 维度设定：问题维度 $d = 20$ ，上下文样本数 $k = 40$ 。
 - 生成逻辑：每次迭代前，随机采样一个真实的参数 $w \sim \mathcal{N}(0, I_d)$ 。
 - 采样输入：生成 $k + 1$ 个输入样本（包含测试样本） $x_i \sim \mathcal{N}(0, I_d)$ 。

- 计算标签: $y_i = w^T x_i$ 。
- 拼接成 Prompt: 最终喂给模型的序列 P 是一个交替的序列 $(x_1, y_1, x_2, y_2, \dots, x_k, y_k, x_{test}, y_{dummy})$ 。
- 线性回归可以进一步改成非线性回归 (比如 ReLU 与 Linear 叠加), 或者更复杂的物理系统。

把 Prompt 映射到高维 Embedding 空间 双通道映射与“拉链式”交织

由于 x 和 y 代表不同的物理量 (一个是空间坐标/特征, 一个是目标观测值), 我们需要赋予它们独立的投影空间:

1. 定义两个独立的线性层:

```
read_in_x = nn.Linear(20, 256)
read_in_y = nn.Linear(1, 256)
```

2. 把前 k 个 x_i 丢给 `read_in_x`, 变成张量 X_{emb} , 形状为 $[batch, k, 256]$ 。
3. 把前 k 个 y_i 丢给 `read_in_y`, 变成张量 Y_{emb} , 形状为 $[batch, k, 256]$ 。
4. 像拉拉链一样, 在 `seq_len` 维度上把 X_{emb} 、 Y_{emb} 交织 (Interleave) 起来, 拼接成最终长度为 $2k$ 的提示向量 P 。

在工程实现中, 我们在 x_{test} 后填充了一个 y_{dummy} (如 0), 使实际拼接长度变为 $2k+2$ (偶数)。因此, 真正对 x_{test} 的预测结果位于输出序列的倒数第二个位置, 即索引 $[-2]$ 处。

1.2 架构各模块代码

1.2.1 import

```
[178]: import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import time
from operator import sub
import copy
import numpy as np
import math
import concurrent.futures
import threading
import os
```

1.2.2 位置编码 (PE, Position Encoding)

绝对位置编码 (APE, Absolute Position Encoding)

```
[179]: class APE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096, b=10000):
        super().__init__()
        theta_i = 1 / (b ** (torch.arange(0, d_model, 2).float() / d_model)) # [d_model/2]
        m = torch.arange(max_seq_len).float() # [max_seq_len]
        m_theta_i = torch.outer(m, theta_i) # [max_seq_len, d_model/2]
        # APE 核心: 直接生成一个完整的 pe 矩阵, 偶数维度放 sin, 奇数维度放 cos
        stacked=torch.stack((torch.sin(m_theta_i), torch.cos(m_theta_i)), dim=-1) # [max_seq_len, d_model/2, 2]
        pe=stacked.flatten(1,2) # [max_seq_len, d_model]
        self.register_buffer('pe', pe.unsqueeze(0)) # shape: [1, max_seq_len, d_model]

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        return x + self.pe[:, :x.shape[1], :]
```

Learned APE

```
[180]: class LearnedAPE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096):
        super().__init__()
        self.pe = nn.Embedding(max_seq_len, d_model)

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        seq_len = x.shape[1]
        positions = torch.arange(seq_len, device=x.device)
        return x + self.pe(positions).unsqueeze(0)
```

ALiBi (Attention with Linear Biases) ALiBi 通过更改内积的方式引入线性位置偏置, 从而实现了位置编码的功能。

$$\text{Score}(q_i, k_j) = q_i \cdot k_j - m \cdot (i - j) \quad (i \geq j)$$

其中 m 是一个与头数 (H) 相关的斜率参数，硬编码为

$$m_h = 2^{-\frac{8h}{H}} \quad h = 0, 1, \dots, H-1$$

```
[181]: class ALiBi(nn.Module):
    def __init__(self, num_heads, max_seq_len=4096):
        super().__init__()
        dist = torch.arange(max_seq_len) # [max_seq_len]
        dist_matrix = torch.abs(dist.view(-1, 1) - dist.view(1, -1)) #
        ↪ [max_seq_len, max_seq_len]
        bias = -(2**-(8*torch.arange(1,num_heads+1) / num_heads)).view(-1, 1,
        ↪ 1) * dist_matrix.unsqueeze(0) # [num_heads, max_seq_len, max_seq_len]
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
        ↪ tril (triangular lower) 提取下三角元素, 并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
        ↪ float('-inf')) # 下三角元素置 0, 上三角元素置 -inf
        self.register_buffer('fused_mask', bias + mask) # [num_heads,
        ↪ max_seq_len, max_seq_len]

    def forward(self, seq_len):
        # score shape: [batch_size, num_heads, seq_len, seq_len]
        return self.fused_mask[:, :seq_len, :seq_len]
```

原来的 RoPE (Rotary Position Encoding)

```
[182]: class RoPE(nn.Module):
    def __init__(self, d_k, max_seq_len = 4096, b=10000):
        super().__init__()
        theta_i = 1/(b**(torch.arange(0,d_k,2).float()/d_k)) # \theta_i =
        ↪ b^{-\frac{2i}{d}}, \quad i \in \{0, 1, \dots, \frac{d}{2}-1\}
        m = torch.arange(max_seq_len).float() # m = [0, 1, ..., seq_len-1]
        m_theta_i = torch.outer(m, theta_i) # [seq_len, d_k/2]
        cos = torch.cos(torch.cat((m_theta_i, m_theta_i), dim=-1)) # [seq_len,
        ↪ d_k]
        sin = torch.sin(torch.cat((m_theta_i, m_theta_i), dim=-1)) # [seq_len,
        ↪ d_k]
```

```

        self.register_buffer('cos', cos[None, None, :, :]) # 好处: cos 和 sin
        不需要更新参数, 注册为 buffer 后会自动放到正确的设备上

        self.register_buffer('sin', sin[None, None, :, :]) # 扩充维度以适应后续
        计算

```

```

def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
    # x: [batch_size, num_heads, seq_len, d_k]
    seq_len = x.shape[2]
    d_2 = x.shape[-1] // 2
    cos = self.cos[:, :, :seq_len, :] # [1, 1, seq_len, d_k]
    sin = self.sin[:, :, :seq_len, :]
    x_first_half = x[..., :d_2]
    x_second_half = x[..., d_2:]
    x_flip = torch.cat((-x_second_half, x_first_half), dim=-1)
    x_out = x * cos + x_flip * sin # 旋转位置编码
    return x_out

```

自创: Multi-Scale Untied Positional Encoding (MS-UPE)

```

[183]: class MS_UPE(nn.Module):
    def __init__(self, num_heads, d_k, max_seq_len = 4096, b_0=10000,
    ↪ head_ratio=2):
        super().__init__()
        b=(b_0*(head_ratio**((torch.arange(0, num_heads))))).float() # [num_heads]
        # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0, 1, \dots, \frac{d}{2}-1\}
        theta_i = 1/(b.unsqueeze(1)**(torch.arange(0, d_k, 2).float()/d_k).
    ↪ unsqueeze(0)) # [num_heads, d_k/2]
        m = torch.arange(max_seq_len).float() # m = [0, 1, ..., seq_len-1]
        m_theta_i = torch.einsum('m, hk->hmk', m, theta_i) # [num_heads,
    ↪ seq_len, d_k/2]
        pe = torch.cat((torch.cos(m_theta_i), torch.sin(m_theta_i)), dim=-1) #
    ↪ [num_heads, seq_len, d_k]
        self.register_buffer('pe', pe[None, :, :, :]) # 好处: pe 不需要更新参数,
        注册为 buffer 后会自动放到正确的设备上

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]

```

```

seq_len = x.shape[2]
pe = self.pe[:, :, :seq_len, :] # [1, num_heads, seq_len, d_k]
x_out = x + pe
return x_out

```

1.2.3 Multi-Head Attention

注：APE 应该加在 ToyModel 上而不是 TransformerBlock 上，因为我们只需要最开头输入处编码位置，而不是叠加 num_blocks 次。

```

[184]: class MultiHeadAttention(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096,
        ↪pe_type='learned_ape', b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        assert d_model % num_heads == 0, "d_model must be divisible by
        ↪num_heads"
        # 维度属性
        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads
        # 权重矩阵
        self.W_Q = nn.Linear(d_model, d_model, bias=False)
        self.W_K = nn.Linear(d_model, d_model, bias=False)
        self.W_V = nn.Linear(d_model, d_model, bias=False)
        self.W_O = nn.Linear(d_model, d_model, bias=False)
        # 位置编码模块
        self.pe_type = [pe_type] if isinstance(pe_type, str) else pe_type
        self.qk_pe_modules = nn.ModuleList()
        self.sc_pe_modules = nn.ModuleList()
        for pe in self.pe_type:
            if pe == 'rope':
                pe_module = RoPE(d_k=self.d_k, max_seq_len=max_seq_len,
                ↪b=b_rope_or_upe)
                self.qk_pe_modules.append(pe_module) # RoPE 应用于 Q 和 K
            elif pe == 'ms_upe':
                pe_module = MS_UPE(num_heads=num_heads, d_k=self.d_k,
                ↪max_seq_len=max_seq_len, b_0=b_rope_or_upe, head_ratio=head_ratio_upe)

```

```

        self.qk_pe_modules.append(pe_module) # MS-UPE 应用于 Q 和 K
    elif pe == 'alibi':
        pe_module = ALiBi(num_heads=num_heads, max_seq_len=max_seq_len)
        self.sc_pe_modules.append(pe_module) # ALiBi 应用于 score 矩阵
    elif pe in ['ape', 'learned_ape']:
        pass # APE 和 Learned APE 在输入时直接加到 x 上, 不需要单独模块
# 因果掩码
tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
↪tril (triangular lower) 提取下三角元素, 并把上三角元素置 0
mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
↪float('-inf')) # 下三角元素置 0, 上三角元素置 -inf
self.register_buffer('mask', mask[None, None, :, :]) # [1, 1, seq_len,
↪seq_len]

def forward(self, x): # x: [batch_size, seq_len, d_model]
    # 线性投影与分头
    batch_size, seq_len, _ = x.shape
    Q = self.W_Q(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # [batch_size, num_heads, seq_len, d_k]
    K = self.W_K(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # 交换 (1, 2) 是因为之后 softmax 时要进行矩阵乘法 (只乘后两维)
    V = self.W_V(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # 在 seq_len 维度上分配注意力
    # 根据选择的 PE 类型对 Q 和 K 进行位置编码
    for pe_module in self.qk_pe_modules:
        Q = pe_module(Q)
        K = pe_module(K)
    if not self.training:
        # 缩放点积注意力与因果掩码 Causal Masking
        ## Query-Key 点积计算注意力得分
        scores = torch.matmul(Q, K.transpose(-2, -1)) / (self.d_k ** 0.5)
↪# [batch_size, num_heads, seq_len, seq_len]
        if self.sc_pe_modules:
            fused_mask = self.sc_pe_modules[0](seq_len) # [num_heads,
↪seq_len, seq_len]
            scores = scores + fused_mask # 将 ALiBi 的偏置加到得分上

```

```

        attention = F.softmax(scores, dim=-1) # [batch_size,
↪ num_heads, seq_len, seq_len]
    else:
        ## 因果掩码和 softmax 计算注意力权重
        attention = F.softmax(scores + self.mask[..., :seq_len, :
↪ seq_len], dim=-1) # [batch_size, num_heads, seq_len, seq_len]
        self.captured_attention = attention # ** 新增 **: 捕获注意力矩阵, 用
hook 记录下来, 方便后续分析
        ## 乘以 Value 完成加权求和
        out = torch.matmul(attention, V) # [batch_size, num_heads,
↪ seq_len, d_k]
    else:
        self.captured_attention = None # 训练模式下不捕获注意力矩阵, 节省内存
        if self.sc_pe_modules:
            fused_mask = self.sc_pe_modules[0](seq_len) # [num_heads,
↪ seq_len, seq_len]
            out = F.scaled_dot_product_attention(Q, K, V,
↪ attn_mask=fused_mask, is_causal=False) # [batch_size, num_heads, seq_len,
↪ d_k]
        else:
            out = F.scaled_dot_product_attention(Q, K, V, is_causal=True)
↪ # [batch_size, num_heads, seq_len, d_k]
            # 多头拼接并乘以输出矩阵 (必须先内存连续化 (contiguous) 再 view, 否则会报错)
            out = out.transpose(1, 2).contiguous().view(batch_size, seq_len, -1)
↪ # [batch_size, seq_len, d_model]
            H = self.W_O(out) # [batch_size, seq_len, d_model]
        return H

```

1.2.4 SwiGLU 和单个 Transformer Block 架构不变

```

[185]: class SwiGLU(nn.Module):
    def __init__(self, d_model, d_hidden):
        super().__init__()
        self.W = nn.Linear(d_model, d_hidden, bias=False)
        self.V = nn.Linear(d_model, d_hidden, bias=False)

```



```

        self.W2 = nn.Linear(d_hidden,d_model,bias=False)
    def forward(self,x):
        x1 = F.silu(self.W(x))
        x2 = self.V(x)
        x_out = self.W2(x1 * x2)
        return x_out

```

```

[186]: class TransformerBlock(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096, multiplier=4,
        ↪norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
        ↪b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.norm1 = nn.RMSNorm(d_model) # 第一次归一化
            self.norm2 = nn.RMSNorm(d_model) # 第二次归一化
        elif norm_type == 'layernorm':
            self.norm1 = nn.LayerNorm(d_model) # 第一次归一化
            self.norm2 = nn.LayerNorm(d_model) # 第二次归一化
        self.attention = MultiHeadAttention(num_heads=num_heads,
        ↪d_model=d_model, max_seq_len=max_seq_len, pe_type=pe_type,
        ↪b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
        if ffn_type == 'swiglu':
            self.ffn = SwiGLU(d_model=d_model, d_hidden=d_model * multiplier)
        elif ffn_type == 'gelu':
            self.ffn = nn.Sequential(
                nn.Linear(d_model, d_model * multiplier),
                nn.GELU(),
                nn.Linear(d_model * multiplier, d_model)
            )

    def forward(self, x):
        # 第一次归一化 Pre-Norm
        x_norm1 = self.norm1(x)
        # 全局信息交互 Multi-Head Attention (MHA)
        h = self.attention(x_norm1)
        # 第一次残差连接

```

```

x1 = x + h
# 第二次归一化
x_norm2 = self.norm2(x1)
# 前馈网络 Feed-Forward Network (FFN)
x_out = self.ffn(x_norm2)
# 第二次残差连接
x2 = x1 + x_out
return x2

```

1.2.5 Attention Probe

返回全量 attention 的 probe

```

[187]: class AttentionProbe:
    def __init__(self):
        self.captured_data = []
    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            attention = module.captured_attention.detach().cpu().numpy()
            self.captured_data.append(attention)
        else:
            self.captured_data.append(None) # 如果没有捕获到注意力矩阵, 记录一个
None 占位符
    def reset(self):
        self.captured_data = []

```

计算 sink score 和 sink rate 的 probe 针对层 l 的第 k 个 token 位置, sink rate 的计算公式定义如下: - Sink Score (汇聚得分):

$$\text{sink score}_k^{(l,h)} = \frac{1}{T} \sum_{t=0}^{T-1} \alpha_{tk}^{(l,h)}$$

- Sink Rate (汇聚率):

$$\text{sink rate}_k^{(l)} = \frac{1}{H} \sum_{h=1}^H \mathbb{I}(\text{sink score}_k^{(l,h)} \geq \tau)$$

公式参数说明: - $\alpha_{tk}^{(l,h)}$ 表示在层 l 的第 h 个注意力头中, 从 token t 到 token k 的注意力权重。 - T 是序列的总长度。 - H 是每层注意力头的总数量。 - $\mathbb{I}$ 是指示函数 (当条件 $\text{sink score}_k^{(l,h)} \geq \tau$ 成立时取值为 1, 否则为 0)。 - 阈值 τ 通常被默认设置为 0.3。

```
[188]: class SinkMetricsProbe:
    def __init__(self, threshold=0.3):
        self.threshold = threshold
        self.captured_scores = []
        self.captured_rates = []

    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            with torch.no_grad(): # attention 矩阵仍在 GPU 上
                attention = module.captured_attention # [batch_size, num_heads,
↪seq_len, seq_len]
                sink_attention = attention[:, :, :, 0] # [batch_size,
↪num_heads, seq_len]
                score_head = sink_attention.mean(dim=2).mean(dim=0) #
↪[num_heads]
                score = score_head.mean().item() # 平均所有头的 sink score
                rate = (score_head >= self.threshold).float().mean().item() #
↪大于等于 threshold 的头占总头数的比例
                self.captured_scores.append(score)
                self.captured_rates.append(rate)
            else:
                self.captured_scores.append(None) # 如果没有捕获到注意力矩阵, 记录一个 None 占位符
                self.captured_rates.append(None)

    def reset(self):
        self.captured_scores = []
        self.captured_rates = []
```

1.2.6 Toy Model: Looped Transformer

```
[189]: class ToyModel(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, max_seq_len=4096,
↪x_init='prompt',
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
↪b_rope_or_upe=10000, head_ratio_upe=2, sink_threshold=0.3,
```

```

        loop=True, residual_gate=(1,1), residual_gate_type='fixed',
↪residual_random=(1,0.1)):
    super().__init__()
    self.loop = loop
    self.x_init = x_init
    if residual_gate==(0,0):
        raise ValueError("residual_gate cannot be (0,0) because then the
↪model would be unable to learn anything (the input would be completely
↪blocked)")
    if residual_gate[1] ==0 and x_init=='zero':
        self.x_init = 'prompt'
        print("Warning: x_init is set to 'prompt' because residual_gate[1]
↪is 0, which would make the model completely blind.")

    if loop:
        self.transformer_block = TransformerBlock(num_heads=num_heads,
↪d_model=d_model, max_seq_len=max_seq_len, norm_type=norm_type,
↪ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
↪head_ratio_upe=head_ratio_upe) # 这里保证了各层权重始终相同
        self.probe = AttentionProbe()
        self.sink_metrics_probe = SinkMetricsProbe(threshold=sink_threshold)
        self.transformer_block.attention.register_forward_hook(self.probe)
        self.transformer_block.attention.register_forward_hook(self.
↪sink_metrics_probe)
        self.captured_attention = None
        self.captured_sink_scores = None
        self.captured_sink_rates = None
    else:
        self.transformer_block = nn.ModuleList([
            TransformerBlock(num_heads=num_heads, d_model=d_model,
↪max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type,
↪pe_type=pe_type, b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
            for _ in range(num_blocks)
        ])
        self.probe = [AttentionProbe() for _ in range(num_blocks)]

```

```

        self.sink_metrics_probe = □
        ↪ [SinkMetricsProbe(threshold=sink_threshold) for _ in range(num_blocks)]
        for block, probe, sink_probe in zip(self.transformer_block, self.
        ↪ probe, self.sink_metrics_probe):
            block.attention.register_forward_hook(probe)
            block.attention.register_forward_hook(sink_probe)

        residual_random = torch.tensor(residual_random, dtype=torch.float32)
        if residual_gate == 'random':
            residual_gate_tensor = torch.
            ↪ randn(2)*residual_random[1]+residual_random[0] # 随机初始化门控参数, 初始正态分
            布
        else:
            residual_gate_tensor = torch.tensor(residual_gate, dtype=torch.
            ↪ float32) # 将输入的元组转换为张量, 方便后续计算
            if residual_gate_type == 'fixed':
                self.register_buffer('residual_gate', residual_gate_tensor) # 固定
                的门控参数, 注册为 buffer 使其成为模型状态的一部分, 但不参与梯度更新
            elif residual_gate_type == 'learnable_scalar':
                self.residual_gate = nn.Parameter(torch.
                ↪ ones(2)*residual_gate_tensor) # 可学习的标量门控参数, 初始值为用户指定的值
            elif residual_gate_type == 'learnable_vector':
                if residual_gate == 'random':
                    self.residual_gate = nn.Parameter(torch.randn(d_model, □
                    ↪ 2)*residual_random[1]+residual_random[0]) # 可学习的向量门控参数, 初始值为随机
                    向量
                else:
                    self.residual_gate = nn.Parameter(torch.ones(d_model, □
                    ↪ 2)*residual_gate_tensor) # 可学习的向量门控参数, 初始值为全 1 向量

        pe_type = self.transformer_block.attention.pe_type
        self.ape = nn.ModuleList()
        for pe in pe_type:
            if pe == 'ape':
                self.ape.append(APE(d_model, max_seq_len=max_seq_len))
            elif pe == 'learned_ape':

```

```

        self.ape.append(LearnedAPE(d_model, max_seq_len=max_seq_len))
    elif pe in ['rope', 'ms_upe']:
        pass # RoPE 和 MS_UPE 在 MultiHeadAttention 内部处理位置编码，不
需要单独的模块在这里处理

    self.num_blocks = num_blocks
    self.num_heads = num_heads
    self.d_model = d_model
    self.max_seq_len = max_seq_len
    self.pe_type = pe_type

def forward(self, x_0, num_eff=15, current_blocks=None):
    # num_eff 是需要加入误差计算的有效层数，也即理论公式中的  $T$ 
    # x 的形状为 [batch_size, seq_len, d_model]
    a=self.residual_gate[...,0]
    b=self.residual_gate[...,1]
    for ape_module in self.ape:
        x_0 = ape_module(x_0) # 绝对位置编码只加在最开始输入处
    if self.x_init == 'prompt':
        x = x_0
    elif self.x_init == 'zero':
        x = torch.zeros_like(x_0) # 直接用全零向量作为初始输入，测试模型是否能
从零基础学习
    else:
        raise ValueError(f"Invalid x_init value: {self.x_init}. Expected_
↪ 'prompt' or 'zero'.")
    if current_blocks is None:
        current_blocks = self.num_blocks
    b_0 = max(0, current_blocks - num_eff) # 计算需要加入误差计算的有效层数对
应的起始层索引
    outputs = []
    if self.loop:
        self.probe.reset() # 每次前向传播前重置捕获的数据，避免混淆不同次前向
传播的注意力矩阵
        self.sink_metrics_probe.reset() # 同样重置 sink metrics 的捕获数据
        for i in range(current_blocks):

```

```

        if i == b_0:
            x = x.detach() # 释放前面的计算图
            x.requires_grad_(True)
            x = self.transformer_block(a*x+b*x_0)
            if i >= b_0:
                outputs.append(x)
            self.captured_attention = list(self.probe.captured_data) # 显式 copy
一份数据，避免后续被修改
            self.captured_sink_scores = list(self.sink_metrics_probe.
↪captured_scores)
            self.captured_sink_rates = list(self.sink_metrics_probe.
↪captured_rates)
        else:
            for probe in self.probe:
                probe.reset()
            for sink_probe in self.sink_metrics_probe:
                sink_probe.reset()
            for i, block in enumerate(self.transformer_block[:current_blocks]):
↪# 只迭代当前有效层数，避免不必要的计算
                if i == b_0:
                    x = x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                    x = block(a*x+b*x_0) # x: [batch_size, seq_len, d_model]
                    if i >= b_0:
                        outputs.append(x)
                    self.captured_attention = [probe.captured_data[-1] if probe.
↪captured_data else None for probe in self.probe]
                    self.captured_sink_scores = [sink_probe.captured_scores[-1] if
↪sink_probe.captured_scores else None for sink_probe in self.
↪sink_metrics_probe]
                    self.captured_sink_rates = [sink_probe.captured_rates[-1] if
↪sink_probe.captured_rates else None for sink_probe in self.
↪sink_metrics_probe]
                outputs = torch.stack(outputs, dim=1) # [batch_size, num_eff, seq_len,
↪d_model]
            return outputs

```

1.2.7 Task: Regression

转换头

```
[190]: class RegressionHead(nn.Module):
    def __init__(self, d_model, d_x, d_y=1, bias=False, init_scale=None):
        super().__init__()
        self.read_in_x = nn.Linear(d_x, d_model, bias=bias)
        self.read_in_y = nn.Linear(d_y, d_model, bias=bias)
        if init_scale is not None:
            nn.init.normal_(self.read_in_x.weight, mean=0.0, std=init_scale)
            nn.init.normal_(self.read_in_y.weight, mean=0.0, std=init_scale)
    def forward(self, x_data, y_data):
        # k 相当于 seq_len 的一半, 因为 x 和 y 交织在一起, 所以总的 seq_len 是 2k
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        x_emb = self.read_in_x(x_data) # [batch_size, k, d_model]
        y_emb = self.read_in_y(y_data) # [batch_size, k, d_model]
        # 交织成 (x_1, y_1, x_2, y_2, ..., x_k, y_k)
        stacked = torch.stack((x_emb, y_emb), dim=2) # [batch_size, k, 2, d_model]
        Prompt = stacked.flatten(1,2) # [batch_size, 2k, d_model]
        return Prompt
```

损失函数

```
[191]: class PredictionLoss(nn.Module):
    def __init__(self, d_model, d_y=1, loss_type='mse', layer_weight_decay=1.0, seq_weight_decay=1.0, norm_type='rmsnorm'):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.ln_f = nn.RMSNorm(d_model)
        elif norm_type == 'layernorm':
            self.ln_f = nn.LayerNorm(d_model)
        self.read_out = nn.Linear(d_model, d_y)
        self.layer_weight_decay = layer_weight_decay
        self.seq_weight_decay = seq_weight_decay
        if loss_type == 'mse':
            self.loss_fn = nn.MSELoss(reduction='none')
        elif loss_type == 'l1':
```



```

        self.loss_fn = nn.L1Loss(reduction='none')

    def forward(self, outputs, y_true, is_eval=False, sink_padding=None):
        # outputs: [batch_size, num_eff, seq_len, d_model]
        # y_true: [batch_size, seq_len/2, 1]
        if is_eval:
            y_outputs = outputs[:, -1, -2, :] # 只取最后一层的最后一个位置的输出
            # 作为预测值
            y_outputs = self.ln_f(y_outputs)
            y_pred_final = self.read_out(y_outputs) # [batch_size, d_y]
            return y_pred_final

            y_outputs = outputs[:, :, 0::2, :] # [batch_size, num_eff, seq_len/2, d_model]
            y_outputs = self.ln_f(y_outputs)
            y_preds = self.read_out(y_outputs) # [batch_size, num_eff, seq_len/2, d_y]

            if sink_padding is not None:
                y_preds = y_preds[:, :, sink_padding:, :]
                y_true = y_true[:, sink_padding:, :]
                y_pred_norm = torch.sqrt(torch.mean(y_preds.detach() ** 2)).item()
                y_true_norm = torch.sqrt(torch.mean(y_true.detach() ** 2)).item()
                loss_unreduced = self.loss_fn(y_preds, y_true.unsqueeze(1).
                expand_as(y_preds)) # [batch_size, num_eff, seq_len/2, d_y]

                if self.layer_weight_decay != 1.0:
                    num_eff = outputs.shape[1]
                    layer_weights = self.layer_weight_decay ** torch.arange(num_eff-1,
                    -1, -1, device=outputs.device) # 从最后一层到第一层递减的权重
                    layer_weights = layer_weights / layer_weights.mean()
                    loss_unreduced = loss_unreduced * layer_weights.view(1, num_eff, 1,
                    1) # 按层加权

                if self.seq_weight_decay != 1.0:
                    k=loss_unreduced.shape[2]
                    seq_weights = self.seq_weight_decay ** torch.arange(k-1, -1, -1,
                    device=outputs.device) # 从最后一次预测到第一次预测递减的权重
                    seq_weights = seq_weights / seq_weights.mean()

```

```

        loss_unreduced = loss_unreduced * seq_weights.view(1, 1, k, 1) # 按
位置加权

    return loss_unreduced.mean(), y_pred_norm, y_true_norm

```

拼装

```

[192]: class RegressionSolver(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, d_x, d_y, max_seq_len,
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
↪b_rope_or_upe=10000, head_ratio_upe=2,
        loop=True, loss_type='mse', sink_threshold=0.3,
        residual_gate=(1,1), residual_gate_type='fixed',
↪residual_random=(1,0.1),
        bias=False, init_scale=None, init_std=0.02, x_init='prompt',
        layer_weight_decay=1.0, seq_weight_decay=1.0):
    super().__init__()
    if init_std == 'auto':
        init_std = d_model ** -0.5
    if init_std is not None:
        init_scale = None # 如果用户指定了 init_std, 则忽略 init_scale, 使用
init_std 进行权重初始化

    self.toy_model = ToyModel(num_blocks=num_blocks, num_heads=num_heads,
↪d_model=d_model, max_seq_len=max_seq_len,
        norm_type=norm_type, ffn_type=ffn_type,
↪pe_type=pe_type, sink_threshold=sink_threshold,
        loop=loop, b_rope_or_upe=b_rope_or_upe,
↪head_ratio_upe=head_ratio_upe,
        residual_gate=residual_gate,
↪residual_gate_type=residual_gate_type, residual_random=residual_random,
↪x_init=x_init)

    self.head = RegressionHead(d_model=d_model, d_x=d_x, d_y=d_y,
↪bias=bias, init_scale=init_scale)

    self.loss_fn = PredictionLoss(d_model=d_model, d_y=d_y,
↪loss_type=loss_type, layer_weight_decay=layer_weight_decay,
↪seq_weight_decay=seq_weight_decay, norm_type=norm_type)

    if init_std is not None:
        self.apply(lambda module: self._init_weights(module, init_std))

```

```

std_res = init_std / math.sqrt(2 * num_blocks)
# 遍历所有参数，通过名称匹配出残差输出层
for name, param in self.named_parameters():
    # W_0 是 MultiHeadAttention 的输出
    # W2 是 SwiGLU 的输出
    # ffn.2 是普通 GELU 序列中的最后一个 Linear
    if name.endswith('W_0.weight') or name.endswith('W2.weight') or
↪name.endswith('ffn.2.weight'):
        torch.nn.init.normal_(param, mean=0.0, std=std_res)

def _init_weights(self, module, init_std):
    if isinstance(module, (nn.Linear, nn.Embedding)):
        nn.init.normal_(module.weight, mean=0.0, std=init_std)
        if isinstance(module, nn.Linear) and module.bias is not None:
            nn.init.zeros_(module.bias)

    def forward(self, x_data, y_data, num_eff, current_blocks=None,
↪is_eval=False, sink_padding=None):
        Prompt = self.head(x_data, y_data) # [batch_size, seq_len, d_model]
        outputs = self.toy_model(Prompt, num_eff=num_eff,
↪current_blocks=current_blocks) # [batch_size, num_eff, seq_len, d_model]
        return self.loss_fn(outputs, y_data, is_eval=is_eval,
↪sink_padding=sink_padding)

```

1.2.8 数据生成与加载

数据生成

线性回归数据

```

[193]: def linear_data_generator(batch_size, seq_len, valid_d_x=None, d_x=20, d_y=1,
↪device='cpu', generator=None, ood_kwargs=None):
    with torch.no_grad():
        if ood_kwargs is None:
            ood_kwargs = {}
        if valid_d_x is None:
            valid_d_x = d_x

```

```

w = torch.randn(batch_size, d_x, d_y, device=device,
generator=generator)/(valid_d_x ** 0.5) # 真实线性函数 w: [batch_size, d_x,
d_y]

if valid_d_x < d_x:
    w[:, valid_d_x:, :] = 0.0 # 将多余的输入维度对应的权重置零，保证它们
对输出没有影响

x_distribution = ood_kwargs.get('x_distribution', 'gaussian')
if x_distribution == 'gaussian':
    x_data = torch.randn(batch_size, seq_len//2, d_x, device=device,
generator=generator) # 输入特征 x: [batch_size, seq_len//2, d_x]
elif x_distribution == 'uniform':
    x_data = (torch.rand(batch_size, seq_len//2, d_x, device=device,
generator=generator) * 2 - 1)/(3**0.5) # 均匀分布，方差归一化为 1
elif x_distribution == 'laplace':
    x_data = torch.distributions.Laplace(0, 1/(2**0.5)).
sample((batch_size, seq_len//2, d_x)).to(device) # 拉普拉斯分布，方差归一化为
1

if 'x_scale' in ood_kwargs:
    x_data = x_data * ood_kwargs['x_scale'] # 调整输入特征的尺度，制造
OOD 数据

if 'x_mean_shift' in ood_kwargs:
    x_data = x_data + ood_kwargs['x_mean_shift'] # 平移输入特征的均值，
制造 OOD 数据

if 'x_noise_std' in ood_kwargs:
    x_data = x_data + torch.randn_like(x_data) *
ood_kwargs['x_noise_std'] # 添加输入特征的噪声，制造 OOD 数据

if valid_d_x < d_x:
    x_data[..., valid_d_x:] = 0.0 # 将多余的输入维度对应的数据置零，保证
它们对输出没有影响

y_data = x_data @ w # 计算标签 y = x @ w: [batch_size, seq_len//2, d_y]

if 'y_scale' in ood_kwargs:

```

```

        y_data = y_data * ood_kwargs['y_scale'] # 调整标签的尺度, 制造 OOD
数据
        if 'y_mean_shift' in ood_kwargs:
            y_data = y_data + ood_kwargs['y_mean_shift'] # 平移标签的均值, 制造
OOD 数据
        if 'y_noise_std' in ood_kwargs:
            y_data = y_data + torch.randn_like(y_data) *
            ↪ood_kwargs['y_noise_std'] # 添加标签的噪声, 制造 OOD 数据

        return x_data, y_data

```

非线性回归数据

```

[194]: def nonlinear_data_generator(batch_size, seq_len, valid_d_x=None, d_x=20,
            ↪d_y=1, d_hidden=None, function_callable=None, device='cpu', generator=None,
            ↪ood_kwargs=None):
    '''
        linear(d_x->d_hidden) + nonlinear_func + linear(d_hidden->d_y) 的形式生成数
据
        function_callable: 非线性函数, 如 lambda x: F.relu(2*torch.sin(x))+torch.
            ↪cos(x)
        d_hidden: 隐藏层维度, 控制非线性函数的输入输出维度。
        若 d_hidden=d_x, 则第一层线性变换默认设定为恒等映射, 数据生成过程变为
nonlinear_func + linear 的形式;
        若 d_hidden=d_y, 则第二层线性变换默认设定为恒等映射, 数据生成过程变为
linear + nonlinear_func 的形式。
    '''
    with torch.no_grad():
        if ood_kwargs is None:
            ood_kwargs = {}
        if valid_d_x is None:
            valid_d_x = d_x

        w1 = torch.randn(batch_size, d_x, d_hidden, device=device,
            ↪generator=generator) / (valid_d_x ** 0.5) # 第一层权重 w1: [batch_size, d_x,
            ↪d_hidden]

```

```

w2 = torch.randn(batch_size, d_hidden, d_y, device=device,
↳generator=generator)/(d_hidden ** 0.5) # 第二层权重 w2: [batch_size,
↳d_hidden, d_y]

if valid_d_x < d_x:
    w1[:, valid_d_x:, :] = 0.0 # 将多余的输入维度对应的权重置零, 保证它们
对输出没有影响

x_distribution = ood_kwargs.get('x_distribution', 'gaussian')
if x_distribution == 'gaussian':
    x_data = torch.randn(batch_size, seq_len//2, d_x, device=device,
↳generator=generator) # 输入特征 x: [batch_size, seq_len//2, d_x]
elif x_distribution == 'uniform':
    x_data = (torch.rand(batch_size, seq_len//2, d_x, device=device,
↳generator=generator) * 2 - 1)/(3**0.5) # 均匀分布, 方差归一化为 1
elif x_distribution == 'laplace':
    x_data = torch.distributions.Laplace(0, 1/(2**0.5)).
↳sample((batch_size, seq_len//2, d_x)).to(device) # 拉普拉斯分布, 方差归一化为
1

if 'x_scale' in ood_kwargs:
    x_data = x_data * ood_kwargs['x_scale'] # 调整输入特征的尺度, 制造
OOD 数据

if 'x_mean_shift' in ood_kwargs:
    x_data = x_data + ood_kwargs['x_mean_shift'] # 平移输入特征的均值,
制造 OOD 数据

if 'x_noise_std' in ood_kwargs:
    x_data = x_data + torch.randn_like(x_data) *
↳ood_kwargs['x_noise_std'] # 添加输入特征的噪声, 制造 OOD 数据

if valid_d_x < d_x:
    x_data[..., valid_d_x:] = 0.0 # 将多余的输入维度对应的数据置零, 保证
它们对输出没有影响

if d_x==d_hidden:
    w1=torch.eye(d_x, device=device).unsqueeze(0).expand(batch_size,
↳-1, -1) # 如果输入维度和隐藏层维度相同, 变为 nonlinear_func + linear 的形式

```

```

        if not getattr(nonlinear_data_generator, '_has_warned_w1_identity',
↪False):
            print("Warning: w1 is set to identity matrix because d_x equals
↪d_hidden.")
            nonlinear_data_generator._has_warned_w1_identity = True
        if d_hidden==d_y:
            w2=torch.eye(d_hidden, device=device).unsqueeze(0).
↪expand(batch_size, -1, -1) # 如果隐藏层维度和输出维度相同, 变为 linear +
↪nonlinear_func 的形式
            if not getattr(nonlinear_data_generator, '_has_warned_w2_identity',
↪False):
                print("Warning: w2 is set to identity matrix because d_hidden
↪equals d_y.")
                nonlinear_data_generator._has_warned_w2_identity = True

        hidden = x_data @ w1 # [batch_size, seq_len//2, d_hidden]

        func=ood_kwargs.get('function_callable', function_callable) # Concept
↪Shift
        # 应用非线性函数
        hidden = func(hidden) # [batch_size, seq_len//2, d_hidden]
        # 第二层线性变换
        y_data = hidden @ w2 # [batch_size, seq_len//2, d_y]

        if 'y_scale' in ood_kwargs:
            y_data = y_data * ood_kwargs['y_scale'] # 调整标签的尺度, 制造 OOD
数据
        if 'y_mean_shift' in ood_kwargs:
            y_data = y_data + ood_kwargs['y_mean_shift'] # 平移标签的均值, 制造
OOD 数据
        if 'y_noise_std' in ood_kwargs:
            y_data = y_data + torch.randn_like(y_data) *
↪ood_kwargs['y_noise_std'] # 添加标签的噪声, 制造 OOD 数据

        return x_data, y_data

```

lorenz attractor 数据 理论公式

系统的演化由以下三个一阶非线性微分方程决定：

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

参数物理意义 * $x(t)$: 表示流体运动的强度（对流速度）。* $y(t)$: 表示上升和下降流体的温度差。* $z(t)$: 表示垂直温度剖面的非线性畸变程度。* σ : Prandtl 数, 影响流动特性 * ρ : Rayleigh 数, 决定对流强度 * β : 几何物理尺寸相关参数

经典的混沌参数 $\sigma = 10, \rho = 28, \beta = \frac{8}{3}$

此时，系统的轨迹会收敛到一个分形结构上，表现出极度敏感的混沌特性。任何微小的初始条件差异，都会导致后续轨迹的巨大分歧。

```
[195]: def lorenz_derivative(state, sigma=10.0, rho=28.0, beta=8.0/3.0):  
    # state: [batch_size, 3]  
    x, y, z = state[:, 0], state[:, 1], state[:, 2]  
    dx = sigma * (y - x)  
    dy = x * (rho - z) - y  
    dz = x * y - beta * z  
    return torch.stack((dx, dy, dz), dim=1) # [batch_size, 3]
```

四阶龙格-库塔法（简称 RK4）

- 核心思想：多看几步，求个平均
 - k1: 先看一眼当前位置的速度。
 - k2: 试探性地往前走半步，看看那里的速度。
 - k3: 根据上一步的结果，再在半步的位置看一眼速度。
 - k4: 试探性地往前走一步，看看终点附近的速度。
- 最后，把这四个速度按照不同的权重（通常是两头轻、中间重）加权平均，作为最终的“靠谱速度”来更新位置。

```
[196]: def rk4_step(state, dt=0.01, sigma=10.0, rho=28.0, beta=8.0/3.0):  
    k1 = lorenz_derivative(state, sigma, rho, beta)  
    k2 = lorenz_derivative(state + 0.5 * dt * k1, sigma, rho, beta)  
    k3 = lorenz_derivative(state + 0.5 * dt * k2, sigma, rho, beta)
```



```

    k4 = lorenz_derivative(state + dt * k3, sigma, rho, beta)
    return state + (dt / 6.0) * (k1 + 2*k2 + 2*k3 + k4)
try:
    rk4_step = torch.compile(rk4_step)
except Exception:
    pass

```

lorenz 数据计算

```

[197]: def lorenz_kernel(init_state, total_steps:int, dt:float, sigma:float, rho:
    ↪float, beta:float):
    # init_state: [batch_size, 3]
    batch_size = init_state.shape[0]
    device = init_state.device
    dtype = init_state.dtype
    trajectory = torch.empty((total_steps, batch_size, 3), device=device, ↪
    ↪dtype=dtype)
    current_state = init_state
    for t in range(total_steps):
        current_state = rk4_step(state=current_state, dt=dt, sigma=sigma, ↪
    ↪rho=rho, beta=beta)
        trajectory[t] = current_state
    trajectory = trajectory.transpose(0, 1) # 转置为 [batch_size, total_steps, ↪
    ↪3]
    return trajectory # [batch_size, total_steps, 3]
'''try:
    lorenz_kernel = torch.compile(lorenz_kernel)
except Exception:
    pass'''

```

```

[197]: 'try:\n    lorenz_kernel = torch.compile(lorenz_kernel)\nexcept Exception:\n
pass'

```

把数据变成 data_generator

```

[198]: _lorenz_cache_lock = threading.Lock() # 定义一个锁对象，确保线程安全

```

```

def lorenz_data_generator(batch_size, seq_len, valid_d_x=None, d_x=3, d_y=3,
    ↪lorenz_kwargs=None, device='cpu', generator=None, ood_kwargs=None,
    ↪load_path=None):
    with torch.no_grad(): # 数据生成过程中不需要计算梯度, 使用 no_grad 加速
        if ood_kwargs is None:
            ood_kwargs = {}
        if valid_d_x is None:
            valid_d_x = d_x
        if lorenz_kwargs is None:
            lorenz_kwargs = {}
        burn_in = lorenz_kwargs.get('burn_in', 500) # 预热步数, 默认为 500 步, 确
        保系统状态进入混沌吸引子
        if load_path is not None and not os.path.exists(load_path):
            print(f"Warning: load_path {load_path} does not exist. Generating
            ↪new Lorenz data.")
            load_path = None
        if load_path is not None:
            with _lorenz_cache_lock:
                if not hasattr(lorenz_data_generator, "_cache") or
            ↪lorenz_data_generator._cache_path != load_path:
                    print(f"Loading Lorenz data from {load_path}...")
                    checkpoint = torch.load(load_path, map_location='cpu')
                    lorenz_data_generator._cache = checkpoint['trajectory']
                    lorenz_data_generator._cache_path = load_path
                    if ood_kwargs.get('init_distribution', 'gaussian') !=
            ↪'gaussian' or ood_kwargs.get('lorenz_noise_std', 0.0) > 0.0 or ood_kwargs.
            ↪get('sigma_shift', 0.0) != 0.0 or ood_kwargs.get('rho_shift', 0.0) != 0.0 or
            ↪ood_kwargs.get('beta_shift', 0.0) != 0.0:
                        print("Warning: The loaded Lorenz data was generated
            ↪with a Gaussian distribution , no noise and default parameters. If you want
            ↪to generate OOD data with different distribution or noise, please set
            ↪load_path to None.")
                    pool = lorenz_data_generator._cache # [pool_size, traj_len, 3]
                    pool_size = pool.shape[0]
                    indices = torch.randint(0, pool_size, (batch_size,), device='cpu')
                    batch_traj = pool[indices] # [batch_size, traj_len, 3]

```

```

    if pool.shape[1] < burn_in + seq_len//2 + 1:
        old_burn_in = burn_in
        burn_in = max(0, pool.shape[1] - seq_len//2 - 1)
        print(f"Warning: The trajectories in the pool have length {pool.
↪shape[1]}, which is less than burn_in + seq_len//2 + 1 = {old_burn_in +
↪seq_len//2 + 1}. Burn-in will be automatically reduced to {burn_in}.")
        batch_traj = batch_traj[:, burn_in:burn_in+seq_len//2+1, :].
↪to(device) # [batch_size, seq_len//2+1, 3]
        x_data = batch_traj[:, :seq_len//2, :] # [batch_size, seq_len//2,
↪3]
        y_data = batch_traj[:, 1:seq_len//2+1, :] # [batch_size, seq_len//
↪2, 3]

    else:
        dt = lorenz_kwargs.get('dt', 0.01) # 时间步长默认为 0.01
        # 初始化
        total_steps = burn_in + seq_len//2 + 1
        init_distribution = ood_kwargs.get('init_distribution', 'gaussian')
        if init_distribution == 'gaussian':
            init_state = torch.randn(batch_size, 3, device=device,
↪generator=generator) * 15 + torch.tensor([0, 0, 25.0], device=device) # 初始
状态 (放大 15 倍并平移 (0, 0, 25) 以确保在洛伦兹吸引子中心附近)
        elif init_distribution == 'uniform':
            init_state = (torch.rand(batch_size, 3, device=device,
↪generator=generator) * 30 - 15) + torch.tensor([0, 0, 25.0], device=device)
↪# 均匀分布, 范围 [-15, 15], 并平移 (0, 0, 25)
        elif init_distribution == 'laplace':
            u = torch.rand(batch_size, 3, device=device,
↪generator=generator) - 0.5
            init_state = torch.tensor([0, 0, 25.0], device=device) - (15 /
↪(2**0.5)) * torch.sign(u) * torch.log(1 - 2 * torch.abs(u))
            noise_std = ood_kwargs.get('lorenz_noise_std', 0.0)
            sigma = ood_kwargs.get('sigma_shift', 0.0)+10.0
            rho = ood_kwargs.get('rho_shift', 0.0)+28.0
            beta = ood_kwargs.get('beta_shift', 0.0)+8.0/3.0
            # 生成轨迹

```

```

        trajectory = lorenz_kernel(init_state=init_state,
↪total_steps=total_steps, dt=dt, sigma=sigma, rho=rho, beta=beta)[: , burn_in:
↪, :] # [batch_size, seq_len//2+1, 3]
        if noise_std > 0.0:
            trajectory = trajectory + torch.randn_like(trajectory) *
↪noise_std # 在观测轨迹上添加噪声, 制造 OOD 数据
        x_data = trajectory[:, :-1, :] # x_data: [batch_size, seq_len//2,
↪3]
        y_data = trajectory[:, 1:, :] # y_data: [batch_size, seq_len//2,
↪3]

        # 裁剪或填充至给定维度
        if valid_d_x > 3:
            x_data = F.pad(x_data, (0, valid_d_x - 3), value=0.0) # 将输入特征
            维度扩展到 valid_d_x, 新增的维度填充为 0
        elif valid_d_x < 3:
            x_data = x_data[:, :, :valid_d_x] # 将输入特征维度裁剪到 valid_d_x
        if d_y > 3:
            y_data = F.pad(y_data, (0, d_y - 3), value=0.0) # 将标签维度扩展到
↪d_y, 新增的维度填充为 0
        elif d_y < 3:
            y_data = y_data[:, :, :d_y] # 将标签维度裁剪到 d_y
        return x_data, y_data

```

```

[199]: def clear_lorenz_cache():
        if hasattr(lorenz_data_generator, "_cache"):
            del lorenz_data_generator._cache
            del lorenz_data_generator._cache_path
            print("Lorenz data cache cleared.")
        else:
            print("No Lorenz data cache found.")

```

由于 lorenz 数据生成较慢, 故预先生成离线数据池, 训练时从中随机采样, 评估时再在线生成。

```

[200]: def create_lorenz_pool(pool_size=10000, traj_len=500, dt=0.01, sigma=10.0,
↪beta=8/3, rho=28.0, save_path='auto'):
        if save_path == 'auto':

```

```

    save_path = f'data/lorenz/length_{traj_len}_dt{dt}_sigma{sigma:.
↪1f}_beta{beta:.1f}_rho{rho:.1f}.pth'

    with torch.no_grad():
        # 初始化
        if torch.cuda.is_available():
            device = 'cuda'
        elif torch.backends.mps.is_available():
            device = 'mps'
        else:
            device = 'cpu'

        generator = torch.Generator(device=device)
        init_state = torch.randn(pool_size, 3, device=device,
↪generator=generator) * 15 + torch.tensor([0, 0, 25], device=device) # 初始状态 (放大 15 倍并平移 (0, 0, 25) 以确保在洛伦兹吸引子中心附近)
        # 生成轨迹
        trajectory = lorenz_kernel(init_state=init_state, total_steps=traj_len,
↪dt=dt, sigma=sigma, rho=rho, beta=beta) # [pool_size, traj_len, 3]
        trajectory = trajectory.cpu() # 将数据移回 CPU, 准备保存

    if save_path and os.path.exists(save_path):
        print(f"检测到已有数据文件, 正在读取并进行增量拼接...")
        old_data = torch.load(save_path, map_location='cpu')
        old_trajectory = old_data['trajectory']
        if old_trajectory.shape[1] != trajectory.shape[1]:
            raise ValueError(f"拼接失败! 本地老数据的时序长度为
↪{old_trajectory.shape[1]}, 而你当前设定的时序长度要求为 {trajectory.shape[1]}。
请保持参数一致! ")
        # 在 Batch 维度 (dim=0) 进行无限追加拼接
        final_trajectory = torch.cat([old_trajectory, trajectory], dim=0)
        print(f"增量拼接成功! 数据池规模由 {old_trajectory.shape[0]} 条扩大至
↪{final_trajectory.shape[0]} 条。")
    else:
        print(f"未检测到历史文件, 正在创建全新的离线数据池...")
        final_trajectory = trajectory

    if save_path:

```

```

os.makedirs(os.path.dirname(save_path), exist_ok=True)
torch.save({'trajectory': final_trajectory}, save_path)
print(f"数据池已安全写入本地: {save_path}")

```

数据加载器

```

[201]: def dataloader(batch_size=64, seq_len=80, valid_d_x=None, d_x=20, d_y=1,
    ↪device='cpu', data_type='linear', d_hidden=None, function_callable=None,
    ↪sink_padding=1, generator=None, ood_kwargs=None, lorenz_kwargs=None,
    ↪load_lorenz_from=None):
    if valid_d_x is None:
        valid_d_x = d_x
    while True:
        if data_type == 'linear':
            x_data, y_data = linear_data_generator(batch_size=batch_size,
    ↪seq_len=seq_len, valid_d_x=valid_d_x, d_x=d_x, d_y=d_y, device=device,
    ↪generator=generator, ood_kwargs=ood_kwargs)
            elif data_type == 'nonlinear':
                x_data, y_data = nonlinear_data_generator(batch_size=batch_size,
    ↪seq_len=seq_len, valid_d_x=valid_d_x, d_x=d_x, d_y=d_y, d_hidden=d_hidden,
    ↪function_callable=function_callable, device=device, generator=generator,
    ↪ood_kwargs=ood_kwargs)
            elif data_type == 'lorenz':
                x_data, y_data = lorenz_data_generator(batch_size=batch_size,
    ↪seq_len=seq_len, valid_d_x=valid_d_x, d_x=d_x, d_y=d_y, device=device,
    ↪generator=generator, ood_kwargs=ood_kwargs, lorenz_kwargs=lorenz_kwargs,
    ↪load_path=load_lorenz_from)
            if sink_padding is not None:
                x_data = F.pad(x_data, (0, 0, sink_padding, 0), value=0) # 在
seq_len 维度最前面添加 sink_padding 个全 0 向量
                y_data = F.pad(y_data, (0, 0, sink_padding, 0), value=0)
            yield x_data, y_data

```

1.2.9 显存检测

```
[202]: def print_vram_usage(tag="", peak=None):
    if torch.backends.mps.is_available():
        # MPS 专用 API (Apple Silicon)
        allocated = torch.mps.current_allocated_memory() / (1024 ** 3)
        driver_alloc = torch.mps.driver_allocated_memory() / (1024 ** 3)
        if peak is not None:
            print(f"[{tag}] 显存占用 -> PyTorch 分配: {peak[0]:.2f} GB | Metal
驱动分配: {peak[1]:.2f} GB")
        else:
            print(f"[{tag}] 显存占用 -> PyTorch 分配: {allocated:.2f} GB | Metal
驱动分配: {driver_alloc:.2f} GB")

    elif torch.cuda.is_available():
        # CUDA 专用 API
        allocated = torch.cuda.memory_allocated() / (1024 ** 3)
        peak = torch.cuda.max_memory_allocated() / (1024 ** 3)
        reserved = torch.cuda.memory_reserved() / (1024 ** 3)
        total = torch.cuda.get_device_properties(0).total_memory / (1024 ** 3)
        free = total - reserved
        print(f"[{tag}] 显存占用 -> 已分配: {allocated:.2f} GB | 峰值: {peak:.
↪2f} GB | 缓存池: {reserved:.2f} GB | 剩余可用: {free:.2f} GB")

    else:
        print(f"[{tag}] 当前运行在 CPU, 占用主板内存。")
```

1.3 实验模块拼装

```
[203]: class LoopedTransformerExperiment:
    def __init__(self, num_blocks, num_heads=8, d_model=256, lr=1e-4, ↵
↪gate_lr_ratio=100, max_seq_len=4096, d_x=20, d_y=1,
        seed=None, experiment_name='Experiment',
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape', ↵
↪b_rope_or_upe=10000, head_ratio_upe=2,
        optimizer_type='adam', wd_adamw=0.01,
```

```

        loop=True, loss_type='mse', bias=False, init_scale=None,
↪init_std=0.02, x_init='prompt', sink_threshold=0.3,
        residual_gate=(1,1), residual_gate_type='fixed',
↪residual_random=(1,0.1),
        task='regression', timing=True, load_path=None,
        layer_weight_decay=1.0, seq_weight_decay=1.0):
    if timing:
        start_time = time.time()
        # 设备选择: 优先使用 MPS (适用于 Apple Silicon), 其次是 CUDA (适用于
↪NVIDIA GPU), 最后退回 CPU
        if torch.backends.mps.is_available():
            self.device = torch.device("mps")
        elif torch.cuda.is_available():
            self.device = torch.device("cuda")
        else:
            self.device = torch.device("cpu")
        self.is_offloaded = False
        self.generator = torch.Generator(device=self.device)
        if seed is not None:
            torch.manual_seed(seed)
            self.generator.manual_seed(seed)
        # 模型和优化器
        self.task = task
        if task == 'regression':
            self.model = RegressionSolver(num_blocks=num_blocks,
↪num_heads=num_heads, d_model=d_model, d_x=d_x, d_y=d_y,
↪max_seq_len=max_seq_len,
                                norm_type=norm_type,
↪ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
                                head_ratio_upe=head_ratio_upe,
↪loop=loop, loss_type=loss_type,
                                bias=bias, init_scale=init_scale,
↪init_std=init_std, x_init=x_init, sink_threshold=sink_threshold,
                                residual_gate=residual_gate,
↪residual_gate_type=residual_gate_type, residual_random=residual_random,

```



```

        layer_weight_decay=layer_weight_decay, seq_weight_decay=seq_weight_decay
    ).to(self.device)

    gate_params=[]
    base_params=[]
    self.eval_results={}
    for name, param in self.model.named_parameters():
        if 'residual_gate' in name:
            gate_params.append(param)
        else:
            base_params.append(param)
    param_groups = [{'params': base_params},
                    {'params': gate_params, 'lr': lr * gate_lr_ratio}]
    if optimizer_type == 'adam':
        self.optimizer = torch.optim.Adam(param_groups, lr=lr)
    elif optimizer_type == 'sgd':
        self.optimizer = torch.optim.SGD(param_groups, lr=lr)
    elif optimizer_type == 'adamw':
        self.optimizer = torch.optim.AdamW(param_groups, lr=lr,
        weight_decay=wd_adamw)
    self.d_x = d_x
    self.d_y = d_y
    self.num_blocks = num_blocks
    self.init_time = None
    self.loss_history = []
    self.y_pred_norm_history = []
    self.y_true_norm_history = []
    self.residual_gate_history = []
    self.sink_score_history = []
    self.sink_rate_history = []
    self.experiment_name = experiment_name
    if load_path is not None:
        self.load_checkpoint(load_path)
    if experiment_name is not None:
        if timing:
            self.init_time = time.time() - start_time

```

```

        print(f"{experiment_name} initialized in {self.init_time:.2f}␣
↪seconds. Using device: {self.device}")
    else:
        print(f"{experiment_name} initialized. Using device: {self.
↪device}")

    def train(self, batch_size=64, seq_len=80, epochs=20, steps_per_epoch=1,
        data_type='linear', d_hidden=None, function_callable=None,␣
↪lorenz_kwargs=None, load_lorenz_from=None,
        num_eff=15, sink_padding=None, scheduled_training=False,
        curriculum=None, scheduler_type=None, lr_scale=0.1,␣
↪step_size_scheduler=10,
        print_every=None, timing=False, save_path=None):
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        self.model.train()
        self.max_torch_vram = 0
        self.max_metal_vram = 0
        self.train_time = None
        dataloader_static_kwargs = dict(batch_size=batch_size, d_x=self.d_x,␣
↪d_y=self.d_y, device=self.device, data_type=data_type,␣
↪sink_padding=sink_padding, generator=self.generator, d_hidden=d_hidden,␣
↪function_callable=function_callable, ood_kwargs=None,␣
↪lorenz_kwargs=lorenz_kwargs, load_lorenz_from=load_lorenz_from)
        active_seq_len = seq_len
        active_d_x = self.d_x
        active_data_loader = dataloader(seq_len=active_seq_len,␣
↪valid_d_x=active_d_x, **dataloader_static_kwargs)
        if scheduler_type == 'cosine':
            scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(self.
↪optimizer, T_max=epochs, eta_min=lr_scale*self.optimizer.
↪param_groups[0]['lr'])
        elif scheduler_type == 'step':
            scheduler = torch.optim.lr_scheduler.StepLR(self.optimizer,␣
↪step_size=step_size_scheduler, gamma=lr_scale)
        num_eff = min(num_eff, self.num_blocks) # 确保 num_eff 不超过总层数

```

```

    if curriculum is None:
        curriculum={}
    total_steps = epochs * steps_per_epoch
    if timing:
        start_time = time.time()
    for epoch in range(epochs):
        epoch_loss = 0.0
        epoch_y_pred_norm = 0.0
        epoch_y_true_norm = 0.0
        if scheduled_training:
            current_blocks = min(num_eff + epoch, self.num_blocks) # 随着训练的
            # 进行, 逐渐增加参与误差计算的有效层数
        else:
            current_blocks = self.num_blocks
        for step in range(steps_per_epoch):
            self.optimizer.zero_grad()
            global_step = epoch * steps_per_epoch + step
            if curriculum:
                progress = min(1.0, global_step / (total_steps*curriculum.
↪get('duration_ratio',0.5)))
                if progress==1.0:
                    curriculum={} # 课程结束, 重置为普通训练
                    print(f"Curriculum ended at epoch {epoch+1}, step_
↪{step+1}.")
                else:
                    d_x_start = curriculum.get('d_x', self.d_x)
                    seq_len_start = curriculum.get('seq_len', seq_len)
                    current_d_x = int(d_x_start + progress * (self.d_x -
↪d_x_start))
                    current_seq_len = int(seq_len_start + progress *
↪(seq_len - seq_len_start))
                else:
                    current_d_x = self.d_x
                    current_seq_len = seq_len
            if current_seq_len != active_seq_len or current_d_x !=
↪active_d_x:

```

```

        active_seq_len = current_seq_len
        active_d_x = current_d_x
        active_data_loader = dataloader(seq_len=active_seq_len,
↪valid_d_x=active_d_x, **dataloader_static_kwargs)
        x_data, y_data = next(active_data_loader) # 从数据加载器中获取一个批次的数据

        loss, y_pred_norm, y_true_norm = self.model(x_data, y_data,
↪num_eff=num_eff, current_blocks=current_blocks, is_eval=False,
↪sink_padding=sink_padding) # 前向传播计算损失 [batch_size, num_eff, seq_len/
↪2, d_y]

        loss = loss.mean() # 对所有维度求平均得到一个标量损失值
        loss.backward() # 反向传播计算梯度
        if torch.backends.mps.is_available():
            self.max_metal_vram = max(self.max_metal_vram, torch.mps.
↪driver_allocated_memory() / (1024 ** 3))
            self.max_torch_vram = max(self.max_torch_vram, torch.mps.
↪current_allocated_memory() / (1024 ** 3))
            self.optimizer.step() # 更新模型参数
            epoch_loss += loss.item()
            epoch_y_pred_norm += y_pred_norm
            epoch_y_true_norm += y_true_norm

        self.model.eval() # 切换到评估模式, 记录注意力分数 (这样不会进行
FlashAttention 跳过 attention 矩阵生成)
        with torch.no_grad():
            _ = self.model(x_data, y_data, num_eff=num_eff,
↪current_blocks=current_blocks, is_eval=False, sink_padding=sink_padding) #
↪再次前向传播以获取当前的残差门控参数和注意力分数

            scores = self.model.toy_model.captured_sink_scores
            rates = self.model.toy_model.captured_sink_rates
            if scores and rates:
                valid_scores = [score for score in scores if score is not
↪None]

                valid_rates = [rate for rate in rates if rate is not None]
                avg_sink_score = np.mean(valid_scores) if valid_scores else
↪0.0

```

```

        avg_sink_rate = np.mean(valid_rates) if valid_rates else 0.0
    else:
        avg_sink_score, avg_sink_rate = 0.0, 0.0
    self.model.train() # 切换回训练模式

    if scheduler_type is not None:
        scheduler.step() # 按 epoch 级别更新学习率
    avg_loss = epoch_loss / steps_per_epoch
    avg_y_pred_norm = epoch_y_pred_norm / steps_per_epoch
    avg_y_true_norm = epoch_y_true_norm / steps_per_epoch
    self.y_pred_norm_history.append(avg_y_pred_norm)
    self.y_true_norm_history.append(avg_y_true_norm)
    self.loss_history.append(avg_loss) # 记录损失值
    self.sink_score_history.append(avg_sink_score)
    self.sink_rate_history.append(avg_sink_rate)
    self.residual_gate_history.append(self.model.toy_model.
↪residual_gate.detach().cpu().numpy() if hasattr(self.model.toy_model,
↪'residual_gate') else None) # 记录门控参数值

    if print_every is not None and (epoch + 1) % print_every == 0:
        print(f"Epoch {epoch+1}/{epochs}, Loss: {avg_loss:.4f}, Norm_
↪Ratio: {avg_y_pred_norm/avg_y_true_norm if avg_y_true_norm != 0 else 0.0:.
↪4f}, Avg Sink Score: {avg_sink_score:.4f}, Avg Sink Rate: {avg_sink_rate:.
↪4f}")

    if timing:
        self.train_time = time.time()-start_time
        print(f"Training completed in {self.train_time:.2f} seconds.")
    self.residual_gate_values = self.residual_gate_history[-1]
    self.final_loss = self.loss_history[-1]
    if save_path is not None:
        self.save_checkpoint(save_path)

    def evaluate(self, eval_name='eval', batch_size=64, seq_len=80,
↪data_type='linear', loss_type='mse', sink_padding=1, d_hidden=None,
↪function_callable=None, ood_kwargs=None, lorenz_kwargs=None):
        self.model.eval()
        if ood_kwargs is None:

```

```

        ood_kwargs = {}
        if 'seq_len_scale' in ood_kwargs:
            seq_len = int(seq_len * ood_kwargs['seq_len_scale']) # 调整序列长度,
制造 OOD 数据

        with torch.no_grad():
            # 在评估模式下, 直接使用所有层的输出进行预测 (因为已经 no_grad 了, 所以
            # 不需要防止梯度爆炸了)
            # 生成一个评估用的数据批次, 注意这里的 seq_len 要比训练时大 2, 因为我们
            # 需要预测最后一个位置的 y 值

            x_eval, y_eval = next(dataloader(batch_size=batch_size,
↪seq_len=seq_len+2, d_x=self.d_x, d_y=self.d_y, device=self.device,
↪data_type=data_type, sink_padding=sink_padding,
                                                    generator=self.generator,
↪d_hidden=d_hidden, function_callable=function_callable,
↪ood_kwargs=ood_kwargs, lorenz_kwargs=lorenz_kwargs, load_lorenz_from=None))
            # Lorenz 在评估时不使用离线数据池, 而是直接生成新的数据

            y_pred = self.model(x_eval, y_eval, num_eff=self.num_blocks,
↪current_blocks=self.num_blocks, is_eval=True, sink_padding=sink_padding) #
↪[batch_size, d_y]

            # 计算 loss
            if loss_type == 'mse':
                loss=F.mse_loss(y_pred, y_eval[:, -1, :]).item()
            elif loss_type == 'l1':
                loss=F.l1_loss(y_pred, y_eval[:, -1, :]).item()
            # 计算预测值和真实值的范数以及它们的比值
            y_pred_norm = torch.sqrt(torch.mean(y_pred ** 2)).item()
            y_true_norm = torch.sqrt(torch.mean(y_eval[:, -1, :] ** 2)).item()
            y_norm_ratio = y_pred_norm / y_true_norm if y_true_norm != 0 else 0.

↪0

            print(f"[{eval_name}] Loss: {loss:.4f}, Norm Ratio: {y_norm_ratio:.
↪4f}")

            self.eval_results.update({
                f'{eval_name}_loss': loss,
                f'{eval_name}_y_pred_norm': y_pred_norm,
                f'{eval_name}_y_true_norm': y_true_norm,
                f'{eval_name}_y_norm_ratio': y_norm_ratio,

```

```

        f'{eval_name}_sink_scores': self.model.toy_model.
        ↪captured_sink_scores, # list[float]: 长度为 num_blocks, 每个元素是对应的一层中,
        所有注意力头分配给第 0 个位置的平均注意力权重

        f'{eval_name}_sink_rates': self.model.toy_model.
        ↪captured_sink_rates, # list[float]: 长度为 num_blocks, 每个元素是对应的一层
        中, 分配给第 0 个位置的注意力权重比例 > sink_threshold 的注意力头的比例

        f'{eval_name}_captured_attention': self.model.toy_model.
        ↪captured_attention
    })

def get_results(self, result_list: list[str]):
    all_results = {
        'init_time': getattr(self, 'init_time', None),
        ↪ # float: 实验初始化时间 (秒)

        'train_time': getattr(self, 'train_time', None),
        ↪ # float: 训练时间 (秒)

        'loss_history': getattr(self, 'loss_history', None),
        ↪ # list[float]: 训练过程中每个 epoch 的损失值

        'y_pred_norm_history': getattr(self, 'y_pred_norm_history', None),
        ↪ # list[float]: 训练过程中每个 epoch 的预测值范数

        'y_true_norm_history': getattr(self, 'y_true_norm_history', None),
        ↪ # list[float]: 训练过程中每个 epoch 的真实值范数

        'y_norm_error_history': [pred - true for pred, true in
        ↪zip(getattr(self, 'y_pred_norm_history', []), getattr(self,
        ↪'y_true_norm_history', []))] if getattr(self, 'y_pred_norm_history', None)
        ↪is not None and getattr(self, 'y_true_norm_history', None) is not None else
        ↪None, # list[float]: 训练过程中每个 epoch 的预测值范数与
        真实值范数之差的绝对值

        'y_norm_ratio_history': [pred / true if true != 0 else None for
        ↪pred, true in zip(getattr(self, 'y_pred_norm_history', []), getattr(self,
        ↪'y_true_norm_history', []))] if getattr(self, 'y_pred_norm_history', None)
        ↪is not None and getattr(self, 'y_true_norm_history', None) is not None else
        ↪None, # list[float]: 训练过程中每个 epoch 的预测值范数与真实值范数之比
    }

```

```

        'sink_score_history': getattr(self, 'sink_score_history', None),
        ↪ # list[float]: 长度为 epochs, 每个元素为每个 epoch 的平均 (对
num_blocks, steps_per_epoch 取平均) sink score
        'sink_rate_history': getattr(self, 'sink_rate_history', None),
        ↪ # list[float]: 长度为 epochs, 每个元素为每个 epoch 的平均 (对
num_blocks, steps_per_epoch 取平均) sink rate
        'final_loss': self.loss_history[-1] if getattr(self,
        ↪ 'loss_history', None) else None, # float: 最终的损失值
        'final_y_pred_norm': self.y_pred_norm_history[-1] if getattr(self,
        ↪ 'y_pred_norm_history', None) else None, # float: 最终的预测值范
数
        'final_y_true_norm': self.y_true_norm_history[-1] if getattr(self,
        ↪ 'y_true_norm_history', None) else None, # float: 最终的真实值范
数
        'residual_gate_history': getattr(self, 'residual_gate_history',
        ↪ None), # list[np.array[2] 或 np.array[d_model, 2]]: 训练过程中每
个 epoch 的残差门控参数值
        'final_residual_gate': getattr(self, 'residual_gate_values', None),
        ↪ # np.array[2] 或 np.array[d_model, 2]: 最终的残差门控参数值
        'final_sink_score': self.sink_score_history[-1] if getattr(self,
        ↪ 'sink_score_history', None) else None, # float: 最后一个 epoch
的 sink score
        'final_sink_rate': self.sink_rate_history[-1] if getattr(self,
        ↪ 'sink_rate_history', None) else None, # float: 最后一个
epoch 的 sink rate
        'captured_sink_scores': getattr(self.model.toy_model,
        ↪ 'captured_sink_scores', None), # list[float]: 同 eval_results 中的
sink_scores, 只不过取自训练过程中最后一次前向传播的捕获值
        'captured_sink_rates': getattr(self.model.toy_model,
        ↪ 'captured_sink_rates', None), # list[float]: 同 eval_results 中的
sink_rates, 只不过取自训练过程中最后一次前向传播的捕获值
        'captured_attention': getattr(self.model.toy_model,
        ↪ 'captured_attention', None) # list[np.array[batch_size, num_heads,
        ↪ seq_len, seq_len]]: 训练时最后一次前向传播捕获的注意力权重
    }

```



```

        if hasattr(self, 'eval_results'):
            all_results.update(self.eval_results) # 将评估结果添加到
↪all_results 中
            if any("residual_gate" in key for key in result_list) and self.
↪residual_gate_values is not None:
                if len(self.residual_gate_values.shape) == 1:
                    all_results['residual_gate_history_a'] = [gate[0] for gate in
↪self.residual_gate_history]
                    all_results['residual_gate_history_b'] = [gate[1] for gate in
↪self.residual_gate_history]
                    all_results['residual_gate_history_a_relative'] = [gate[0]-self.
↪residual_gate_history[0][0] for gate in self.residual_gate_history]
                    all_results['residual_gate_history_b_relative'] = [gate[1]-self.
↪residual_gate_history[0][1] for gate in self.residual_gate_history]
                    # 标量门控的 _mean 就是自身，便于与向量门控混合绘图
                    all_results['residual_gate_history_a_mean'] =
↪all_results['residual_gate_history_a']
                    all_results['residual_gate_history_b_mean'] =
↪all_results['residual_gate_history_b']
                    all_results['residual_gate_history_a_mean_relative'] =
↪all_results['residual_gate_history_a_relative']
                    all_results['residual_gate_history_b_mean_relative'] =
↪all_results['residual_gate_history_b_relative']
                    all_results['final_residual_gate_a'] = self.
↪residual_gate_values[0]
                    all_results['final_residual_gate_b'] = self.
↪residual_gate_values[1]
                    all_results['final_residual_gate_a_mean'] =
↪all_results['final_residual_gate_a']
                    all_results['final_residual_gate_b_mean'] =
↪all_results['final_residual_gate_b']
                    all_results['final_residual_gate_b']
                elif len(self.residual_gate_values.shape) == 2:
                    all_results['residual_gate_history_a_mean'] = [gate[:,0].mean()
↪for gate in self.residual_gate_history]

```

```

        all_results['residual_gate_history_b_mean'] = [gate[:,1].mean()
↪for gate in self.residual_gate_history]
        all_results['residual_gate_history_a_mean_relative'] = [gate[:
↪,0].mean()-self.residual_gate_history[0][:,0].mean() for gate in self.
↪residual_gate_history]
        all_results['residual_gate_history_b_mean_relative'] = [gate[:
↪,1].mean()-self.residual_gate_history[0][:,1].mean() for gate in self.
↪residual_gate_history]
        all_results['residual_gate_history_a_std'] = [gate[:,0].std()
↪for gate in self.residual_gate_history]
        all_results['residual_gate_history_b_std'] = [gate[:,1].std()
↪for gate in self.residual_gate_history]
        all_results['final_residual_gate_a_mean'] = self.
↪residual_gate_values[:,0].mean()
        all_results['final_residual_gate_b_mean'] = self.
↪residual_gate_values[:,1].mean()
        all_results['final_residual_gate_a_std'] = self.
↪residual_gate_values[:,0].std()
        all_results['final_residual_gate_b_std'] = self.
↪residual_gate_values[:,1].std()
        if result_list is not None:
            return {key: all_results.get(key) for key in result_list}

def offload_to_cpu(self):
    self.model.to('cpu')
    for state in self.optimizer.state.values():
        for k, v in state.items():
            if isinstance(v, torch.Tensor):
                state[k] = v.cpu()
    self.is_offloaded = True

def load_to_device(self):
    self.model.to(self.device)
    for state in self.optimizer.state.values():
        for k, v in state.items():

```

```

        if isinstance(v, torch.Tensor):
            state[k] = v.to(self.device)
self.is_offloaded = False

def save_checkpoint(self, path='auto'):
    if path == 'auto':
        exp_name = self.experiment_name if self.experiment_name is not None
else "Experiment"
        safe_name = exp_name.replace(' ', '_').replace('(', '').
replace(')', '')
        path = f'saved_checkpoints/{safe_name}.pth'
        """ 保存实验的完整状态 """
        os.makedirs(os.path.dirname(path) if os.path.dirname(path) else '.',
exist_ok=True)
        checkpoint = {
            'model_state_dict': self.model.state_dict(),
            'loss_history': getattr(self, 'loss_history', []),
            'y_pred_norm_history': getattr(self, 'y_pred_norm_history', []),
            'y_true_norm_history': getattr(self, 'y_true_norm_history', []),
            'sink_score_history': getattr(self, 'sink_score_history', []),
            'sink_rate_history': getattr(self, 'sink_rate_history', []),
            'residual_gate_history': getattr(self, 'residual_gate_history', []),
        }
        if hasattr(self, 'optimizer'):
            checkpoint['optimizer_state_dict'] = self.optimizer.state_dict()
        torch.save(checkpoint, path)
        print(f" Checkpoint 已安全保存至: {path}")

def load_checkpoint(self, path='auto'):
    """ 加载实验的完整状态 """
    if path == 'auto':
        exp_name = self.experiment_name if self.experiment_name is not None
else "Experiment"
        safe_name = exp_name.replace(' ', '_').replace('(', '').
replace(')', '')
        path = f'saved_checkpoints/{safe_name}.pth'

```

```

if not os.path.exists(path):
    print(f"  找不到文件 {path}, 跳过加载。")
    return
# 统一先加载到 CPU, 后续再通过 load_to_device() 转移
checkpoint = torch.load(path, map_location='cpu', weights_only=False)
if isinstance(checkpoint, dict) and 'model_state_dict' in checkpoint:
    self.model.load_state_dict(checkpoint['model_state_dict'])
    # 恢复历史记录 (方便画图断点续联)
    self.loss_history = checkpoint.get('loss_history', [])
    self.y_pred_norm_history = checkpoint.get('y_pred_norm_history', [])
    self.y_true_norm_history = checkpoint.get('y_true_norm_history', [])
    self.sink_score_history = checkpoint.get('sink_score_history', [])
    self.sink_rate_history = checkpoint.get('sink_rate_history', [])
    self.residual_gate_history = checkpoint.
↪get('residual_gate_history', [])
    # 恢复优化器状态
    if 'optimizer_state_dict' in checkpoint and hasattr(self, '
↪'optimizer'):
        self.optimizer.
↪load_state_dict(checkpoint['optimizer_state_dict'])
    else:
        print("检测到纯权重 state_dict 文件 (或历史版本), 直接加载
↪model_state_dict...")
        self.model.load_state_dict(checkpoint)
        self.loss_history = []
        self.y_pred_norm_history = []
        self.y_true_norm_history = []
        self.sink_score_history = []
        self.sink_rate_history = []
        self.residual_gate_history = []
    print(f"  成功从 {path} 恢复实验状态。")

```

1.4 默认配置参数

```
[204]: def default_setup(manual=None):
    init_parameters = dict(
        # RoPE/MS-UPE
        b_rope_or_upe=10000,          # float: RoPE 或 UPE 的基数
        head_ratio_upe=2,             # float: UPE 头比例

        # MultiHeadAttention
        num_heads=8,                  # int: 注意力头数 H
        d_model=256,                   # int: Transformer 的维度 D
        max_seq_len=100,               # int: 模型支持的最大序列长度 (位置编码相
        关)

        pe_type=['learned_ape'],       # list: 位置编码类型 ('ape',
        ↪ 'learned_ape', 'rope', 'ms_upe', 'alibi')

        # SinkMetricsProbe
        sink_threshold=0.3,             # float: 判断阈值, sink rate 表示所有
        ↪ token 在第 0 个位置的注意力的平均值超过这个阈值的头数比例

        # TransformerBlock
        norm_type='layernorm',          # str: transformer_block 中的归一化类型
        ↪ ('layernorm' 或 'rmsnorm')
        ffn_type='gelu',               # str: transformer_block 中前馈网络激活函
        数类型 ('gelu' 或 'swiglu')

        # ToyModel
        num_blocks=20,                 # int: Transformer 的层数 b
        loop=True,                     # bool: 是否权重共享
        residual_gate=(1,1),            # tuple or str: transformer_block 之间传
        递残差的门控参数初始值, 使  $x=a*x+b*x_0$  ((a,b) 或 'random')
        residual_gate_type='fixed',     # str: 残差门控类型 ('fixed',
        ↪ 'learnable_scalar', 'learnable_vector')
        residual_random=(1,0.1),        # tuple (mean, std): 当
        ↪ residual_gate='random' 时满足高斯分布  $N(mean, std)$ 
        x_init='zero',                  # str: x 在所有 block 之前的初始化方式
        ↪ ('prompt'(x=x_0) 或 'zero'(x=0))
```

```

# RegressionHead
d_x=20, # int: 数据输入的特征维度
d_y=1, # int: 数据输出的特征维度
bias=False, # bool: Regression Head 是否使用偏置
init_scale=None, # float or None: Regression Head 的初始
                化缩放因子 (None 表示使用默认初始化)。仅当 init_std 为 None 时有效, 如果 init_std
                不为 None, 则使用 init_std 进行权重初始化, 并忽略 init_scale。

# RegressionSolver
init_std=0.02, # float or 'auto' or None: 所有 nn.
               ↳ Linear 和 nn.Embedding 权重初始化的标准差 ('auto'表示直接取 1/sqrt(d_model),
               None 表示使用默认初始化)

# PredictionLoss
loss_type='mse', # str: 损失函数类型 ('mse' 或 'l1')
layer_weight_decay=0.8, # float: 层权重衰减因子, 控制不同层的损失
                        权重 (默认 0.8 表示从最后一层到第一层递减的权重, 1.0 表示不加权)
seq_weight_decay=0.8, # float: 序列权重衰减因子, 控制不同时间步
                       的损失权重 (默认 0.8 表示从最后一次预测到第一次预测递减的权重, 1.0 表示不加权)

# LoopedTransformerExperiment
lr=1e-4, # float: 学习率
gate_lr_ratio=100, # float: 当使用残差门控时, 门控参数的学习
                  率相对于模型其他参数的倍数
seed=42, # int or None: 手动随机种子, 确保实验可复
         现
experiment_name='Experiment', # str or None: 实验名称 (None 表示不打印
                             设备和实验初始化信息)
timing=True, # bool: 是否打印初始化时间
optimizer_type='adamw', # str: 优化器类型 ('adam' 或 'sgd'或↳
↳ 'adamw')
wd_adamw=0.01, # float: AdamW 优化器的权重衰减系数, 仅当
optimizer_type='adamw'时有效
task='regression', # str: 任务类型 ('regression')

```

```

load_path=None, # str or 'auto'(指'saved_checkpoints/
↳<safe_experiment_name>.pth') or None: 预训练模型路径 (None 表示不加载预训练模
型)

)

train_parameters = dict(
# MultiHeadAttention

batch_size=64, # int: 数据批次大小
seq_len=80, # int: 数据序列长度

# ToyModel
num_eff=15, # int: 有效层数, 即参与误差计算、参与梯度
反向传播的层数  $T=b-b_0$ 

# dataloader
sink_padding=None, # int or None: seq 最前面填充的 sink_
↳token 组数 (None 表示不使用 sink padding)

d_hidden=64, # int: 非线性数据生成器中隐藏层的维度, 仅
当 data_type='nonlinear'时有效

function_callable=lambda x: 2**0.5 * F.relu(x), # function: 非线性数据生
成器中使用的非线性函数, 仅当 data_type='nonlinear'时有效 (系数 2**0.5 是为了使方差
期望变为 1)

lorenz_kwargs=dict(dt=0.01, burn_in=500), # dict: 洛伦兹系统数据生
成的额外参数, 仅当 data_type='lorenz'时有效

# lorenz_kwargs 的可选键包括:
# 'dt': float, 洛伦兹系统的时间步长
# 'burn_in': int, 洛伦兹系统的预热步数,
即在正式生成训练数据之前, 先运行系统 burn_in 步以达到吸引子区域

load_lorenz_from=None, # str or None: 洛伦兹系统数据生成的初始状
态加载路径 (None 表示随机初始化)

# LoopedTransformerExperiment
epochs=100, # int: 训练轮数
steps_per_epoch=1, # int: 每轮训练的步数
data_type='linear', # str: 回归数据类型_
↳('linear'或'nonlinear'或'lorenz')

```

```

# 注意：当 data_type='lorenz'时，务必令
d_x=3,d_y=3 以匹配洛伦兹系统的状态空间维度

    scheduler_type=None, # str or None: 学习率调节器类型
    ↪('cosine' 或 'step' 或 None)

    lr_scale=0.1, # float: 学习率调节器的缩放因子，仅当
scheduler_type 不为 None 时有效

    step_size_scheduler=10, # int: StepLR 调度器的步长，仅当
scheduler_type='step'时有效

    scheduled_training=False, # bool: 是否使用 kick-start，即随着训练的
进行逐渐增加参与的层数 b=epoch+num_eff，直到达到最大层数 num_blocks（用处并不大）

    curriculum=None, # dict or None: 课程学习配置（None 表示不
使用课程学习）

# curriculum 的可选键包括：
# 'd_x': int, curriculum learning 的初始
有效 d_x

# 'seq_len': int, curriculum learning 的
初始有效 seq_len

    ↪'duration_ratio': float, curriculum_
learning 的持续时间占比，取值范围为 0 到 1，表示在训练的前 duration_ratio 比例的
时间内逐渐增加有效 d_x 或 seq_len，之后保持不变。

    print_every=20, # int or None: 打印频率/epoch（None 表示
不打印训练过程中的损失信息）

    timing=True, # bool: 是否打印训练时间

    save_path=None # str or 'auto' (指'saved_checkpoints/
↪<safe_experiment_name>.pth') or None: 训练完成后模型的保存路径（None 表示不保存
模型）

)

eval_config=dict(
    eval_name='eval', # str: 评估名称
    ood_kwargs=None, # dict or None: 用于 OOD 数据生成的额外参
数，仅 evaluate 时有效。

# ood_kwargs 的可选键包括：
# 'x_distribution': str, 输入特征的分布类
型 ('gaussian', 'uniform', 'laplace') # x 分布偏移

```



```

子
    # 'x_scale': float, 输入特征的尺度调整因子
    # x 尺度偏移
    # 'x_mean_shift': float, 输入特征的均值平
移量    # x 均值偏移
    # 'x_noise_std': float, 输入特征的噪声标
准差    # x 噪声鲁棒性
    # 'y_scale': float, 标签的尺度调整因子
    # y 尺度偏移
    # 'y_mean_shift': float, 标签的均值平移
    # y 均值偏移
    # 'y_noise_std': float, 标签的噪声标准差
    # y 噪声鲁棒性
    # 'function_callable': function, 数据生
成函数, 仅当 data_type='nonlinear'时有效    # 函数偏移
    # 'seq_len_scale': float, evaluate 时序
列长度缩放因子    # 序列长度外推
    # 'init_distribution': str,
    # (for lorenz attractor) 初始状态分布类型
    # 'lorenz_noise_std': float, 洛伦兹系统
噪声标准差    # (for lorenz attractor) 噪声鲁棒性
    # 'sigma_shift': float, 洛伦兹系统的
sigma 参数偏移量    # (for lorenz attractor) 参数偏移
    # 'rho_shift': float, 洛伦兹系统的 rho
参数偏移量    # (for lorenz attractor) 参数偏移
    # 'beta_shift': float, 洛伦兹系统的 beta
参数偏移量    # (for lorenz attractor) 参数偏移
)

if manual is not None:
    for key,value in manual.items():
        if key in init_parameters:
            init_parameters[key]=value
        elif key in train_parameters:
            train_parameters[key]=value
        else:

```

```

        print(f"Warning:key {key} do not exist in init_parameters and_
↪train_parameters.")
    return init_parameters, train_parameters

```

1.5 实验台搭建

```

[205]: class ExperimentTable:
    '''
    Attributes:
        num_experiments: int, 实验数量
        init_parameters: list of dict, 在 __init__() 方法中传入, 用于指定每个实验
        的初始化参数, 可以覆盖默认参数
        train_parameters: list of dict, 在 __init__() 方法中传入, 用于指定每个实
        验的训练参数, 可以覆盖默认参数
        result_lists: list of tuple of list, 在 run() 方法中传入, 用于指定需要展
        示的结果名称和对应的横坐标类型。

        results: list of dict, 所有实验的结果, 每个 dict 是一个实验的结果, 为
        metric 和 value 的键值对。
        如 [{'loss_history': [...], 'final_loss': 0.123,
↪'residual_gate_history_a_mean': [...], 'final_residual_gate_a_mean': 0.45, ..
↪.}, {...}, ...]
    '''
    def __init__(self, params_groups:list[dict], manual=None):
        '''
        Args:
            params_groups: list of dict, 每个 dict 包含一个实验的参数设置, 可以覆
            盖默认参数,
            如 [{residual_gate: (0.5, 0.5), experiment_name: 'Experiment 1'},
↪{residual_gate: (0.8, 0.2), experiment_name: 'Experiment 2'}]

            manual: dict, 全局默认参数修改

        Attributes:
            num_experiments: int, 实验数量
            init_parameters: list of dict, 每个 dict 包含一个实验的初始化参数设置

```

```

        train_parameters: list of dict, 每个 dict 包含一个实验的训练参数设置
        '''
        init, train = default_setup(manual=manual)
        self.num_experiments = len(params_groups)
        self.init_parameters = [copy.deepcopy(init) for _ in range(self.
↪num_experiments)]
        self.train_parameters = [copy.deepcopy(train) for _ in range(self.
↪num_experiments)]
        # 自动分类传入的参数以更新默认参数
        for i, params in enumerate(params_groups):
            for key, value in params.items():
                if key in self.init_parameters[i]:
                    self.init_parameters[i][key] = value
                elif key in self.train_parameters[i]:
                    self.train_parameters[i][key] = value
                else:
                    raise ValueError(f"Unknown parameter: {key}")
        self.experiments=[None]*self.num_experiments
        for i in range(self.num_experiments):
            experiment = LoopedTransformerExperiment(**self.init_parameters[i])
            experiment.offload_to_cpu() # 初始化后将模型移回 CPU, 节省 GPU 显存
            self.experiments[i] = experiment

    def run(self, result_lists:list[tuple[list[str], str, int]],
↪modes=['train'], parallel_workers=1, eval_configs=None):
        '''
        Args:
            result_lists: list of tuple of list, 每个 tuple 包含一组实验测量的数
            据、对应的横坐标类型和 baseline 索引（不写表示不使用 baseline），
            每个 list 中的字符串表示需要展示的结果名称，必须是
            ↪LoopedTransformerExperiment.get_results() 中返回的结果名称，用 '/' 作为双 y 轴分
            割符

```

横坐标类型可以从 *'epoch'*（表示以训练轮数为横坐标的折线图，*metric* 必须是列表）、*'block'*（表示以模型深度为横坐标的折线图，*metric* 必须是列表，仅限 *eval* 和 *captured* 的 *sink* 指标）、*'experiment'*（表示以实验编号为横坐标的柱状图，*metric* 是单个值）中选择。

```
如 [(['loss_history', '/', 'residual_gate_history_a_mean',
↪ 'residual_gate_history_b_mean'], 'epoch', 0),
↪ (['final_residual_gate_a_mean', 'final_residual_gate_b_mean', '/'],
↪ 'experiment')]
```

modes: list of str, 表示需要运行的模式, 可以是 'train' 和 'evaluate' 的任意组合

parallel_workers: int, 表示并行运行实验的工作线程数。

eval_configs: list of dict (长度最好与 *modes* 中 'evaluate' 的数量相同), 每个 dict 的形式为 {'eval_name': str, 'ood_kwargs': dict} (详见 default_setup() 中的 eval_config 注释)

```
如 [{'eval_name': 'id'}, {'eval_name': 'ood_x_scale', 'ood_kwargs':
↪ {'x_scale': 2.0}}]
```

需要注意的是, *eval_configs* 中的 *eval_name* 将直接作为评估结果中损失值、范数等指标的前缀, 如 *id_loss*、*ood_x_scale_y_pred_norm*、*ood_x_scale_y_norm_ratio*

Attributes:

result_lists: 同 Args

results: list of dict, 所有实验的结果, 每个 dict 是一个实验的结果, 为 *metric* 和 *value* 的键值对。

experiments: list of LoopedTransformerExperiment 对象, 已转移到 CPU 以节省 GPU 显存

```
'''
# 创建实验对象并运行
result_set = set(key for item in result_lists for key in item[0] if key
↪ != '|') # 兼容长度为 2 或 3 的 tuple, 获取所有需要展示的结果名称的集合
self.result_lists = result_lists
self.results = [None]*self.num_experiments
if eval_configs is None:
    eval_configs = []
print_lock = threading.Lock() # 用于在线程中安全地打印日志
vram_printed = False
def single_train(i, parallel=False):
```

```

nonlocal vram_printed
eval_count=0
experiment = self.experiments[i]
experiment.load_to_device() # 训练前将模型加载到 GPU
if parallel:
    self.train_parameters[i]['timing']=False
    self.train_parameters[i]['print_every']=None
if parallel:
    with print_lock:
        if not vram_printed:
            print_vram_usage(tag=f"Parallel 峰值显存检测")
            vram_printed = True
else:
    print_vram_usage(tag=f"Experiment {i+1} 训练前")
for mode in modes:
    if mode == 'train':
        experiment.train(**self.train_parameters[i])
        self.results[i] = experiment.get_results(result_set)
        if parallel:
            with print_lock:
                print_vram_usage(tag=f"Experiment {i+1} 训练峰值",
↪peak=(experiment.max_torch_vram, experiment.max_metal_vram))
        else:
            print_vram_usage(tag=f"Experiment {i+1} 训练峰值",
↪peak=(experiment.max_torch_vram, experiment.max_metal_vram))
    elif mode == 'evaluate':
        eval_parameters = {k:v for k,v in self.train_parameters[i].
↪items() if k in ['batch_size', 'seq_len', 'data_type', 'sink_padding',
↪'d_hidden', 'function_callable', 'lorenz_kwargs']}
        if 'loss_type' in self.init_parameters[i]:
            eval_parameters['loss_type'] = self.
↪init_parameters[i]['loss_type'] # 确保评估使用与训练相同类型的损失函数
        if eval_configs and len(eval_configs) > 0:
            for k,current_config in enumerate(eval_configs):
                current_eval_params = eval_parameters.copy()
                current_eval_params.update(current_config)

```

```

        experiment.evaluate(**current_eval_params)
    else: # 万一没有传入 eval_configs 的底
        print(f"  No eval_configs provided for evaluation,
↪using default eval parameters for Experiment {i+1}.")
        default_config = {'eval_name': f'eval_{eval_count}'}
        current_eval_params = eval_parameters.copy()
        current_eval_params.update(default_config)
        experiment.evaluate(**current_eval_params)
        eval_count+=1

    self.results[i] = experiment.get_results(result_set)
experiment.offload_to_cpu() # 训练完成后将模型和结果移回 CPU, 节省 GPU
显存

self.experiments[i] = experiment
if torch.cuda.is_available():
    torch.cuda.empty_cache()
if torch.backends.mps.is_available():
    torch.mps.empty_cache()
if not parallel:
    print_vram_usage(tag=f"Experiment {i+1} 清理后")
    print(f"Experiment {i+1}/{self.num_experiments} completed.\n")

print_vram_usage(tag=f"本底显存检测")
if parallel_workers == 1:
    for i in range(self.num_experiments):
        single_train(i, parallel=False)
elif isinstance(parallel_workers, int) and parallel_workers > 1:
    print(f"Running {self.num_experiments} experiments in parallel with
↪{parallel_workers} workers...")
    with concurrent.futures.
↪ThreadPoolExecutor(max_workers=parallel_workers) as executor:
        executor.map(lambda i: single_train(i, parallel=True),
↪range(self.num_experiments))
    else:
        raise ValueError("parallel_workers must be a positive integer.")

if self.results[0] is not None:

```

```

        invalid_keys = [key for key in result_set if key not in self.
↪results[0]]
        if invalid_keys:
            print(f"\n[Final Check Warning] 以下请求的指标在所有流程结束后仍
未找到，请检查拼写是否正确或数据是否生成失败：\n{invalid_keys}\n")

    def plot(self, compare_experiments=True,
        subplot_shape=(1,-1),figure_size=None,suptitle='Looped Transformer_
↪Experiment Results',
        colors=['blue', 'red', 'green', 'yellow', 'cyan', 'orange', '
↪purple', 'brown']):
        '''
        Args:
            compare_experiments: bool, True 表示在一张图中比较不同实验的结果，每张
图仅比较一个指标；False 表示每个实验单独成图，每张图仅比较一个实验的多个指标（第一个
指标使用左轴，第二个及以后的指标使用共享 x 轴的右轴）

            subplot_shape: tuple of int, 表示子图的（行数，列数），(-1 表示自动计
算），如（1，-1）表示所有图在一行，（-1，2）表示每行两个图，自动计算行数

            figure_size: tuple of float, 画布大小，单位为英寸，如（15，10）。默认
值 None 表示使用（8*subplot_shape[1], 6*subplot_shape[0]）的大小

            suptitle: str, 整体图的标题

            colors: list of str, 画图使用的颜色列表，每张图从头开始循环使用这些颜色
        '''
        # 画图
        num_plots = 0
        for item in self.result_lists:
            x_type = item[1]
            clean_list = [m for m in item[0] if m != '|']
            if x_type in ['epoch', 'block']:
                if compare_experiments:
                    num_plots += len(clean_list)
                else:

```

```

        num_plots += self.num_experiments
    elif x_type == 'experiment':
        num_plots += 1
    if subplot_shape == (-1, -1):
        raise ValueError("Invalid subplot_shape: both dimensions cannot be -1. Please specify at least one dimension or provide a valid shape.")
    if subplot_shape[1] == -1:
        subplot_shape = (subplot_shape[0], math.ceil(num_plots / subplot_shape[0])) # 自动计算列数
    elif subplot_shape[0] == -1:
        subplot_shape = (math.ceil(num_plots / subplot_shape[1]), subplot_shape[1]) # 自动计算行数
    if subplot_shape[0] * subplot_shape[1] < num_plots:
        raise ValueError(f"subplot_shape {subplot_shape} is too small for the number of plots {num_plots}. Please increase the dimensions.")
    if figure_size is None:
        figure_size = (8*subplot_shape[1], 6*subplot_shape[0])

    if self.results[0] is None:
        print("No results to plot. Please run the experiments first.")
        return
    results = self.results
    fig, axes = plt.subplots(*subplot_shape, figsize=figure_size)
    axes_flat = np.atleast_1d(axes).flatten() # 将 axes 转换为一维数组，方便索引

    index=0
    for item in self.result_lists:
        result_list = item[0] # 需要展示的结果名称列表
        x = item[1] # 横坐标类型 ('epoch' 或 'experiment')
        baseline_index = item[2] if len(item) > 2 else None # 基线索引 (可选), 表示用于计算相对值的基线实验索引
        baseline_name=None
        if x in ['epoch', 'block']: # 表示画横坐标是 epoch 或 block 的曲线
            if compare_experiments:
                clean_result_list = [m for m in result_list if m != '|']
                for metric in clean_result_list:

```



```

        if baseline_index is not None and 0 <= baseline_index <
↪self.num_experiments:
            baseline_name = self.
↪init_parameters[baseline_index].get('experiment_name', f'Experiment_
↪{baseline_index+1}')
            baseline_data = np.
↪array(results[baseline_index][metric])
            else:
                baseline_name = None
                baseline_data = None
            for i in range(self.num_experiments):
                exp_name = self.init_parameters[i].
↪get('experiment_name', f'Experiment {i+1}')
                y_data = np.array(results[i][metric])
                label_name = exp_name # 专门给图例起名字, 保护原变
量
                if baseline_name is not None:
                    min_length = min(len(y_data),
↪len(baseline_data))
                    y_data = y_data[:min_length] - baseline_data[:
↪min_length] # 计算相对于基线的差值
                    if i == baseline_index:
                        label_name += ' (Baseline)'
                    axs_flat[index].plot(y_data, label=label_name,
↪color=colors[i%len(colors)])
                    axs_flat[index].set_xlabel('Epoch' if x == 'epoch' else
↪'Block (Depth)')
                if baseline_name is not None:
                    # 加上 \n 换行, 防止 ylabel 太长挤出画布边界
                    axs_flat[index].set_ylabel(f"{metric}\n(relative to_
↪{baseline_name})")
                else:
                    axs_flat[index].set_ylabel(metric)
                    axs_flat[index].tick_params(axis='y')
                    axs_flat[index].set_title('Comparison of ' + metric)

```

```

        axs_flat[index].legend(loc='upper right')
        axs_flat[index].grid(True)
        # 为顶部留出 35% 的空间防止遮挡图例
        y_min, y_max = axs_flat[index].get_ylim()
        axs_flat[index].set_ylim(y_min, y_max + 0.35 * (y_max -
↪y_min))

        index += 1

    else:
        if baseline_index is not None and 0 <= baseline_index <
↪self.num_experiments:
            baseline_name = self.init_parameters[baseline_index].
↪get('experiment_name', f'Experiment {baseline_index+1}')
            baseline_data_dict = {m: np.
↪array(results[baseline_index][m]) for m in result_list if m!='|'}
        else:
            baseline_name = None
            baseline_data_dict = None
            for i in range(self.num_experiments):
                color_index = 0
                exp_name = self.init_parameters[i].
↪get('experiment_name', f'Experiment {i+1}')
                if '|' in result_list:
                    split_idx = result_list.index('|')
                    clean_result_list = [metric for metric in
↪result_list if metric != '|']
                else:
                    split_idx = len(result_list)
                    clean_result_list = result_list
                num_metrics = len(clean_result_list)
                for j in range(split_idx):
                    metric_j = clean_result_list[j]
                    y_data = np.array(results[i][metric_j])
                    label_name = metric_j
                    if baseline_name is not None:
                        baseline_data = baseline_data_dict[metric_j]

```

```

        min_length = min(len(y_data),
↪len(baseline_data))

        y_data = y_data[:min_length] - baseline_data[:
↪min_length] # 计算相对于基线的差值

        if i == baseline_index:
            label_name += ' (Baseline)'
            axs_flat[index].plot(y_data, label=label_name,
↪color=colors[color_index%len(colors)])
            color_index += 1
            axs_flat[index].set_xlabel('Epoch' if x == 'epoch' else
↪'Block (Depth)')

            ylabel_left = '\n'.join(clean_result_list[:split_idx])
            if baseline_name is not None:
                ylabel_left += f"\n(relative to {baseline_name})"
            axs_flat[index].set_ylabel(ylabel_left, color=colors[0])
            axs_flat[index].tick_params(axis='y',
↪labelcolor=colors[0])

            if num_metrics > split_idx:
                ax_twin = axs_flat[index].twinx() # 创建共享 x 轴但
独立 y 轴的第二个坐标轴

                primary_color = colors[split_idx%len(colors)]
                for j in range(split_idx, num_metrics):
                    metric_j = clean_result_list[j]
                    y_data = np.array(results[i][metric_j])
                    label_name = metric_j
                    if baseline_name is not None:
                        baseline_data = baseline_data_dict[metric_j]
                        min_length = min(len(y_data),
↪len(baseline_data))

                        y_data = y_data[:min_length] -
↪baseline_data[:min_length] # 计算相对于基线的差值

                        if i == baseline_index:
                            label_name += ' (Baseline)'
                            ax_twin.plot(y_data, label=label_name,
↪color=colors[color_index%len(colors)])

```

```

        color_index += 1
        ax_twin.set_ylabel('\n'.
↪join(clean_result_list[split_idx:]), color=primary_color)
        ax_twin.tick_params(axis='y',
↪labelcolor=primary_color)
        ax_twin.legend(loc='upper right')
        axs_flat[index].set_title(exp_name + ' - ' + ' and\n'.
↪join(clean_result_list))
        axs_flat[index].legend(loc='upper left')
        axs_flat[index].grid(True)
        # 为顶部留出 35% 的空间防止遮挡图例
        y1_min, y1_max = axs_flat[index].get_ylim()
        axs_flat[index].set_ylim(y1_min, y1_max + 0.35 *
↪(y1_max - y1_min))
        if num_metrics > split_idx:
            y2_min, y2_max = ax_twin.get_ylim()
            ax_twin.set_ylim(y2_min, y2_max + 0.35 * (y2_max -
↪y2_min))

        index += 1
        elif x == 'experiment': # 表示画不同实验对比的柱状图
            exp_names = [self.init_parameters[i].get('experiment_name',
↪f'Experiment {i+1}')] for i in range(self.num_experiments)]
            if baseline_index is not None and 0 <= baseline_index < self.
↪num_experiments:
                baseline_name = self.init_parameters[baseline_index].
↪get('experiment_name', f'Experiment {baseline_index+1}')
                exp_names[baseline_index] += '\n(Baseline)'

            x_pos = np.arange(len(exp_names))
            if '|' in result_list:
                split_idx = result_list.index('|')
                clean_result_list = [metric for metric in result_list if
↪metric != '|']
            else:
                split_idx = len(result_list)

```

```

        clean_result_list = result_list
        num_metrics = len(clean_result_list)
        width = 0.8 / num_metrics # 确保所有柱子加起来不超过 0.8 的宽度,
        留下 0.2 的组间空白

        color_index = 0
        # 左侧坐标轴
        for j in range(split_idx):
            metric_j = clean_result_list[j]
            y_data_j = [results[i][metric_j] for i in range(self.
↪num_experiments)]

            if baseline_name is not None:
                base_val = y_data_j[baseline_index]
                y_data_j = [val - base_val for val in y_data_j]
                offset_j = (j - num_metrics / 2 + 0.5) * width
                axs_flat[index].bar(x_pos + offset_j, y_data_j, width,
↪label=metric_j, color=colors[color_index % len(colors)])
                color_index += 1

            ylabel_left = '\n'.join(clean_result_list[:split_idx])
            if baseline_name is not None:
                ylabel_left += f"\n(relative to {baseline_name})"
            axs_flat[index].set_ylabel(ylabel_left, color=colors[0])
            axs_flat[index].tick_params(axis='y', labelcolor=colors[0])

        if num_metrics > split_idx:
            ax_twin = axs_flat[index].twinx() # 创建共享 X 轴但独立 Y 轴
            的坐标系

            primary_color = colors[split_idx % len(colors)]
            for j in range(split_idx, num_metrics):
                metric_j = clean_result_list[j]
                y_data_j = [results[i][metric_j] for i in range(self.
↪num_experiments)]

                if baseline_name is not None:
                    base_val = y_data_j[baseline_index]
                    y_data_j = [val - base_val for val in y_data_j]

                offset_j = (j - num_metrics / 2 + 0.5) * width

```

```

        ax_twin.bar(x_pos + offset_j, y_data_j, width,
↪label=metric_j, color=colors[color_index % len(colors)])
        color_index += 1
        ylabel_right = '\n'.join(clean_result_list[split_idx:])
        if baseline_name is not None:
            ylabel_right += f"\n(relative to {baseline_name})"
            ax_twin.set_ylabel(ylabel_right, color=primary_color)
            ax_twin.tick_params(axis='y', labelcolor=primary_color)
            ax_twin.legend(loc='upper right')
            axs_flat[index].legend(loc='upper left')
        else:
            axs_flat[index].legend(loc='best')
        y1_min, y1_max = axs_flat[index].get_ylim()
        pos1 = max(y1_max, 0)
        neg1 = max(-y1_min, 0)
        s1 = max(pos1, neg1) if max(pos1, neg1) > 0 else 1.0
        pos_norm1 = pos1 / s1
        neg_norm1 = neg1 / s1
        if num_metrics > split_idx:
            y2_min, y2_max = ax_twin.get_ylim()
            pos2 = max(y2_max, 0)
            neg2 = max(-y2_min, 0)
            s2 = max(pos2, neg2) if max(pos2, neg2) > 0 else 1.0
            pos_norm2 = pos2 / s2
            neg_norm2 = neg2 / s2
            max_pos_norm = max(pos_norm1, pos_norm2)
            max_neg_norm = max(neg_norm1, neg_norm2)
        else:
            max_pos_norm = pos_norm1
            max_neg_norm = neg_norm1
            s2 = 1.0 # 占位
        final_pos_norm = max(max_pos_norm * 1.35, 0.35)
        needs_zero_alignment = (baseline_name is not None) or (y1_min <
↪0) or (num_metrics > split_idx and y2_min < 0)
        if needs_zero_alignment:
            # 按照统一计算好的归一化比例，乘以各自的真实尺度

```

```

        axs_flat[index].set_ylim(-max_neg_norm * s1, final_pos_norm *
↪* s1)

        if num_metrics > split_idx:
            ax_twin.set_ylim(-max_neg_norm * s2, final_pos_norm *
↪s2)

        else:
            axs_flat[index].set_ylim(0, final_pos_norm * s1)
            if num_metrics > split_idx:
                ax_twin.set_ylim(0, final_pos_norm * s2)
            axs_flat[index].axhline(0, color='gray', linewidth=1,
↪linestyle='-', zorder=0) # 画一条贯穿整个图表的辅助零线，充当视觉上的“地平线”
            axs_flat[index].set_xticks(x_pos)
            axs_flat[index].set_xticklabels(exp_names, rotation=45,
↪ha='right')

            axs_flat[index].set_title(f"Comparison of {' and\n'.
↪join(clean_result_list)}")

            axs_flat[index].grid(axis='y', linestyle='--', alpha=0.7)
            index += 1

clear_lorenz_cache() # 清理洛伦兹系统数据缓存，释放内存
# 如果绘制的图的数量少于子图的总数量，则隐藏多余的子图
for i in range(index, len(axs_flat)):
    axs_flat[i].set_visible(False)
if suptitle is not None:
    plt.suptitle(suptitle)
plt.tight_layout()
plt.show()

```

2 Linear Regression Experiments

2.1 实验数据观测

2.1.1 绘制 Loss 下降曲线

```

[182]: loss_table = ExperimentTable(params_groups=[{'experiment_name':
↪↪'loss_history', 'curriculum': {'d_x': 5, 'seq_len': 10, 'duration_ratio': 0.
↪↪5}}])
loss_table.run(result_lists=[(['loss_history'], 'epoch')])

```

```
loss_table.plot(figure_size=(8,6))
```

loss_history initialized in 0.03 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.41 GB | Metal 驱动分配: 2.06 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.42 GB | Metal 驱动分配: 2.06 GB

Epoch 20/100, Loss: 1.1422, Norm Ratio: 0.1630, Avg Sink Score: 0.1154, Avg Sink Rate: 0.0000

Epoch 40/100, Loss: 0.9943, Norm Ratio: 0.0928, Avg Sink Score: 0.0737, Avg Sink Rate: 0.0000

Curriculum ended at epoch 51, step 1.

Epoch 60/100, Loss: 0.9526, Norm Ratio: 0.0989, Avg Sink Score: 0.0618, Avg Sink Rate: 0.0000

Epoch 80/100, Loss: 0.8416, Norm Ratio: 0.1008, Avg Sink Score: 0.0618, Avg Sink Rate: 0.0000

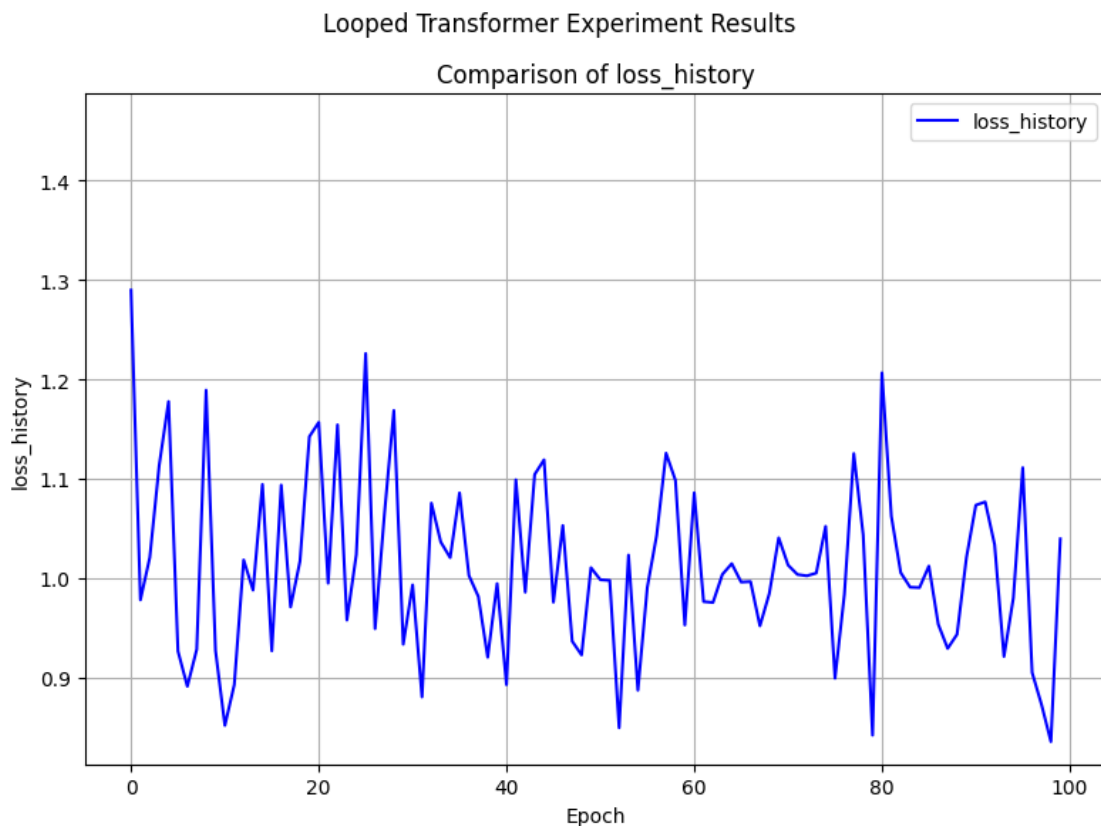
Epoch 100/100, Loss: 1.0393, Norm Ratio: 0.1031, Avg Sink Score: 0.0617, Avg Sink Rate: 0.0000

Training completed in 48.43 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.44 GB | Metal 驱动分配: 6.72 GB

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.43 GB | Metal 驱动分配: 2.11 GB

Experiment 1/1 completed.



2.1.2 绘制 y_{pred} 和 y_{true} 的变化曲线

```
[75]: norm_table = ExperimentTable(params_groups=[{'epochs': 1000, 'experiment_name': 'Looped Transformer', 'y_norm_error_history': 'y_norm_error_history', 'print_every': 200}])
norm_table.run(result_lists=[(['y_norm_error_history'], 'epoch')])
norm_table.plot(figsize=(8,6),compare_experiments=False)
```

y_norm_error_history initialized in 0.02 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB

Epoch 200/1000, Loss: 18.8248, current_blocks: 20

Epoch 400/1000, Loss: 19.7988, current_blocks: 20

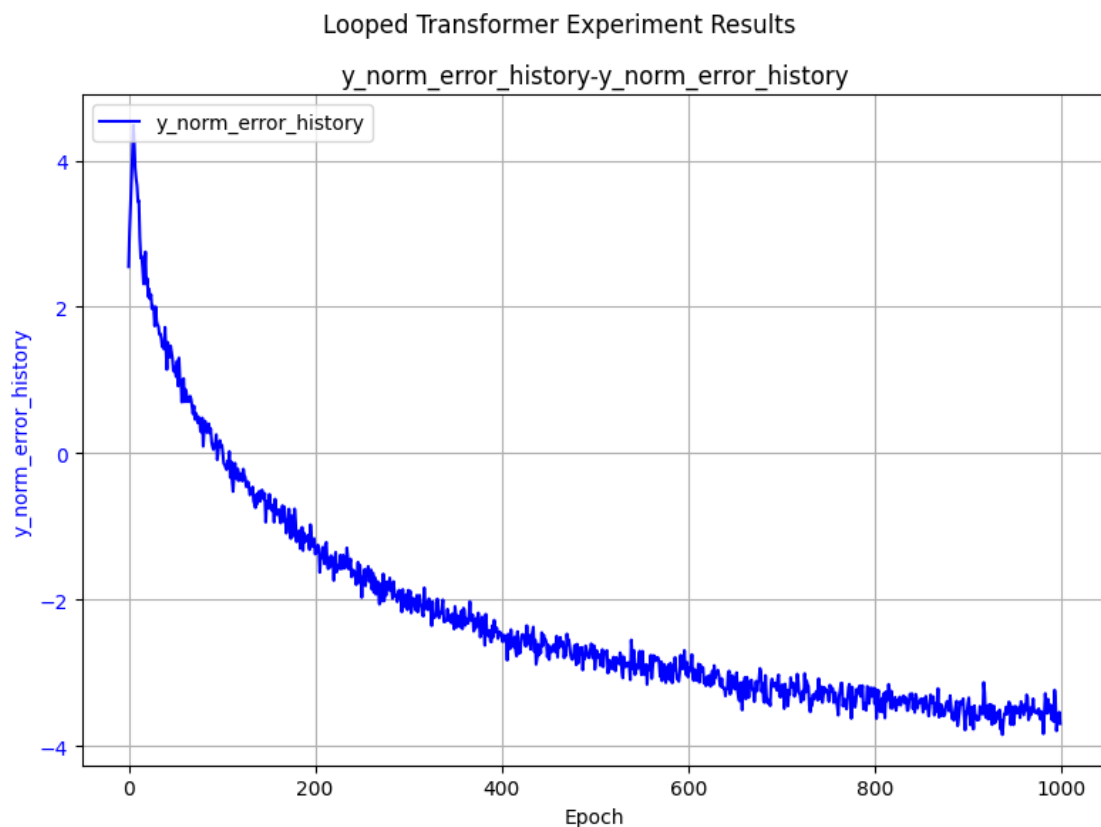
Epoch 600/1000, Loss: 17.7491, current_blocks: 20

Epoch 800/1000, Loss: 20.9858, current_blocks: 20

Epoch 1000/1000, Loss: 20.9834, current_blocks: 20

Training completed in 451.35 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 1/1 completed.



2.2 对比实验

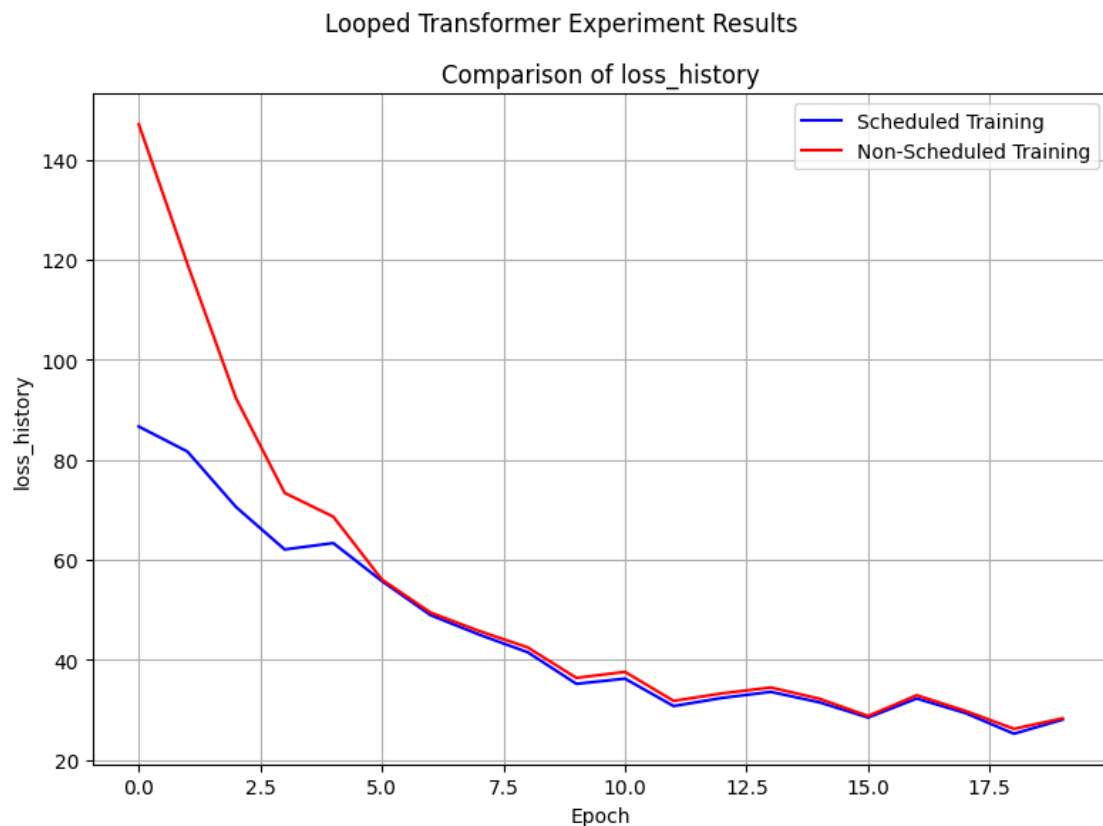
2.2.1 Scheduled Training vs Non-Scheduled Training 对比实验

```
[76]: table = ExperimentTable(params_groups=[{'scheduled_training': True,
↪ 'experiment_name': 'Scheduled Training'},
{'scheduled_training': False,
↪ 'experiment_name': 'Non-Scheduled Training'}])
table.run(result_lists=[(['loss_history'], 'epoch')])
table.plot(figure_size=(8, 6), compare_experiments=True)
```

Scheduled Training initialized in 0.02 seconds. Using device: mps

Non-Scheduled Training initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
Epoch 5/20, Loss: 63.3171, current_blocks: 19
Epoch 10/20, Loss: 35.1716, current_blocks: 20
Epoch 15/20, Loss: 31.4720, current_blocks: 20
Epoch 20/20, Loss: 28.0077, current_blocks: 20
Training completed in 9.75 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
Epoch 5/20, Loss: 68.5955, current_blocks: 20
Epoch 10/20, Loss: 36.3633, current_blocks: 20
Epoch 15/20, Loss: 32.1849, current_blocks: 20
Epoch 20/20, Loss: 28.2660, current_blocks: 20
Training completed in 9.81 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.13 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 2/2 completed.



2.2.2 不同 Positional Embedding 对比实验

```
[79]: pe_table = ExperimentTable(params_groups=[
    {'pe_type': ['learned_ape'], 'experiment_name': 'Learned APE'},
    {'pe_type': ['ape'], 'experiment_name': 'APE'},
    {'pe_type': ['alibi'], 'experiment_name': 'ALiBi'},
    {'pe_type': ['rope'], 'experiment_name': 'RoPE'},
    {'pe_type': ['ms_upe'], 'experiment_name': 'MS-UPE'},
    {'pe_type': ['ms_upe', 'learned_ape'], 'experiment_name': 'MS-UPE + Learned_
    ↪APE'},
], manual={'epochs': 100, 'print_every': 20})
pe_table.run(result_lists=[
    (['loss_history'], 'epoch'),
    (['loss_history'], 'epoch', 2) # 以 ALiBi 为基线
])
```

```
pe_table.plot(figure_size=(12, 6))
```

Learned APE initialized in 0.02 seconds. Using device: mps

APE initialized in 0.01 seconds. Using device: mps

ALiBi initialized in 0.01 seconds. Using device: mps

RoPE initialized in 0.01 seconds. Using device: mps

MS-UPE initialized in 0.01 seconds. Using device: mps

MS-UPE + Learned APE initialized in 0.01 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.19 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.19 GB

Epoch 20/100, Loss: 28.0077, current_blocks: 20

Epoch 40/100, Loss: 20.8999, current_blocks: 20

Epoch 60/100, Loss: 20.2006, current_blocks: 20

Epoch 80/100, Loss: 18.3025, current_blocks: 20

Epoch 100/100, Loss: 21.6682, current_blocks: 20

Training completed in 47.76 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB

Experiment 1/6 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Epoch 20/100, Loss: 25.4152, current_blocks: 20

Epoch 40/100, Loss: 20.2102, current_blocks: 20

Epoch 60/100, Loss: 19.7146, current_blocks: 20

Epoch 80/100, Loss: 17.6053, current_blocks: 20

Epoch 100/100, Loss: 21.3824, current_blocks: 20

Training completed in 48.05 seconds.

[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.21 GB

[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB

Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Epoch 20/100, Loss: 24.0522, current_blocks: 20

Epoch 40/100, Loss: 20.8352, current_blocks: 20

Epoch 60/100, Loss: 19.5621, current_blocks: 20

Epoch 80/100, Loss: 17.1478, current_blocks: 20

Epoch 100/100, Loss: 21.7769, current_blocks: 20

Training completed in 55.24 seconds.

[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.23 GB

[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB

Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Epoch 20/100, Loss: 23.8768, current_blocks: 20

Epoch 40/100, Loss: 20.4678, current_blocks: 20

Epoch 60/100, Loss: 19.6902, current_blocks: 20

Epoch 80/100, Loss: 17.1698, current_blocks: 20

Epoch 100/100, Loss: 21.6961, current_blocks: 20

Training completed in 68.12 seconds.

[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB

[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB

Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Epoch 20/100, Loss: 23.8994, current_blocks: 20

Epoch 40/100, Loss: 20.4356, current_blocks: 20

Epoch 60/100, Loss: 19.6087, current_blocks: 20

Epoch 80/100, Loss: 17.1837, current_blocks: 20

Epoch 100/100, Loss: 21.6859, current_blocks: 20

Training completed in 52.83 seconds.

[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB

[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB

Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Epoch 20/100, Loss: 28.1398, current_blocks: 20

Epoch 40/100, Loss: 20.6823, current_blocks: 20

Epoch 60/100, Loss: 20.1465, current_blocks: 20

Epoch 80/100, Loss: 18.4156, current_blocks: 20

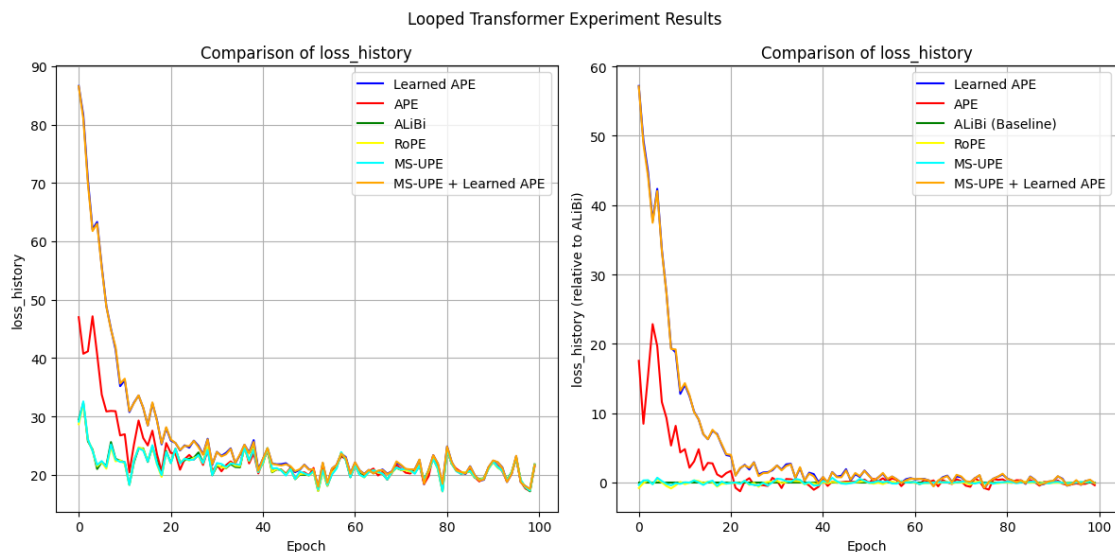
Epoch 100/100, Loss: 21.6613, current_blocks: 20

Training completed in 54.00 seconds.

[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.23 GB

[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB

Experiment 6/6 completed.



2.2.3 sink padding 的有无对比实验

```
[80]: sink_table = ExperimentTable(params_groups=[
    {'experiment_name': 'Without Sink Padding'},
    {'experiment_name': 'With Sink Padding (1)', 'sink_padding': 1},
    {'experiment_name': 'With Sink Padding (2)', 'sink_padding': 2},
    {'experiment_name': 'With Sink Padding (5)', 'sink_padding': 5},
    {'experiment_name': 'With Sink Padding (10)', 'sink_padding': 10},
], manual={'epochs': 100, 'print_every': 20})
sink_table.run(result_lists=[
    (['loss_history'], 'epoch'),
    (['loss_history'], 'epoch', 0), # 以 Without Sink Padding 为基线
])
sink_table.plot(figure_size=(12, 6))
```

Without Sink Padding initialized in 0.02 seconds. Using device: mps
 With Sink Padding (1) initialized in 0.01 seconds. Using device: mps
 With Sink Padding (2) initialized in 0.01 seconds. Using device: mps
 With Sink Padding (5) initialized in 0.01 seconds. Using device: mps
 With Sink Padding (10) initialized in 0.01 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.0077, current_blocks: 20
Epoch 40/100, Loss: 20.8999, current_blocks: 20
Epoch 60/100, Loss: 20.2006, current_blocks: 20
Epoch 80/100, Loss: 18.3025, current_blocks: 20
Epoch 100/100, Loss: 21.6682, current_blocks: 20
Training completed in 50.96 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 1/5 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.1106, current_blocks: 20
Epoch 40/100, Loss: 20.9680, current_blocks: 20
Epoch 60/100, Loss: 20.6434, current_blocks: 20
Epoch 80/100, Loss: 17.8341, current_blocks: 20
Epoch 100/100, Loss: 21.5738, current_blocks: 20
Training completed in 53.92 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 2/5 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 29.5266, current_blocks: 20
Epoch 40/100, Loss: 20.6409, current_blocks: 20
Epoch 60/100, Loss: 20.1771, current_blocks: 20
Epoch 80/100, Loss: 18.3634, current_blocks: 20
Epoch 100/100, Loss: 21.4438, current_blocks: 20
Training completed in 53.87 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 3/5 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 26.8264, current_blocks: 20
Epoch 40/100, Loss: 20.8156, current_blocks: 20

Epoch 60/100, Loss: 20.2012, current_blocks: 20

Epoch 80/100, Loss: 18.7726, current_blocks: 20

Epoch 100/100, Loss: 21.6547, current_blocks: 20

Training completed in 57.91 seconds.

[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 6.22 GB

[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Experiment 4/5 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Epoch 20/100, Loss: 27.0638, current_blocks: 20

Epoch 40/100, Loss: 21.4043, current_blocks: 20

Epoch 60/100, Loss: 20.5190, current_blocks: 20

Epoch 80/100, Loss: 18.5852, current_blocks: 20

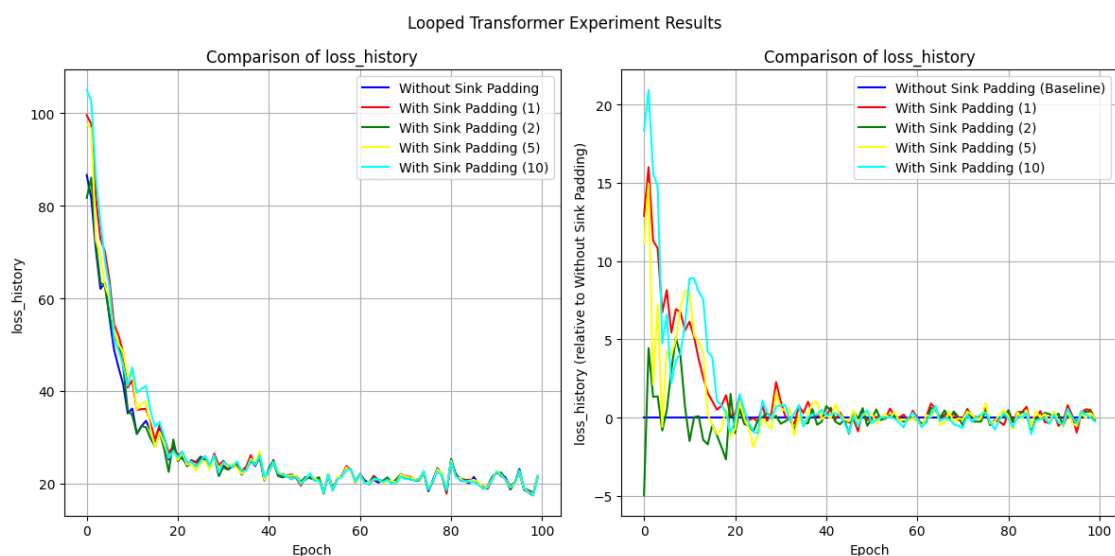
Epoch 100/100, Loss: 21.4375, current_blocks: 20

Training completed in 66.44 seconds.

[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 3.14 GB | Metal 驱动分配: 6.26 GB

[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB

Experiment 5/5 completed.



2.2.4 Residual Gate 对比实验

```
[27]: gate_table = ExperimentTable(params_groups=[
    {'experiment_name': 'No Residual Gate', 'residual_gate': (0, 0),
    ↪ 'residual_gate_type': 'fixed'},
    {'experiment_name': 'Residual Gate (1, 0.5)', 'residual_gate': (1, 0.5),
    ↪ 'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0.5) Vector', 'residual_gate': (1, 0.
    ↪ 5), 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Residual Gate (1, 0)', 'residual_gate': (1, 0),
    ↪ 'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0) Vector', 'residual_gate': (1, 0),
    ↪ 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Random Learnable Scalar', 'residual_gate': 'random',
    ↪ 'residual_gate_type': 'learnable_scalar', 'residual_random': (0.9, 0.1)},
    {'experiment_name': 'Random Learnable Vector', 'residual_gate': 'random',
    ↪ 'residual_gate_type': 'learnable_vector', 'residual_random': (0.9, 0.1)},
], manual={'gate_lr_ratio': 100, 'scheduler_type': 'cosine', 'epochs': 50,
    ↪ 'print_every': 25})
gate_table.run(result_lists=[
    (['loss_history'], 'epoch')
])
gate_table.plot(figure_size=(8, 6))
```

No Residual Gate initialized in 0.01 seconds. Using device: mps

Residual Gate (1, 0.5) initialized in 0.01 seconds. Using device: mps

Residual Gate (1, 0.5) Vector initialized in 0.01 seconds. Using device: mps

Residual Gate (1, 0) initialized in 0.01 seconds. Using device: mps

Residual Gate (1, 0) Vector initialized in 0.01 seconds. Using device: mps

Random Learnable Scalar initialized in 0.01 seconds. Using device: mps

Random Learnable Vector initialized in 0.01 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB

Epoch 25/50, Loss: 20.7067, current_blocks: 20

Epoch 50/50, Loss: 19.7848, current_blocks: 20

Training completed in 19.79 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/7 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 25/50, Loss: 21.0445, current_blocks: 20
Epoch 50/50, Loss: 19.9295, current_blocks: 20
Training completed in 20.28 seconds.

[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 2/7 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 25/50, Loss: 21.1970, current_blocks: 20
Epoch 50/50, Loss: 20.4095, current_blocks: 20
Training completed in 20.98 seconds.

[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 3/7 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 21.1870, current_blocks: 20
Epoch 50/50, Loss: 19.9217, current_blocks: 20
Training completed in 20.64 seconds.

[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 4/7 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.5020, current_blocks: 20
Epoch 50/50, Loss: 19.9380, current_blocks: 20
Training completed in 21.49 seconds.

[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 5/7 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.6064, current_blocks: 20

Epoch 50/50, Loss: 19.8823, current_blocks: 20

Training completed in 21.05 seconds.

[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB

[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB

Experiment 6/7 completed.

[Experiment 7 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB

Epoch 25/50, Loss: 22.2538, current_blocks: 20

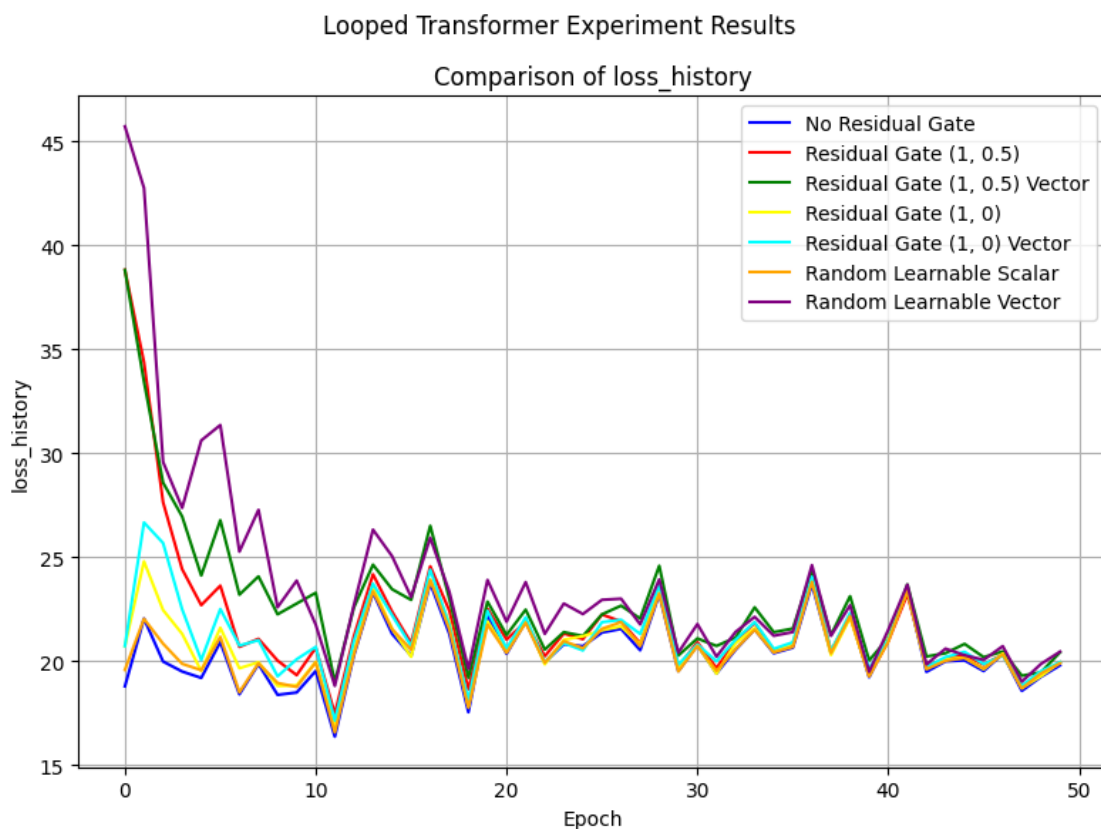
Epoch 50/50, Loss: 20.4417, current_blocks: 20

Training completed in 21.80 seconds.

[Experiment 7 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB

[Experiment 7 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB

Experiment 7/7 completed.



```
[26]: gate_table = ExperimentTable(params_groups=[
    {'experiment_name': 'Residual Gate (1, 0.5)', 'residual_gate': (1, 0.5),
    ↪ 'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0.5) Vector', 'residual_gate': (1, 0.
    ↪ 5), 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Residual Gate (1, 0)', 'residual_gate': (1, 0),
    ↪ 'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0) Vector', 'residual_gate': (1, 0),
    ↪ 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Random Learnable Scalar', 'residual_gate': 'random',
    ↪ 'residual_gate_type': 'learnable_scalar', 'residual_random': (0.9, 0.1)},
    {'experiment_name': 'Random Learnable Vector', 'residual_gate': 'random',
    ↪ 'residual_gate_type': 'learnable_vector', 'residual_random': (0.9, 0.1)},
], manual={'gate_lr_ratio': 100, 'scheduler_type': 'cosine', 'epochs': 100,
    ↪ 'print_every': 25})
gate_table.run(result_lists=[
    (['residual_gate_history_a_mean_relative',
    ↪ 'residual_gate_history_b_mean_relative'], 'epoch')
])
gate_table.plot(figure_size=(20,30),compare_experiments=False,
    ↪ subplot_shape=(-1,2), suptitle='')
```

```
Residual Gate (1, 0.5) initialized in 0.02 seconds. Using device: mps
Residual Gate (1, 0.5) Vector initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) Vector initialized in 0.01 seconds. Using device: mps
Random Learnable Scalar initialized in 0.01 seconds. Using device: mps
Random Learnable Vector initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Epoch 25/100, Loss: 21.1192, current_blocks: 20
Epoch 50/100, Loss: 19.8862, current_blocks: 20
Epoch 75/100, Loss: 21.3765, current_blocks: 20
Epoch 100/100, Loss: 20.4947, current_blocks: 20
Training completed in 34.69 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
```

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/6 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 21.3562, current_blocks: 20
Epoch 50/100, Loss: 20.4467, current_blocks: 20
Epoch 75/100, Loss: 21.5648, current_blocks: 20
Epoch 100/100, Loss: 20.8265, current_blocks: 20
Training completed in 35.89 seconds.

[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 21.0529, current_blocks: 20
Epoch 50/100, Loss: 19.8676, current_blocks: 20
Epoch 75/100, Loss: 21.1713, current_blocks: 20
Epoch 100/100, Loss: 20.4512, current_blocks: 20
Training completed in 35.34 seconds.

[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 20.5109, current_blocks: 20
Epoch 50/100, Loss: 20.1726, current_blocks: 20
Epoch 75/100, Loss: 21.4211, current_blocks: 20
Epoch 100/100, Loss: 20.5051, current_blocks: 20
Training completed in 36.48 seconds.

[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 20.5986, current_blocks: 20
Epoch 50/100, Loss: 19.8349, current_blocks: 20
Epoch 75/100, Loss: 21.2072, current_blocks: 20

Epoch 100/100, Loss: 20.3681, current_blocks: 20

Training completed in 37.95 seconds.

[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB

[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB

Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB

Epoch 25/100, Loss: 22.4328, current_blocks: 20

Epoch 50/100, Loss: 20.2457, current_blocks: 20

Epoch 75/100, Loss: 21.3175, current_blocks: 20

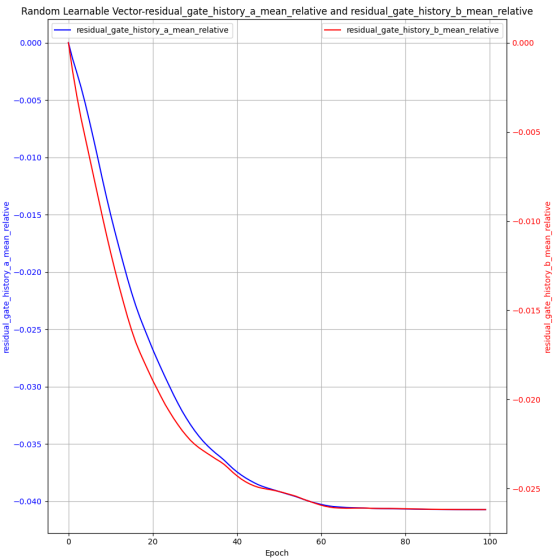
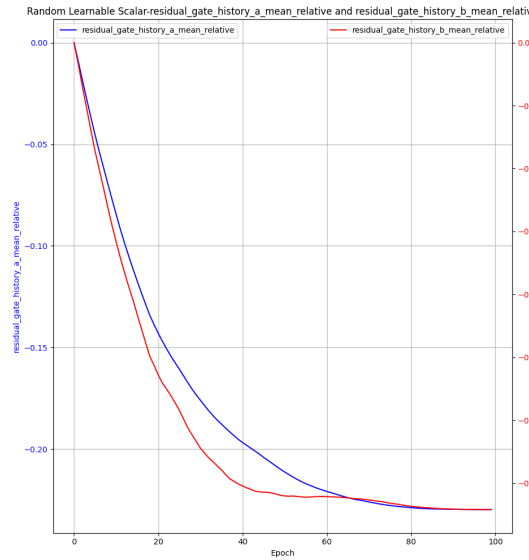
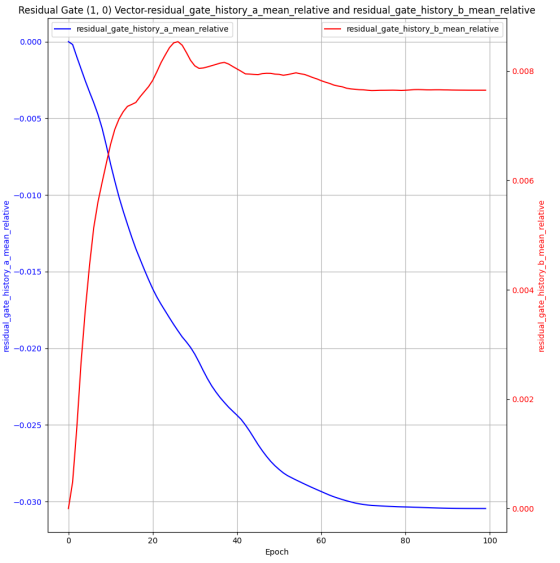
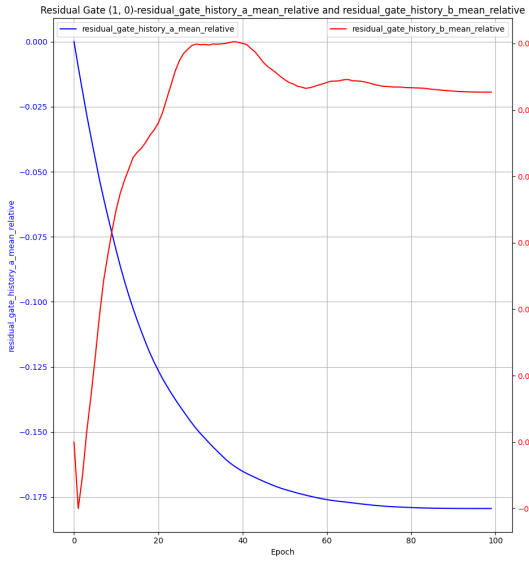
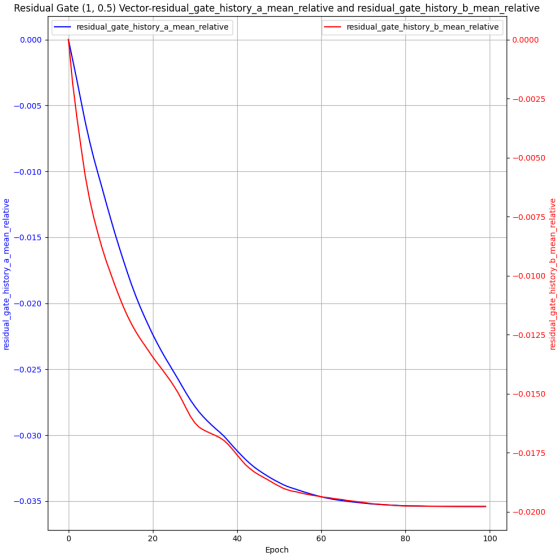
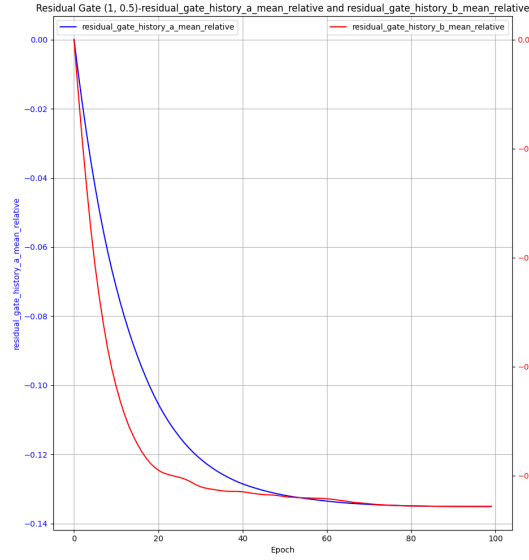
Epoch 100/100, Loss: 20.7065, current_blocks: 20

Training completed in 41.02 seconds.

[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB

[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB

Experiment 6/6 completed.



2.2.5 不同 scheduler 对比实验

```
[28]: experiment_table = ExperimentTable(params_groups=[{'experiment_name': 'No_
    ↪Scheduler'},
                                                    {'scheduler_type': 'cosine',
    ↪'experiment_name': 'Cosine Scheduler', 'lr_scale': 0},
                                                    {'scheduler_type': 'step',
    ↪'experiment_name': 'Step Scheduler', 'lr_scale': 0.1, 'step_size_scheduler':
    ↪90}],
    manual={'epochs':200, 'lr': 1e-3, 'optimizer_type': 'adamw', 'pe_type':
    ↪['alibi'],
            'wd_adamw': 0.2, 'd_model': 32, 'num_heads': 1, 'max_seq_len': 500,
    ↪'seq_len': 400, 'batch_size': 64, 'd_x': 20,
            'num_blocks': 20, 'num_eff': 15,
            'residual_gate': (1,1), 'residual_gate_type': 'learnable_scalar',
    ↪'gate_lr_ratio': 100,
            'scheduler_type': None, 'lr_scale': 0.01, 'step_size_scheduler': 10,
            'print_every': 50,
            'layer_weight_decay': 0.8, 'seq_weight_decay': 0.9
    })
experiment_table.run(result_lists=[(['loss_history'], 'epoch', 0),
    ↪(['y_pred_norm_history'], 'epoch', 0), (['y_true_norm_history'], 'epoch')])
experiment_table.plot(figure_size=(18, 6))
```

No Scheduler initialized in 0.02 seconds. Using device: mps

Cosine Scheduler initialized in 0.01 seconds. Using device: mps

Step Scheduler initialized in 0.01 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB

Epoch 50/200, Loss: 19.8760, current_blocks: 20

Epoch 100/200, Loss: 21.0343, current_blocks: 20

Epoch 150/200, Loss: 19.6615, current_blocks: 20

Epoch 200/200, Loss: 18.4375, current_blocks: 20

Training completed in 66.28 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.07 GB
Experiment 1/3 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Epoch 50/200, Loss: 19.8793, current_blocks: 20
Epoch 100/200, Loss: 21.0404, current_blocks: 20
Epoch 150/200, Loss: 19.6585, current_blocks: 20
Epoch 200/200, Loss: 18.4290, current_blocks: 20
Training completed in 64.98 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Experiment 2/3 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.09 GB
Epoch 50/200, Loss: 19.8760, current_blocks: 20
Epoch 100/200, Loss: 21.0657, current_blocks: 20
Epoch 150/200, Loss: 19.6840, current_blocks: 20
Epoch 200/200, Loss: 18.4393, current_blocks: 20
Training completed in 65.11 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Experiment 3/3 completed.



3 Non-linear Regression Experiments

3.1 实验数据观测

```
[70]: manual=dict(
    data_type='nonlinear',
    function_callable=lambda x: F.relu(x),
    max_seq_len=500,
    scheduled_training=True,
    pe_type=['alibi'],
    residual_gate=(1, 0.5),
    residual_gate_type='learnable_scalar',
    num_eff=5,
    num_blocks=20,
    batch_size=8,
    num_heads=2,
    d_model=128,
    seq_len=400,
    d_hidden=8,
    epochs=100)
```

```
[31]: loss_table = ExperimentTable(params_groups=[{'experiment_name': 'loss_history',
                                                    'epochs': 500,
                                                    'print_every': 100}],
    ↪manual=manual)
loss_table.run(result_lists=[(['loss_history'], 'epoch'),
    ↪(['y_norm_ratio_history', 'residual_gate_history_a',
    ↪'residual_gate_history_b'], 'epoch')])
loss_table.plot(figure_size=(8,12), compare_experiments=False,
    ↪subplot_shape=(-1,1))
```

loss_history initialized in 0.02 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.28 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.28 GB

Epoch 100/500, Loss: 0.4739, Norm Ratio: 0.4741, Current Blocks: 20

Epoch 200/500, Loss: 0.3862, Norm Ratio: 0.6557, Current Blocks: 20

Epoch 300/500, Loss: 0.6595, Norm Ratio: 0.5505, Current Blocks: 20

Epoch 400/500, Loss: 0.3205, Norm Ratio: 0.5234, Current Blocks: 20

Epoch 500/500, Loss: 0.3823, Norm Ratio: 0.5465, Current Blocks: 20

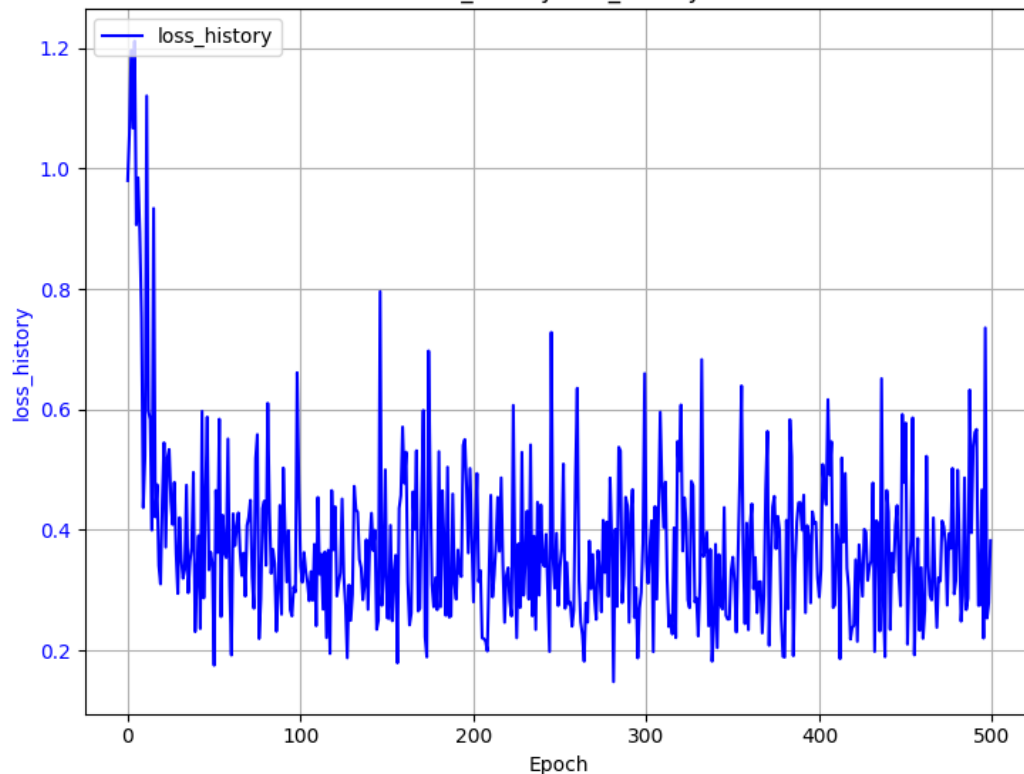
Training completed in 51.75 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.23 GB | Metal 驱动分配: 0.93 GB

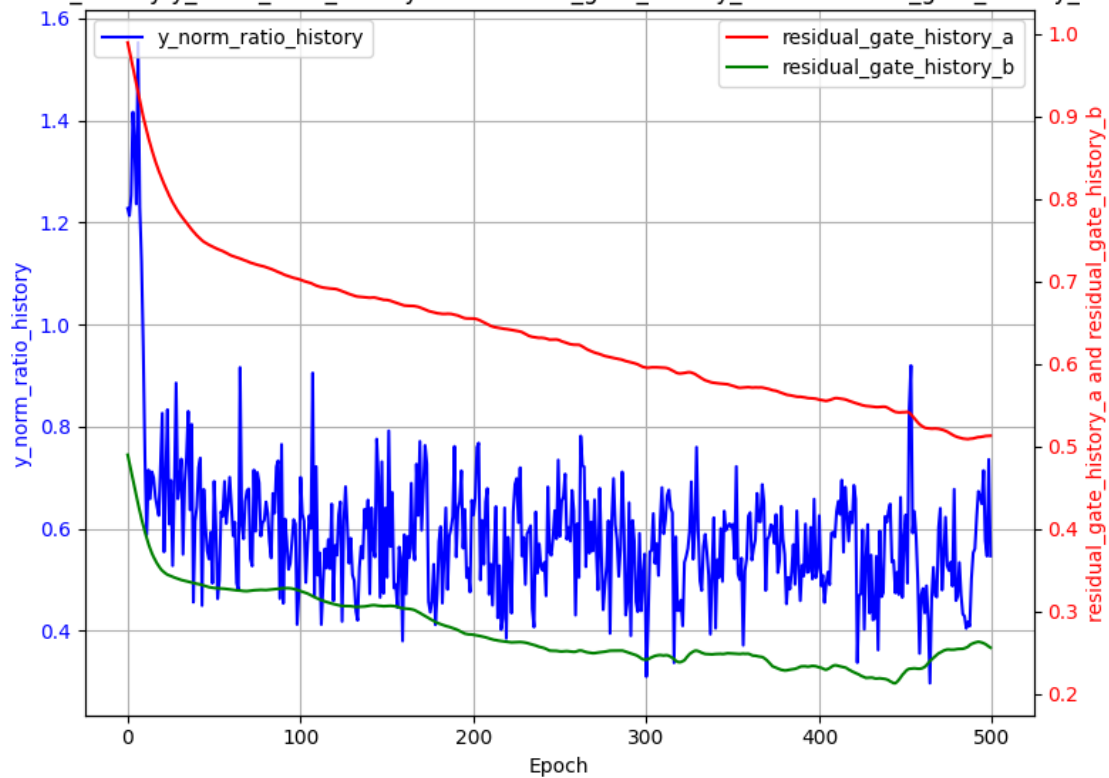
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.29 GB

Experiment 1/1 completed.

Looped Transformer Experiment Results
loss_history-loss_history



loss_history-y_norm_ratio_history and residual_gate_history_a and residual_gate_history_b



```
[95]: loss_table = ExperimentTable(params_groups=[{'experiment_name': 'loss_history',
                                                    'epochs': 50,
                                                    'steps_per_epoch': 200,
                                                    'init_std': 'auto',
                                                    'print_every': 5}], manual=manual)
loss_table.run(result_lists=[(['loss_history'], 'epoch'),
                               ↳(['y_norm_ratio_history'], 'epoch'), (['residual_gate_history_a'],
                               ↳['residual_gate_history_b', '|'], 'epoch'])])
loss_table.plot(compare_experiments=False, subplot_shape=(-1,1))
```

loss_history initialized in 0.02 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.28 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.28 GB

Epoch 5/50, Loss: 0.3561, Norm Ratio: 0.5421, Current Blocks: 9

Epoch 10/50, Loss: 0.3383, Norm Ratio: 0.5416, Current Blocks: 14

Epoch 15/50, Loss: 0.3495, Norm Ratio: 0.5456, Current Blocks: 19

Epoch 20/50, Loss: 0.3471, Norm Ratio: 0.5498, Current Blocks: 20

Epoch 25/50, Loss: 0.3525, Norm Ratio: 0.5475, Current Blocks: 20

Epoch 30/50, Loss: 0.3495, Norm Ratio: 0.5437, Current Blocks: 20

Epoch 35/50, Loss: 0.3371, Norm Ratio: 0.5565, Current Blocks: 20

Epoch 40/50, Loss: 0.3466, Norm Ratio: 0.5229, Current Blocks: 20

Epoch 45/50, Loss: 0.3412, Norm Ratio: 0.5482, Current Blocks: 20

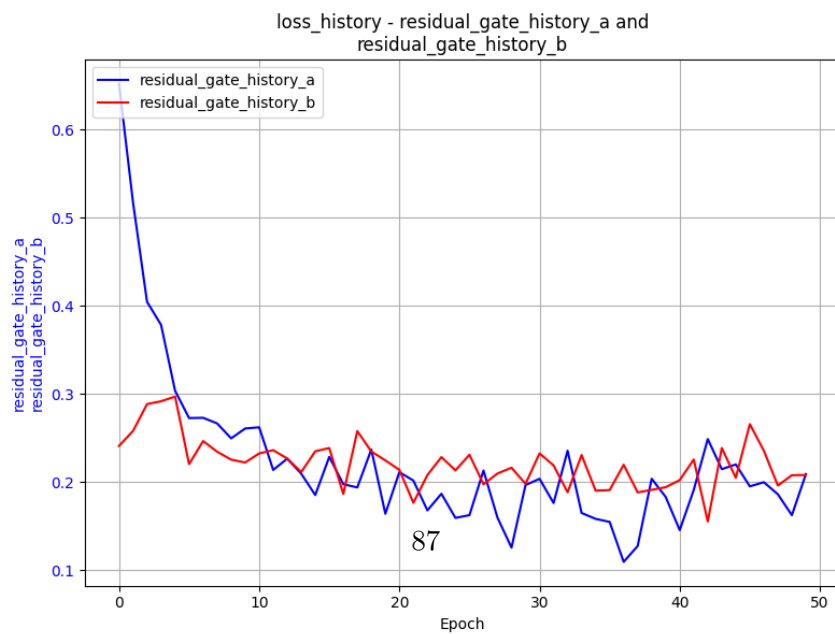
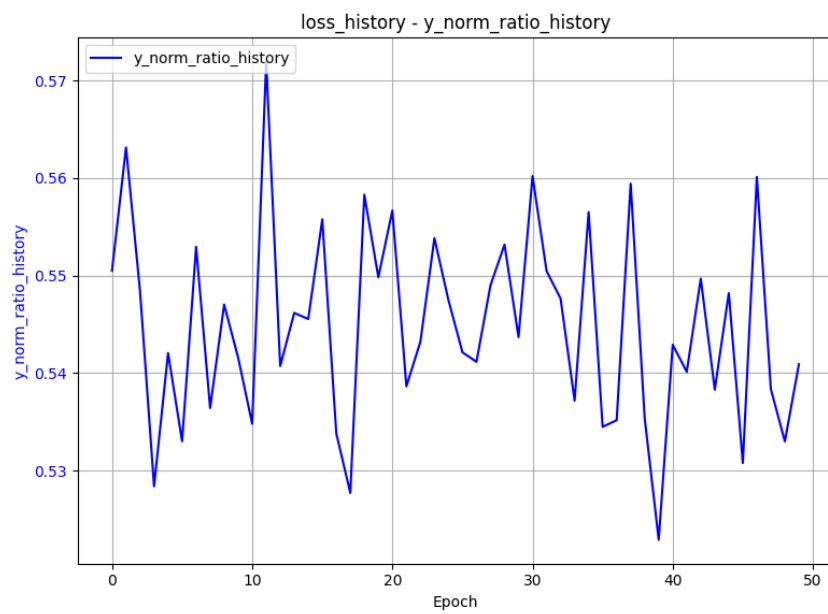
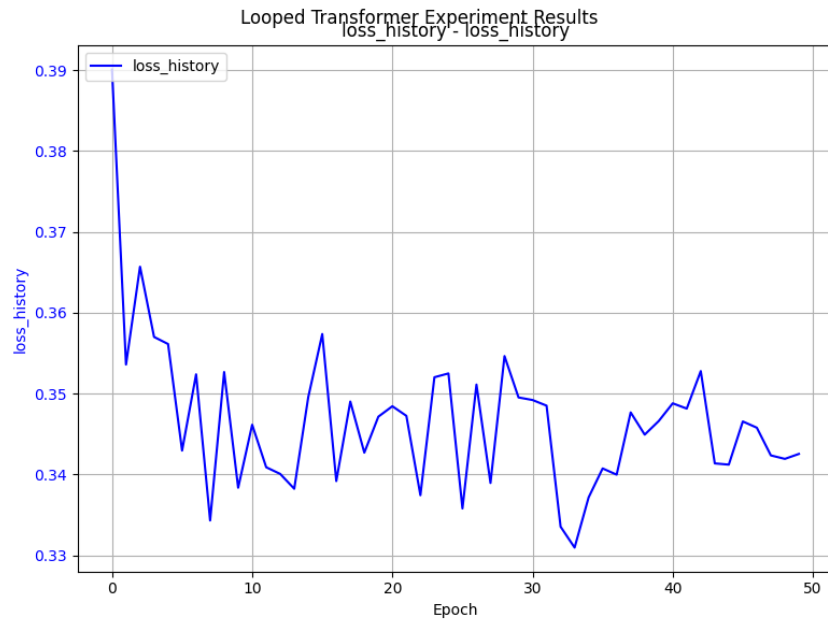
Epoch 50/50, Loss: 0.3425, Norm Ratio: 0.5409, Current Blocks: 20

Training completed in 1039.72 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 0.70 GB

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB

Experiment 1/1 completed.



实验数据分析 1. 参数量分析 2. norm_ratio 与 loss 的关系分析 - 对于 Linear+ReLU+Linear，由于 Linear 矩阵归一化后方差为 1，ReLU 又会丢掉一半的激活，因此 y_{true} 的二阶矩（能量）期望为 $\mathbb{E}[y^2] \approx 0.5$ - 模型在进行 In-Context Learning (ICL) 时，遇到了认知瓶颈。它把复杂的真实信号 y 拆解成了两个部分： $y = y_{\text{learnable}} + y_{\text{noise}}$ - $y_{\text{learnable}}$ ：模型通过现有的上下文能够学明白的“低频/线性”规律。- y_{noise} ：由于非线性过于复杂或上下文信息仍显不足，模型无法解开的“高频/非线性”残差。- 在 MSE 损失函数的逼迫下，模型最聪明的做法就是精准预测自己能看懂的部分，对看不懂的部分直接输出 0。因此它的预测值 $\hat{y} \approx y_{\text{learnable}}$ 。- 根据 Norm Ratio 倒推 loss：- 观察到 Norm Ratio 在 0.54 附近震荡，即

$$\text{NormRatio} = \sqrt{\frac{\mathbb{E}[\hat{y}^2]}{\mathbb{E}[y^2]}} \approx \sqrt{\frac{\mathbb{E}[y_{\text{learnable}}^2]}{\mathbb{E}[y_{\text{learnable}}^2] + \mathbb{E}[y_{\text{noise}}^2]}} \approx 0.54$$

- 代入 $\mathbb{E}[y^2] \approx 0.5$ ，可以推断出 $\mathbb{E}[y_{\text{learnable}}^2] \approx 0.14$ ， $\mathbb{E}[y_{\text{noise}}^2] \approx 0.36$ 。- 由于 $\hat{y} \rightarrow y_{\text{learnable}}$ ，因此最终的 loss 约为

$$\mathbb{E}[(y - \hat{y})^2] = \mathbb{E}[y_{\text{noise}}^2] + \mathbb{E}[(y_{\text{learnable}} - \hat{y})^2] \approx 0.36$$

与实际观察到的 loss 值相近。

4 Curriculum Learning 实验

4.1 linear

4.1.1 训练 10000 步

```
[ ]: manual_config = dict(
    # 任务基础配置
    data_type='linear',
    d_x=20,
    seq_len=80,
    batch_size=64,

    # 模型架构（完全对齐论文）
    num_blocks=20,
    num_eff=15,
    d_model=256,
    num_heads=8,
    pe_type=['learned_ape'],
```



```

# 核心物理机制配置
x_init='zero',          # 绝对零起点
init_std='auto',        # 方差自适应 (1/sqrt(256))
residual_gate=(1, 1),   # 标准注入
residual_gate_type='fixed',

# 优化器剥离外挂
optimizer_type='adam',
lr=1e-4,
layer_weight_decay=1.0,  # 关闭层级惩罚
seq_weight_decay=1.0,   # 关闭时间步惩罚
scheduler_type=None,     # 恒定学习率

# 运行控制
epochs=50,
steps_per_epoch=200,
print_every=1,

scheduled_training=True  # 开启 b 的 Kick-start
)

# === 定义并执行实验表 ===
final_table = ExperimentTable(params_groups=[
    {
        'experiment_name': 'Without Curriculum (Hard Mode)',
        'curriculum': {} # 空字典关闭课程学习
    },
    {
        'experiment_name': 'With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8 # 前 40 个 epoch 完成课程
        }
    }
], manual=manual_config)

```

```

final_table.run(result_lists=[(['loss_history'], 'epoch'),  

    ↳(['y_norm_ratio_history'], 'epoch')])  

final_table.plot(figure_size=(10, 12), compare_experiments=True,  

    ↳subplot_shape=(-1, 1), subtitle='Curriculum Learning Comparison on Looped_  

    ↳Transformer')

```

Without Curriculum (Hard Mode) initialized in 0.03 seconds. Using device: mps
 With Curriculum (Perfect Path) initialized in 0.02 seconds. Using device: mps
 [本底显存检测] 显存占用 -> PyTorch 分配: 0.04 GB | Metal 驱动分配: 1.03 GB
 [Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.05 GB | Metal 驱动分配: 1.03 GB

Epoch 1/50, Loss: 1.0582, Norm Ratio: 0.1946, Avg Sink Score: 0.0655, Avg Sink Rate: 0.0000
 Epoch 2/50, Loss: 1.0064, Norm Ratio: 0.0913, Avg Sink Score: 0.0636, Avg Sink Rate: 0.0000
 Epoch 3/50, Loss: 1.0058, Norm Ratio: 0.0701, Avg Sink Score: 0.0629, Avg Sink Rate: 0.0000
 Epoch 4/50, Loss: 1.0001, Norm Ratio: 0.0598, Avg Sink Score: 0.0622, Avg Sink Rate: 0.0000
 Epoch 5/50, Loss: 1.0026, Norm Ratio: 0.0482, Avg Sink Score: 0.0622, Avg Sink Rate: 0.0000
 Epoch 6/50, Loss: 0.9974, Norm Ratio: 0.0422, Avg Sink Score: 0.0621, Avg Sink Rate: 0.0000
 Epoch 7/50, Loss: 1.0029, Norm Ratio: 0.0469, Avg Sink Score: 0.0620, Avg Sink Rate: 0.0000
 Epoch 8/50, Loss: 0.9991, Norm Ratio: 0.0409, Avg Sink Score: 0.0618, Avg Sink Rate: 0.0000
 Epoch 9/50, Loss: 1.0000, Norm Ratio: 0.0467, Avg Sink Score: 0.0619, Avg Sink Rate: 0.0000
 Epoch 10/50, Loss: 1.0101, Norm Ratio: 0.0459, Avg Sink Score: 0.0619, Avg Sink Rate: 0.0000
 Epoch 11/50, Loss: 1.0005, Norm Ratio: 0.0469, Avg Sink Score: 0.0618, Avg Sink Rate: 0.0000
 Epoch 12/50, Loss: 1.0032, Norm Ratio: 0.0422, Avg Sink Score: 0.0617, Avg Sink Rate: 0.0000
 Epoch 13/50, Loss: 1.0008, Norm Ratio: 0.0384, Avg Sink Score: 0.0617, Avg Sink Rate: 0.0000

Epoch 14/50, Loss: 1.0042, Norm Ratio: 0.0402, Avg Sink Score: 0.0616, Avg Sink Rate: 0.0000

Epoch 15/50, Loss: 1.0014, Norm Ratio: 0.0380, Avg Sink Score: 0.0616, Avg Sink Rate: 0.0000

Epoch 16/50, Loss: 0.9992, Norm Ratio: 0.0409, Avg Sink Score: 0.0616, Avg Sink Rate: 0.0000

Epoch 17/50, Loss: 1.0021, Norm Ratio: 0.0352, Avg Sink Score: 0.0616, Avg Sink Rate: 0.0000

Epoch 18/50, Loss: 0.9987, Norm Ratio: 0.0407, Avg Sink Score: 0.0615, Avg Sink Rate: 0.0000

Epoch 19/50, Loss: 1.0048, Norm Ratio: 0.0314, Avg Sink Score: 0.0616, Avg Sink Rate: 0.0000

Epoch 20/50, Loss: 1.0034, Norm Ratio: 0.0404, Avg Sink Score: 0.0615, Avg Sink Rate: 0.0000

Epoch 21/50, Loss: 1.0021, Norm Ratio: 0.0342, Avg Sink Score: 0.0615, Avg Sink Rate: 0.0000

Epoch 22/50, Loss: 1.0008, Norm Ratio: 0.0289, Avg Sink Score: 0.0615, Avg Sink Rate: 0.0000

Epoch 23/50, Loss: 1.0047, Norm Ratio: 0.0295, Avg Sink Score: 0.0614, Avg Sink Rate: 0.0000

Epoch 24/50, Loss: 1.0013, Norm Ratio: 0.0320, Avg Sink Score: 0.0614, Avg Sink Rate: 0.0000

Epoch 25/50, Loss: 1.0034, Norm Ratio: 0.0348, Avg Sink Score: 0.0614, Avg Sink Rate: 0.0000

Epoch 26/50, Loss: 0.9986, Norm Ratio: 0.0308, Avg Sink Score: 0.0613, Avg Sink Rate: 0.0000

Epoch 27/50, Loss: 1.0008, Norm Ratio: 0.0267, Avg Sink Score: 0.0613, Avg Sink Rate: 0.0000

Epoch 28/50, Loss: 1.0033, Norm Ratio: 0.0325, Avg Sink Score: 0.0613, Avg Sink Rate: 0.0000

Epoch 29/50, Loss: 1.0068, Norm Ratio: 0.0361, Avg Sink Score: 0.0613, Avg Sink Rate: 0.0000

Epoch 30/50, Loss: 1.0005, Norm Ratio: 0.0325, Avg Sink Score: 0.0613, Avg Sink Rate: 0.0000

Epoch 31/50, Loss: 1.0023, Norm Ratio: 0.0311, Avg Sink Score: 0.0612, Avg Sink Rate: 0.0000

Epoch 32/50, Loss: 1.0016, Norm Ratio: 0.0284, Avg Sink Score: 0.0611, Avg Sink

Rate: 0.0000
Epoch 33/50, Loss: 1.0026, Norm Ratio: 0.0264, Avg Sink Score: 0.0611, Avg Sink
Rate: 0.0000
Epoch 34/50, Loss: 1.0028, Norm Ratio: 0.0295, Avg Sink Score: 0.0611, Avg Sink
Rate: 0.0000
Epoch 35/50, Loss: 1.0044, Norm Ratio: 0.0264, Avg Sink Score: 0.0610, Avg Sink
Rate: 0.0000
Epoch 36/50, Loss: 1.0022, Norm Ratio: 0.0256, Avg Sink Score: 0.0609, Avg Sink
Rate: 0.0000
Epoch 37/50, Loss: 1.0007, Norm Ratio: 0.0263, Avg Sink Score: 0.0608, Avg Sink
Rate: 0.0000
Epoch 38/50, Loss: 1.0035, Norm Ratio: 0.0268, Avg Sink Score: 0.0607, Avg Sink
Rate: 0.0000
Epoch 39/50, Loss: 1.0021, Norm Ratio: 0.0262, Avg Sink Score: 0.0605, Avg Sink
Rate: 0.0000
Epoch 40/50, Loss: 1.0019, Norm Ratio: 0.0252, Avg Sink Score: 0.0604, Avg Sink
Rate: 0.0000
Epoch 41/50, Loss: 1.0015, Norm Ratio: 0.0256, Avg Sink Score: 0.0603, Avg Sink
Rate: 0.0000
Epoch 42/50, Loss: 0.9988, Norm Ratio: 0.0248, Avg Sink Score: 0.0602, Avg Sink
Rate: 0.0000
Epoch 43/50, Loss: 1.0032, Norm Ratio: 0.0256, Avg Sink Score: 0.0601, Avg Sink
Rate: 0.0000
Epoch 44/50, Loss: 0.9974, Norm Ratio: 0.0256, Avg Sink Score: 0.0600, Avg Sink
Rate: 0.0000
Epoch 45/50, Loss: 1.0036, Norm Ratio: 0.0226, Avg Sink Score: 0.0599, Avg Sink
Rate: 0.0000
Epoch 46/50, Loss: 0.9997, Norm Ratio: 0.0225, Avg Sink Score: 0.0598, Avg Sink
Rate: 0.0000
Epoch 47/50, Loss: 1.0008, Norm Ratio: 0.0271, Avg Sink Score: 0.0597, Avg Sink
Rate: 0.0000
Epoch 48/50, Loss: 1.0028, Norm Ratio: 0.0248, Avg Sink Score: 0.0595, Avg Sink
Rate: 0.0000
Epoch 49/50, Loss: 0.9979, Norm Ratio: 0.0251, Avg Sink Score: 0.0593, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 1.0009, Norm Ratio: 0.0250, Avg Sink Score: 0.0592, Avg Sink
Rate: 0.0000

Training completed in 5862.61 seconds.

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.07 GB | Metal 驱动分配: 2.16 GB

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.06 GB | Metal 驱动分配: 1.08 GB

Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.06 GB | Metal 驱动分配: 1.08 GB

Epoch 1/50, Loss: 1.0791, Norm Ratio: 0.2363, Avg Sink Score: 0.2882, Avg Sink Rate: 0.3833

Epoch 2/50, Loss: 0.9322, Norm Ratio: 0.2950, Avg Sink Score: 0.2736, Avg Sink Rate: 0.2903

Epoch 3/50, Loss: 0.8603, Norm Ratio: 0.4350, Avg Sink Score: 0.2572, Avg Sink Rate: 0.1875

Epoch 4/50, Loss: 0.8075, Norm Ratio: 0.4584, Avg Sink Score: 0.2414, Avg Sink Rate: 0.1364

Epoch 5/50, Loss: 0.7707, Norm Ratio: 0.5010, Avg Sink Score: 0.2263, Avg Sink Rate: 0.1059

Epoch 6/50, Loss: 0.7786, Norm Ratio: 0.4951, Avg Sink Score: 0.2133, Avg Sink Rate: 0.0857

Epoch 7/50, Loss: 0.7639, Norm Ratio: 0.5090, Avg Sink Score: 0.2016, Avg Sink Rate: 0.0720

Epoch 8/50, Loss: 0.7322, Norm Ratio: 0.5403, Avg Sink Score: 0.1926, Avg Sink Rate: 0.0621

Epoch 9/50, Loss: 0.7622, Norm Ratio: 0.5098, Avg Sink Score: 0.1844, Avg Sink Rate: 0.0545

Epoch 10/50, Loss: 0.7029, Norm Ratio: 0.5579, Avg Sink Score: 0.1771, Avg Sink Rate: 0.0486

Epoch 11/50, Loss: 0.6820, Norm Ratio: 0.5744, Avg Sink Score: 0.1702, Avg Sink Rate: 0.0439

Epoch 12/50, Loss: 0.6731, Norm Ratio: 0.5831, Avg Sink Score: 0.1640, Avg Sink Rate: 0.0400

Epoch 13/50, Loss: 0.6210, Norm Ratio: 0.6213, Avg Sink Score: 0.1583, Avg Sink Rate: 0.0367

Epoch 14/50, Loss: 0.6425, Norm Ratio: 0.6033, Avg Sink Score: 0.1534, Avg Sink Rate: 0.0340

Epoch 15/50, Loss: 0.6188, Norm Ratio: 0.6263, Avg Sink Score: 0.1487, Avg Sink Rate: 0.0316

Epoch 16/50, Loss: 0.5912, Norm Ratio: 0.6490, Avg Sink Score: 0.1448, Avg Sink

Rate: 0.0295
Epoch 17/50, Loss: 0.6237, Norm Ratio: 0.6166, Avg Sink Score: 0.1409, Avg Sink
Rate: 0.0277
Epoch 18/50, Loss: 0.5706, Norm Ratio: 0.6545, Avg Sink Score: 0.1374, Avg Sink
Rate: 0.0261
Epoch 19/50, Loss: 0.5770, Norm Ratio: 0.6559, Avg Sink Score: 0.1340, Avg Sink
Rate: 0.0247
Epoch 20/50, Loss: 0.5845, Norm Ratio: 0.6527, Avg Sink Score: 0.1309, Avg Sink
Rate: 0.0234
Epoch 21/50, Loss: 0.5433, Norm Ratio: 0.6775, Avg Sink Score: 0.1279, Avg Sink
Rate: 0.0222
Epoch 22/50, Loss: 0.5647, Norm Ratio: 0.6624, Avg Sink Score: 0.1251, Avg Sink
Rate: 0.0212
Epoch 23/50, Loss: 0.5379, Norm Ratio: 0.6831, Avg Sink Score: 0.1225, Avg Sink
Rate: 0.0202
Epoch 24/50, Loss: 0.5074, Norm Ratio: 0.7029, Avg Sink Score: 0.1201, Avg Sink
Rate: 0.0194
Epoch 25/50, Loss: 0.5467, Norm Ratio: 0.6778, Avg Sink Score: 0.1178, Avg Sink
Rate: 0.0186
Epoch 26/50, Loss: 0.5018, Norm Ratio: 0.7081, Avg Sink Score: 0.1156, Avg Sink
Rate: 0.0178
Epoch 27/50, Loss: 0.5041, Norm Ratio: 0.7057, Avg Sink Score: 0.1136, Avg Sink
Rate: 0.0171
Epoch 28/50, Loss: 0.5030, Norm Ratio: 0.7082, Avg Sink Score: 0.1115, Avg Sink
Rate: 0.0165
Epoch 29/50, Loss: 0.4671, Norm Ratio: 0.7335, Avg Sink Score: 0.1096, Avg Sink
Rate: 0.0159
Epoch 30/50, Loss: 0.4845, Norm Ratio: 0.7197, Avg Sink Score: 0.1078, Avg Sink
Rate: 0.0154
Epoch 31/50, Loss: 0.4578, Norm Ratio: 0.7373, Avg Sink Score: 0.1060, Avg Sink
Rate: 0.0149
Epoch 32/50, Loss: 0.4337, Norm Ratio: 0.7541, Avg Sink Score: 0.1044, Avg Sink
Rate: 0.0144
Epoch 33/50, Loss: 0.4702, Norm Ratio: 0.7292, Avg Sink Score: 0.1028, Avg Sink
Rate: 0.0140
Epoch 34/50, Loss: 0.4324, Norm Ratio: 0.7545, Avg Sink Score: 0.1013, Avg Sink
Rate: 0.0135

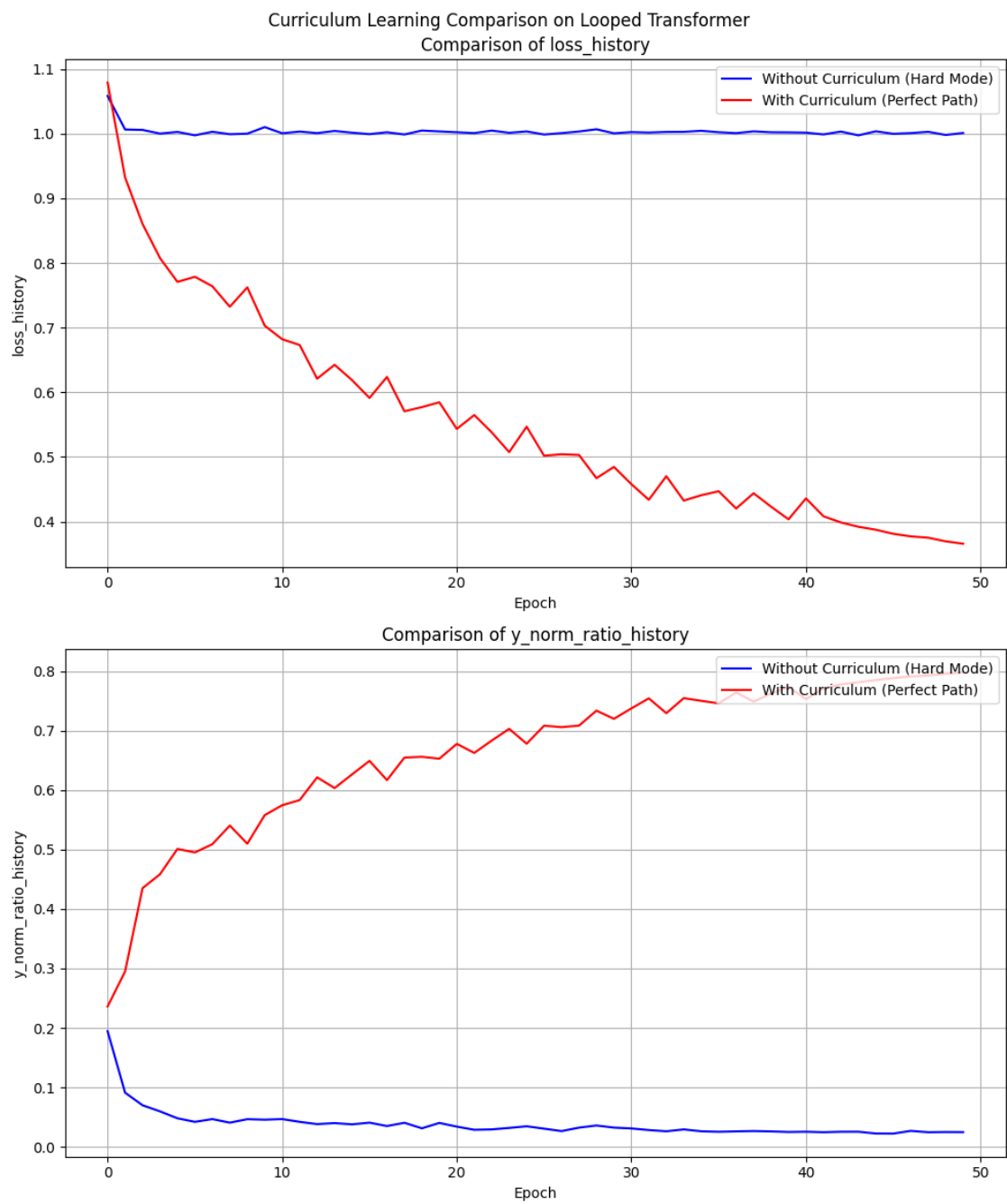
Epoch 35/50, Loss: 0.4407, Norm Ratio: 0.7499, Avg Sink Score: 0.0999, Avg Sink Rate: 0.0131
Epoch 36/50, Loss: 0.4469, Norm Ratio: 0.7456, Avg Sink Score: 0.0986, Avg Sink Rate: 0.0128
Epoch 37/50, Loss: 0.4202, Norm Ratio: 0.7639, Avg Sink Score: 0.0972, Avg Sink Rate: 0.0124
Epoch 38/50, Loss: 0.4438, Norm Ratio: 0.7486, Avg Sink Score: 0.0959, Avg Sink Rate: 0.0121
Epoch 39/50, Loss: 0.4230, Norm Ratio: 0.7621, Avg Sink Score: 0.0947, Avg Sink Rate: 0.0118
Epoch 40/50, Loss: 0.4035, Norm Ratio: 0.7733, Avg Sink Score: 0.0935, Avg Sink Rate: 0.0115
Curriculum ended at epoch 41, step 1.
Epoch 41/50, Loss: 0.4358, Norm Ratio: 0.7527, Avg Sink Score: 0.0923, Avg Sink Rate: 0.0112
Epoch 42/50, Loss: 0.4082, Norm Ratio: 0.7704, Avg Sink Score: 0.0912, Avg Sink Rate: 0.0109
Epoch 43/50, Loss: 0.3986, Norm Ratio: 0.7775, Avg Sink Score: 0.0902, Avg Sink Rate: 0.0107
Epoch 44/50, Loss: 0.3920, Norm Ratio: 0.7811, Avg Sink Score: 0.0892, Avg Sink Rate: 0.0104
Epoch 45/50, Loss: 0.3873, Norm Ratio: 0.7848, Avg Sink Score: 0.0883, Avg Sink Rate: 0.0102
Epoch 46/50, Loss: 0.3810, Norm Ratio: 0.7882, Avg Sink Score: 0.0874, Avg Sink Rate: 0.0099
Epoch 47/50, Loss: 0.3771, Norm Ratio: 0.7910, Avg Sink Score: 0.0865, Avg Sink Rate: 0.0097
Epoch 48/50, Loss: 0.3750, Norm Ratio: 0.7924, Avg Sink Score: 0.0857, Avg Sink Rate: 0.0095
Epoch 49/50, Loss: 0.3694, Norm Ratio: 0.7954, Avg Sink Score: 0.0850, Avg Sink Rate: 0.0093
Epoch 50/50, Loss: 0.3657, Norm Ratio: 0.7977, Avg Sink Score: 0.0843, Avg Sink Rate: 0.0091

Training completed in 3460.71 seconds.

[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.09 GB | Metal 驱动分配: 5.65 GB

[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.07 GB | Metal 驱动分配: 2.09 GB

Experiment 2/2 completed.



4.1.2 保存模型权重

```
[212]: import torch
import os
os.makedirs('saved_checkpoints', exist_ok=True)
print("正在保存模型权重...")
for i, exp in enumerate(final_table.experiments):
    exp_name = final_table.init_parameters[i].get('experiment_name', 'Experiment_{i}')
    # 清理文件名中的特殊字符
    safe_name = exp_name.replace(' ', '_').replace('(', '').replace(')', '')
    save_path = f'saved_checkpoints/{safe_name}.pth'
    checkpoint = {
        'model_state_dict': exp.model.state_dict(),
        'loss_history': getattr(exp, 'loss_history', []),
        'y_pred_norm_history': getattr(exp, 'y_pred_norm_history', []),
        'y_true_norm_history': getattr(exp, 'y_true_norm_history', []),
        'sink_score_history': getattr(exp, 'sink_score_history', []),
        'sink_rate_history': getattr(exp, 'sink_rate_history', []),
        'residual_gate_history': getattr(exp, 'residual_gate_history', []),
    }
    if hasattr(exp, 'optimizer'):
        checkpoint['optimizer_state_dict'] = exp.optimizer.state_dict()
    torch.save(checkpoint, save_path)
    print(f"Checkpoint 已安全保存至: {save_path}")
```

正在保存模型权重...

Checkpoint 已安全保存至:

saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth

Checkpoint 已安全保存至:

saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth

4.1.3 加载模型、评估结果

```
[ ]: manual_config = dict(
    # 任务基础配置
    data_type='linear',
    d_x=20,
```

```

seq_len=80,
batch_size=64,

# 模型架构 (完全对齐论文)
num_blocks=20,
num_eff=15,
d_model=256,
num_heads=8,
pe_type=['learned_ape'],

# 核心物理机制配置
x_init='zero',          # 绝对零起点
init_std='auto',        # 方差自适应 (1/sqrt(256))
residual_gate=(1, 1),    # 标准注入
residual_gate_type='fixed',

# 优化器剥离外挂
optimizer_type='adam',
lr=1e-4,
layer_weight_decay=1.0,  # 关闭层级惩罚
seq_weight_decay=1.0,    # 关闭时间步惩罚
scheduler_type=None,     # 恒定学习率

# 运行控制
epochs=50,
steps_per_epoch=200,
print_every=1,

scheduled_training=True  # 开启 b 的 Kick-start
)

test_table = ExperimentTable(params_groups=[
    {
        'experiment_name': 'Linear Without Curriculum (Hard Mode)',
        'curriculum': {}, # 空字典关闭课程学习
    }
])

```

```

        'load_path': 'saved_checkpoints/Without_Curriculum_Hard_Mode_weights.
    ↪pth'
    },
    {
        'experiment_name': 'Linear With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8 # 前 40 个 epoch 完成课程
        },
        'load_path': 'saved_checkpoints/With_Curriculum_Perfect_Path_weights.
    ↪pth'
    }
], manual=manual_config)

results_lists = [
    ([ '1_ID_Baseline_sink_scores', '2_OOD_Scale_x2_sink_scores', ↪
    ↪ '3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    ([ '1_ID_Baseline_y_norm_ratio', '2_OOD_Scale_x2_y_norm_ratio', ↪
    ↪ '3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    ([ '1_ID_Baseline_loss', '2_OOD_Scale_x2_loss', ↪
    ↪ '3_OOD_Seq_Extrapolation_loss'], 'experiment')
]

eval_configs = [
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}}, # 分布内基准
    {'eval_name': '2_OOD_Scale_x2', 'ood_kwargs': {'x_scale': 2.0}}, # X 尺度放
    大 (测试抗干扰)
    {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
    ↪2}} # 序列长度外推 (测试算法泛化)
]
test_table.run(result_lists=results_lists, modes=['evaluate'], ↪
    ↪ eval_configs=eval_configs)
test_table.plot(compare_experiments=False, subplot_shape=(-1,1),suptitle='')

```

检测到纯权重 state_dict 文件 (或历史版本), 直接加载 model_state_dict...

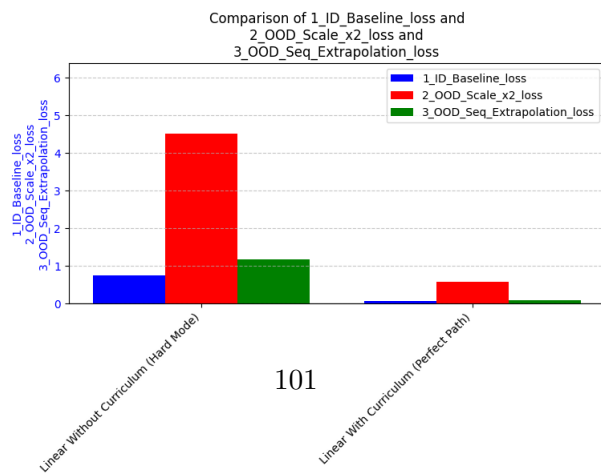
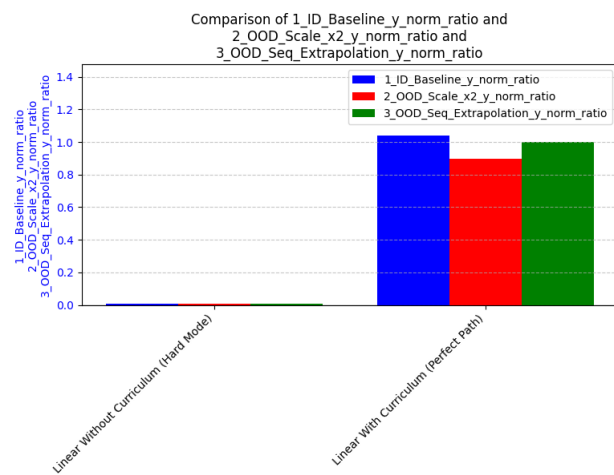
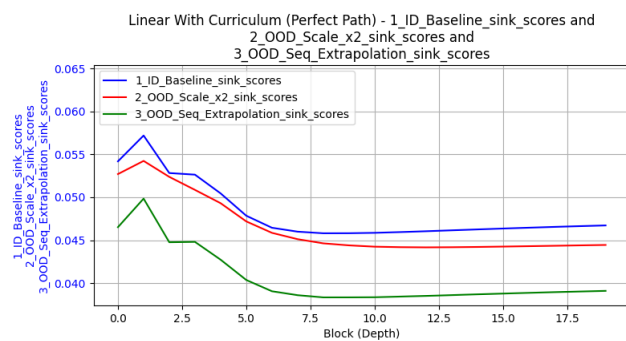
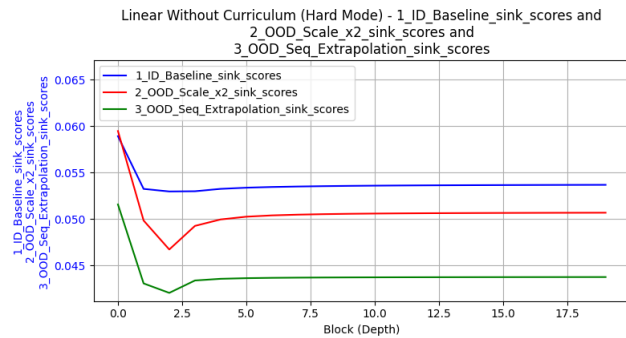
成功从 saved_checkpoints/Without_Curriculum_Hard_Mode_weights.pth 恢复实验状态。
Linear Without Curriculum (Hard Mode) initialized in 0.06 seconds. Using device:
mps

检测到纯权重 state_dict 文件 (或历史版本), 直接加载 model_state_dict...

成功从 saved_checkpoints/With_Curriculum_Perfect_Path_weights.pth 恢复实验状态。
Linear With Curriculum (Perfect Path) initialized in 0.03 seconds. Using device:
mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.11 GB | Metal 驱动分配: 2.07 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.11 GB | Metal 驱动分配: 2.07 GB
[1_ID_Baseline] Loss: 0.7418, Norm Ratio: 0.0097
[2_OOD_Scale_x2] Loss: 4.5082, Norm Ratio: 0.0054
[3_OOD_Seq_Extrapolation] Loss: 1.1665, Norm Ratio: 0.0102
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.13 GB | Metal 驱动分配: 2.07 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.14 GB | Metal 驱动分配: 2.07 GB
[1_ID_Baseline] Loss: 0.0728, Norm Ratio: 1.0402
[2_OOD_Scale_x2] Loss: 0.5680, Norm Ratio: 0.8993
[3_OOD_Seq_Extrapolation] Loss: 0.0907, Norm Ratio: 0.9985
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.16 GB | Metal 驱动分配: 2.07 GB
Experiment 2/2 completed.



4.2 nonlinear

4.2.1 任务配置

```
[60]: manual_config = dict(  
    # 任务基础配置  
    data_type='nonlinear',  
    function_callable=lambda x: 2**0.5 *F.relu(x),  
    max_seq_len=200,  
    d_x=20,  
    seq_len=80,  
    batch_size=64,  
  
    # 模型架构 (完全对齐论文)  
    num_blocks=20,  
    num_eff=15,  
    d_model=256,  
    num_heads=8,  
    pe_type=['learned_ape'],  
  
    # 核心物理机制配置  
    x_init='zero', # 绝对零起点  
    init_std='auto', # 方差自适应 (1/sqrt(256))  
    residual_gate=(1, 1), # 标准注入  
    residual_gate_type='fixed',  
  
    # 优化器剥离外挂  
    optimizer_type='adam',  
    lr=1e-4,  
    layer_weight_decay=1.0, # 关闭层级惩罚  
    seq_weight_decay=1.0, # 关闭时间步惩罚  
    scheduler_type=None, # 恒定学习率  
  
    # 运行控制  
    epochs=50,
```

```

steps_per_epoch=200,
print_every=5,

scheduled_training=True,      # 开启 b 的 Kick-start
save_path='auto',
load_path='auto'
)

param_groups = [
    {
        'experiment_name': 'Nonlinear Without Curriculum (Hard Mode)',
        'curriculum': {} # 空字典关闭课程学习
    },
    {
        'experiment_name': 'Nonlinear With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8 # 前 40 个 epoch 完成课程
        }
    }
]

results_lists = [
    (['1_ID_Baseline_sink_scores', '2_OOD_Scale_x2_sink_scores',
↪ '3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    (['1_ID_Baseline_y_norm_ratio', '2_OOD_Scale_x2_y_norm_ratio',
↪ '3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    (['1_ID_Baseline_loss', '2_OOD_Scale_x2_loss',
↪ '3_OOD_Seq_Extrapolation_loss'], 'experiment')
]

eval_configs = [
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}}, # 分布内基准
    {'eval_name': '2_OOD_Scale_x2', 'ood_kwargs': {'x_scale': 2.0}}, # x 尺度放
大 (测试抗干扰)

```

```

        {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
↪2}} # 序列长度外推 (测试算法泛化)
]

```

4.2.2 训练

```

[26]: nonlinear_table = ExperimentTable(params_groups=param_groups,
↪manual=manual_config)
nonlinear_table.run(result_lists=[(['loss_history'], 'epoch'),
↪(['y_norm_ratio_history'], 'epoch')])
nonlinear_table.plot(figure_size=(10, 12), compare_experiments=True,
↪subplot_shape=(-1, 1), subtitle='Nonlinear Curriculum Learning Comparison on
↪Looped Transformer')

```

找不到文件 saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth, 跳过加载。

Nonlinear Without Curriculum (Hard Mode) initialized in 0.49 seconds. Using device: mps

找不到文件 saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth, 跳过加载。

Nonlinear With Curriculum (Perfect Path) initialized in 0.02 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Epoch 5/50, Loss: 0.7411, Norm Ratio: 0.5209, Avg Sink Score: 0.0620, Avg Sink Rate: 0.0000

Epoch 10/50, Loss: 0.7376, Norm Ratio: 0.5189, Avg Sink Score: 0.0683, Avg Sink Rate: 0.0000

Epoch 15/50, Loss: 0.7305, Norm Ratio: 0.5225, Avg Sink Score: 0.0745, Avg Sink Rate: 0.0000

Epoch 20/50, Loss: 0.7315, Norm Ratio: 0.5204, Avg Sink Score: 0.0926, Avg Sink Rate: 0.0000

Epoch 25/50, Loss: 0.7207, Norm Ratio: 0.5260, Avg Sink Score: 0.0901, Avg Sink Rate: 0.0000

Epoch 30/50, Loss: 0.6593, Norm Ratio: 0.5865, Avg Sink Score: 0.0537, Avg Sink Rate: 0.0000

Epoch 35/50, Loss: 0.5928, Norm Ratio: 0.6362, Avg Sink Score: 0.0412, Avg Sink

Rate: 0.0000
Epoch 40/50, Loss: 0.5718, Norm Ratio: 0.6509, Avg Sink Score: 0.0415, Avg Sink
Rate: 0.0000
Epoch 45/50, Loss: 0.5614, Norm Ratio: 0.6618, Avg Sink Score: 0.0383, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 0.5496, Norm Ratio: 0.6730, Avg Sink Score: 0.0425, Avg Sink
Rate: 0.0000
Training completed in 5794.94 seconds.
Checkpoint 已安全保存至:
saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.23 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 1.05 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 1.07 GB
Epoch 5/50, Loss: 0.7774, Norm Ratio: 0.4921, Avg Sink Score: 0.1907, Avg Sink
Rate: 0.1118
Epoch 10/50, Loss: 0.7188, Norm Ratio: 0.5427, Avg Sink Score: 0.1222, Avg Sink
Rate: 0.0000
Epoch 15/50, Loss: 0.6781, Norm Ratio: 0.5789, Avg Sink Score: 0.0782, Avg Sink
Rate: 0.0000
Epoch 20/50, Loss: 0.6299, Norm Ratio: 0.6089, Avg Sink Score: 0.0646, Avg Sink
Rate: 0.0000
Epoch 25/50, Loss: 0.6095, Norm Ratio: 0.6223, Avg Sink Score: 0.0571, Avg Sink
Rate: 0.0000
Epoch 30/50, Loss: 0.5900, Norm Ratio: 0.6420, Avg Sink Score: 0.0521, Avg Sink
Rate: 0.0000
Epoch 35/50, Loss: 0.5725, Norm Ratio: 0.6545, Avg Sink Score: 0.0479, Avg Sink
Rate: 0.0000
Epoch 40/50, Loss: 0.5549, Norm Ratio: 0.6664, Avg Sink Score: 0.0431, Avg Sink
Rate: 0.0000
Curriculum ended at epoch 41, step 1.
Epoch 45/50, Loss: 0.5380, Norm Ratio: 0.6804, Avg Sink Score: 0.0437, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 0.5280, Norm Ratio: 0.6893, Avg Sink Score: 0.0466, Avg Sink
Rate: 0.0000
Training completed in 5309.81 seconds.

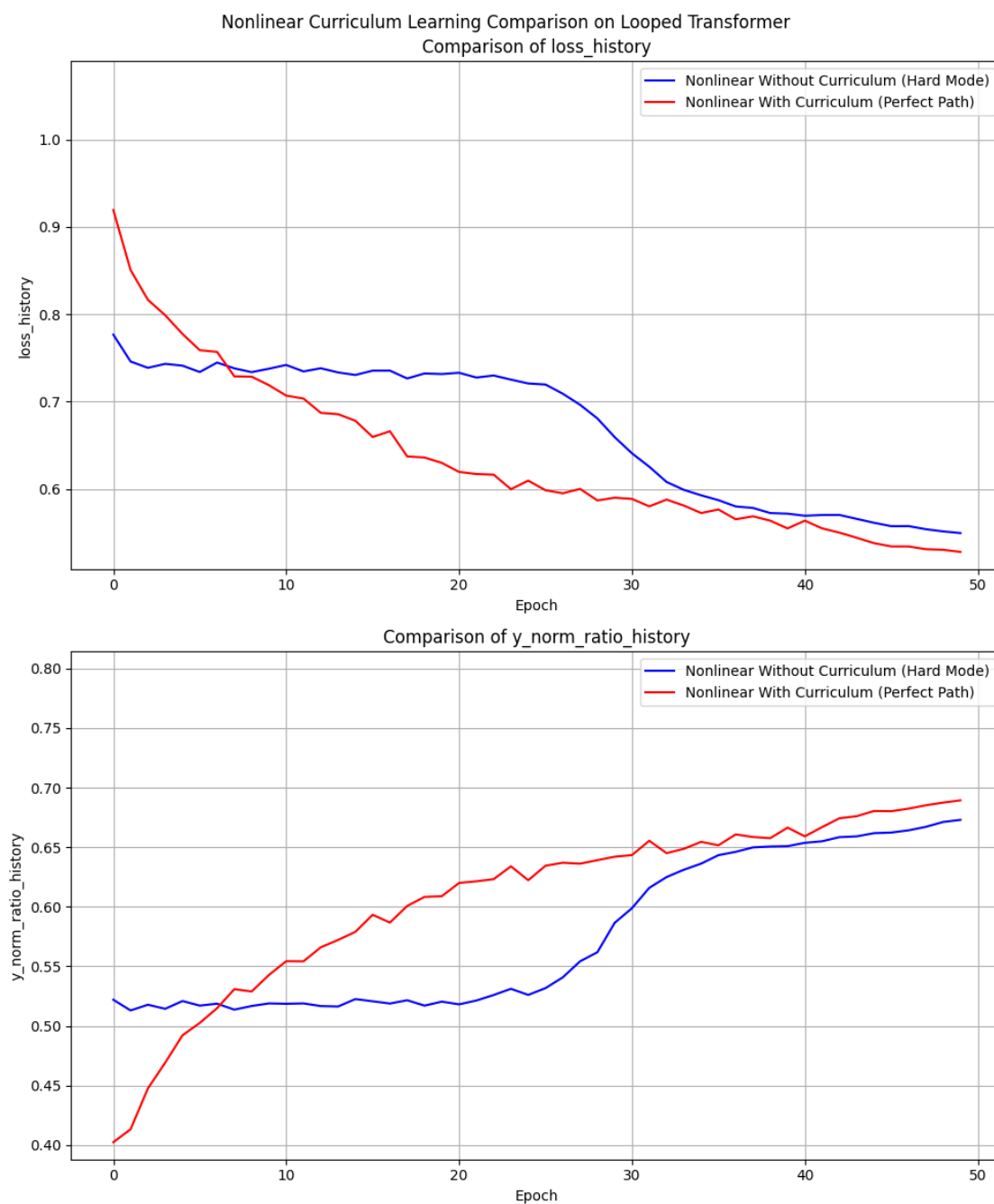
Checkpoint 已安全保存至:

saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth

[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.04 GB | Metal 驱动分配: 5.62 GB

[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.07 GB

Experiment 2/2 completed.



4.2.3 评估

```
[61]: test_table = ExperimentTable(params_groups=param_groups, manual=manual_config)
test_table.run(result_lists=results_lists, modes=['evaluate'],
               eval_configs=eval_configs)
test_table.plot(compare_experiments=False, subplot_shape=(-1,1),suptitle='')
```

成功从 saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth 恢复实验状态。

Nonlinear Without Curriculum (Hard Mode) initialized in 0.10 seconds. Using
device: mps

成功从 saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth 恢复实验状态。

Nonlinear With Curriculum (Perfect Path) initialized in 0.03 seconds. Using
device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.02 GB

[1_ID_Baseline] Loss: 0.4018, Norm Ratio: 0.7298

[2_OOD_Scale_x2] Loss: 2.0818, Norm Ratio: 0.9717

[3_OOD_Seq_Extrapolation] Loss: 0.3924, Norm Ratio: 0.7338

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 1.00 GB

Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 1.02 GB

[1_ID_Baseline] Loss: 0.4543, Norm Ratio: 0.6583

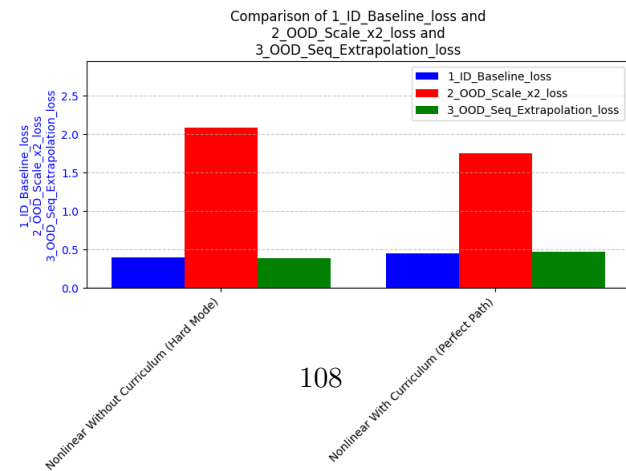
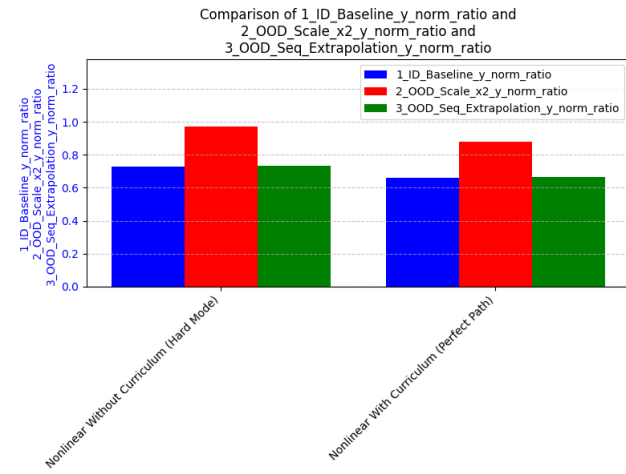
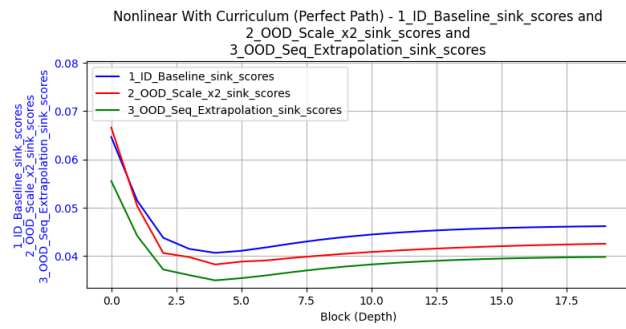
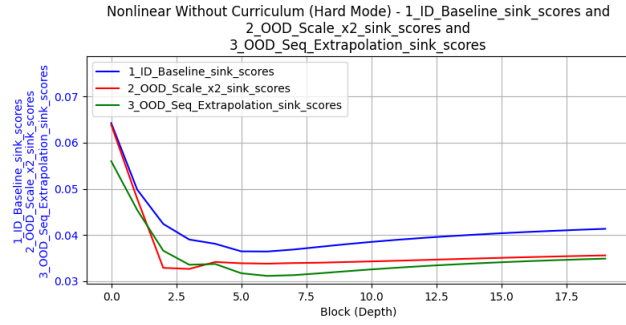
[2_OOD_Scale_x2] Loss: 1.7523, Norm Ratio: 0.8792

[3_OOD_Seq_Extrapolation] Loss: 0.4746, Norm Ratio: 0.6668

[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.05 GB | Metal 驱动分配: 1.00 GB

Experiment 2/2 completed.

No Lorenz data cache found.



5 Lorenz Attractor 实验

```
[210]: create_lorenz_pool(pool_size=800000, traj_len=1000, dt=0.01, sigma=10.0, beta=8/  
      ↪3, rho=28.0, save_path='auto')
```

检测到已有数据文件，正在读取并进行增量拼接...

增量拼接成功！数据池规模由 111200 条扩大至 911200 条。

数据池已安全写入本地：data/lorenz/length_1000_dt0.01_sigma10.0_beta2.7_rho28.0.pth

5.1 实验

```
[ ]: # %% [markdown]  
# ## Lorenz Attractor Looped Transformer 实验  
# %% [markdown]  
# ### 任务配置  
  
# %%  
import torch  
import torch.nn.functional as F  
# 确保在运行此脚本前，你已经在此环境中运行或 import 了你的 ExperimentTable,   
↪ default_setup 等模块  
  
# 提示：在这里填入你刚刚定量生成的离线池本地路径  
MY_LORENZ_POOL_PATH = 'data/lorenz/length_1000_dt0.01_sigma10.0_beta2.7_rho28.0.  
↪pth'  
  
manual_config = dict(  
    # 1. 【核心修改】任务基础配置切换为 Lorenz  
    data_type='lorenz',  
    d_x=3, # 洛伦兹状态空间的绝对物理维度 (X, Y, Z)  
    d_y=3, # 预测目标同样是完整的三维下一时刻状态  
    max_seq_len=500,  
    seq_len=300, # 上下文包含 150 个 (x, y) 对  
    batch_size=64,
```

```

# 模型架构 (完全对齐论文与你的默认优秀设定)
num_blocks=20,
num_eff=15,
d_model=256,
num_heads=8,
pe_type=['learned_ape'],      # 使用你在线性任务中表现优异的 Learned APE

# 核心物理机制配置
x_init='zero',                # 绝对零起点, 迫使模型完全依赖 Prompt 的 ICL 驱动
init_std='auto',              # 方差自适应 ( $1/\sqrt{256}$ )
residual_gate=(1, 1),         # 标准残差注入
residual_gate_type='fixed',

# 【核心修改】接入洛伦兹数据代入的特定物理参数
lorenz_kwargs=dict(dt=0.01, burn_in=500),
load_lorenz_from=MY_LORENZ_POOL_PATH,    # 载入你的高密度离线系综池

# 优化器剥离外挂
optimizer_type='adam',
lr=1e-4,
layer_weight_decay=1.0,      # 关闭层级惩罚
seq_weight_decay=1.0,        # 关闭时间步惩罚
scheduler_type=None,         # 恒定学习率

# 运行控制
epochs=50,
steps_per_epoch=200,
print_every=5,

scheduled_training=True,     # 开启层数 b 的 Kick-start 渐进
save_path='auto',
load_path='auto'
)

param_groups = [
{

```

```

        'experiment_name': 'Lorenz Without Curriculum (Hard Mode)',
        'curriculum': {} # 空字典关闭课程学习，一上来就啃 80 步长序列
    },
    {
        'experiment_name': 'Lorenz With Curriculum (Perfect Path)',
        # 【理论调整点】：删除了带有隐患的 'd_x' 渐变，仅对时序步长进行课程演进
        'curriculum': {
            'seq_len': 20, # 课程从极短的 20 步（10 对点）开始，降低初始积分关联难度
            'duration_ratio': 0.8 # 前 30 个 epoch 随着线性进度逐步拉长到 300 步
        }
    }
]

# 用于最终多场景 OOD 综合性能度量的指标看板
results_lists = [
    (['1_ID_Baseline_sink_scores', '2_OOD_Param_Shift_sink_scores', ↵
    ↵ '3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    (['1_ID_Baseline_y_norm_ratio', '2_OOD_Param_Shift_y_norm_ratio', ↵
    ↵ '3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    (['1_ID_Baseline_loss', '2_OOD_Param_Shift_loss', ↵
    ↵ '3_OOD_Seq_Extrapolation_loss'], 'experiment')
]

# 针对混沌混沌特征量身定制的 OOD 测试集
eval_configs = [
    # 1. 分布内基准（使用在线即时生成，检验模型对流形任一位置的单步预测）
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}},

    # 2. 物理流形参数偏移 (Parameter Shift)
    # 将瑞利数 rho 向上偏移 5.0（变成 33.0），测试网络是否掌握了隐式微分算子，还是只会背吸引子几何外形
    {'eval_name': '2_OOD_Param_Shift', 'ood_kwargs': {'rho_shift': 5.0}},

    # 3. 混沌时序长度外推 (Sequence Extrapolation)
    # 将序列长度外推 20%，跨入更长历史的确定性窗口，测试注意力机制的泛化边界

```

```

        {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
↪2}}
]

# %% [markdown]
# ### 训练环节

```

```

[ ]: # %%
lorenz_table = ExperimentTable(params_groups=param_groups, manual=manual_config)
# 启动多实验并行，开启你的系综平均训练
lorenz_table.run(result_lists=[(['loss_history'], 'epoch'), ↵
↪(['y_norm_ratio_history'], 'epoch')])

# 可视化两个实验在训练过程中的 Loss 收敛轨迹
lorenz_table.plot(subplot_shape=(-1, 1),
                  subtitle='Lorenz Attractor Dynamic Curriculum Learning ↵
↪Comparison')

# %% [markdown]
# ### 评估环节（自动化零线对齐多指标可视化）

```

```

[ ]: # %%
test_table = ExperimentTable(params_groups=param_groups, manual=manual_config)
# 开启评估模式。注意：评估时代码会自动切换为直接在线生成，保证测试样本的“纯净度”
test_table.run(result_lists=results_lists, modes=['evaluate'], ↵
↪eval_configs=eval_configs, parallel_workers=2)

# 打印最终结果，查看在物理参数偏移与长序列外推下，有无课程学习对 Attention Sink 汇
聚率的长期负荷影响
test_table.plot(compare_experiments=False, subplot_shape=(-1,1), subtitle='OOD ↵
↪Evaluation Metrics on Chaotic Systems')

```